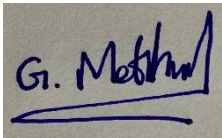


Name: **Samarasinghe Methmal**

Student Reference Number: **10898561**

Module Code: PUSL3190	Module Name: Computing Project
Coursework Title: Final Report	
Deadline Date: 05/05/2025	Member of staff responsible for coursework: Mr. Gayan Perera
Programme: BSc. (Hons) Software Engineering	
Please note that University Academic Regulations are available under Rules and Regulations on the University website www.plymouth.ac.uk/studenthandbook .	
Group work: please list all names of all participants formally associated with this work and state whether the work was undertaken alone or as part of a team. Please note you may be required to identify individual responsibility for component parts.	
<i>We confirm that we have read and understood the Plymouth University regulations relating to Assessment Offences and that we are aware of the possible penalties for any breach of these regulations. We confirm that this is the independent work of the group.</i> Signed on behalf of the group:	
Individual assignment: <i>I confirm that I have read and understood the Plymouth University regulations relating to Assessment Offences and that I am aware of the possible penalties for any breach of these regulations. I confirm that this is my own independent work.</i> Signed: 	
Use of translation software: failure to declare that translation software or a similar writing aid has been used will be treated as an assessment offence.	
I * have used <u>not used</u> translation software.	
If used, please state name of software.....	
Overall mark _____ % Assessors Initials _____ Date _____	



UNIVERSITY OF
PLYMOUTH

PUSL3190 – Computing Project

Final Report

MoodSync – Mental Health Monitoring System

Supervisor: Mr. Gayan Perera

Name: Samarasinghe Methmal

Plymouth Index Number: 10898561

Degree Program: BSc. (Hons) Software Engineering

Acknowledgement

I extend my gratitude to my supervisor, Mr. Gayan Perera, for his expert guidance, unconditional support, and constructive comments, which were key in tackling complex tasks in this project, MoodSync. His mentorship has been invaluable to me in both my academic growth and the successful development of Project MoodSync - Mental Health Monitoring System.

I am deeply indebted to Dr. Saman Weerawardana from the National Institute of Mental Health (NIMH), Angoda, whose mental health clinical expertise and insights poured into developing this mental healthcare framework that shaped the therapy components of this application. The same appreciation goes to Dr. Deshantha Pathinayake from Wellness Medi Clinic, Moratuwa for his contributions to the development and tuning of the ideas of the system for Sri Lankan culture enhancing its clinical relevance.

My sincere thanks go to the administration and staff at Colombo South Teaching Hospital—Kalubowila for providing the thresholds, data collection approval, and education on user privacy, which helped form the backbone of this project's machine learning models.

I would also like to acknowledge the Faculty of Computing at NSBM Green University, especially the module leader Dr. Mohammed Shafriz for providing guidance, resources, and environment to innovate. Finally, I express my heartfelt appreciation to family and friends, whose patience, understanding and encouragement kept me encouraged during this challenging journey.

Abstract

Mental health problems affect millions of people in the world, with barriers like stigma and lack of resources especially in Sri Lanka. This project develops a comprehensive mental health monitoring system that uses machine learning for sentiment analysis in user content to detect problems early and provide timely support. The system uses FARM Stack (FastAPI, React.js and MongoDB) with Natural Language Processing (NLP) techniques like Term Frequency-Inverse Document Frequency (TF-IDF) vectorization and Synthetic Minority Over-Sampling Technique (SMOTE) to handle imbalances in the data. Key innovations include multi-source data collection from user inputs/social media, Machine Learning (ML)-driven sentiment analysis with emotional state classification, anonymous professional support connections to users addressing cultural stigma, a well tiered real-time alert system for crisis prevention, and culturally tuned content that is specific to Sri Lankan users. This software integrates visualizing tools to track mental health trends and connects users and mental health professionals while maintaining anonymity. Testing showed technical performance metrics (93% accuracy, precision, recall) and user experience through a pilot study with participants and also with gathered information from a professional mental health doctor. This project solution addresses critical gaps in mental health support by combining modern advanced machine learning techniques with practical support networks. The cultural fit-in with Sri Lankan users showcases a modern approach to match mental health technology. This system can access mental health resources, reduce stigma through anonymous supporting channels and prevent self-threatening mental health illnesses through early recognition and intervention.

Keywords - Mental Health Informatics, Machine Learning, Sentiment Analysis, Crisis Intervention, Sri Lankan Healthcare.

Table of Contents

Acknowledgement.....	1
Abstract.....	2
1. Introduction	10
2. Background, Objectives and Deliverables	12
2.1. Background	12
2.2. Objectives	13
1. Sentiment Analysis Implementation.....	13
2. Personalized Mental Health Recommendations.....	13
3. Anonymous Communication	13
4. Developing a Real-Time Alert System.....	13
5. Mental Health Trends Visualization	13
6. Integrate Support Channels.....	14
7. Promote Accessibility and Inclusivity.....	14
8. Ensure Data Security and Privacy.....	14
2.3. Deliverables	15
2.3.1. Software Components	15
2.3.2. Additional Resources	18
3. Literature Review.....	19
3.1. Research Gap	23
4. Method of Approach.....	25
4.1. Development Methodology.....	25
4.2. System Architecture Overview.....	26
4.3. Data Collection and Processing.....	29

4.4.	Implementation Strategy	31
5.	Requirements	33
5.1.	Functional Requirements	33
5.2.	Non-Functional Requirements	35
5.3.	Hardware / Software Requirements	37
5.4.	Networking Requirements	38
6.	Design and Implementation	39
6.1.	System Architecture	39
6.2.	Database Design	41
6.3.	User Interface Design	43
6.3.1.	User Interface	43
6.3.2.	Professional Interface	54
6.3.3.	Admin Interface	58
6.4.	Machine Learning Implementation	61
6.5.	Key Components Implementation	63
6.5.1.	Sentiment Analysis Module	63
6.5.2.	Anonymous Communication System	64
6.5.3.	Alert System	65
6.5.4.	Visualization Module	66
6.6.	Security Implementation	67
7.	Testing and Evaluation	69
7.1.	Testing Methodology	69
7.2.	Unit Testing	71
7.3.	Integration Testing	73
7.4.	User Acceptance Testing	74
7.5.	Machine Learning Model Evaluation	75
7.6.	Performance and Security Testing	76

8. End – Project Report	78
8.1. Objective Evaluation	78
8.2. Key Achievements.....	80
8.3. Challenges and Adaptions	81
8.4. Limitations and Future Work.....	82
 9. Project Post – Mortem.....	 83
9.1. Evaluation	83
9.2. Development Process	83
9.3. Tech Stack Evaluation.....	84
9.4. Future Recommendations	84
 10. Conclusions	 86
10.1. Summary of Achievements	86
10.2. Impact Assessment	86
10.3. Final Remarks	87
 References	 88
Bibliography.....	90
Appendices	91
User Guide	91
 Project Initiation Document (PID).....	 96
1. Introduction	96
2. Business Case	97
2.1. Business Need	97
2.2. Business Objectives.....	98
3. Project Objectives.....	99
4. Literature Review	101

4.1. Research Gap.....	107
5. Method of Approach	111
5.1. Technical Approach	111
5.2. Project Management Approach.....	114
5.3. Functional Components Approach.....	115
5.4. System Architecture Approach	116
6. Initial Project Plan.....	117
7. Risk Analysis.....	119
7.1. Technical Risks	119
7.2. Project Management Risks	120
7.3. Financial and Resource Risks.....	120
7.4. Data Privacy and Security Risks.....	121
7.5. User Adoption Risk	122
7.6. Ethical Risks.....	123
8. Additional Sections	124
8.1. Stakeholder Analysis	124
8.2. Ethical Considerations	126
8.3. Communication Plan	126
References.....	127
 Interim Report	 130
1. Introduction	130
1.1. Introduction.....	130
1.2. Problem Definition.....	131
1.3. Project Objectives	132
2. System Analysis	134
2.1. Facts Gathering Techniques.....	134
2.2. Existing System	136
2.3. Use Case Diagram	137
2.4. Drawbacks of the Existing System	138

3. Requirements Specification	139
3.1. Functional Requirements	139
3.2. Non-Functional Requirements	141
3.3. Hardware / Software Requirements	143
3.4. Networking Requirements	144
4. Feasibility Study	145
4.1. Operational Feasibility	145
4.2. Technical Feasibility	147
4.3. Outline Budget	149
5. System Architecture	150
5.1. Class Diagram of Proposed System	150
5.2. Database Diagram	151
5.3. High – Level Architectural Diagram	152
5.4. Networking Diagram	153
6. Development Tools and Techniques	154
6.1. Development Methodology	154
6.2. Programming Languages and Tools	155
6.3. Third Party Components and Libraries	158
6.4. Algorithms	159
7. Discussion	160
7.1. Overview of the Interim Report	160
7.2. Summary of the Report	160
7.3. Challenges Faced	160
7.4. Future Plans / Upcoming Work	161
References	162
 Records of Supervisory Meetings	 163
Other Materials	170
Doctor Approval Letter	170
Meeting Minutes	171

Wireframes UI Designs	175
Test Cases	177
Hyperparameter Tuning.....	178
Project Links	179

List of Figures

Figure 1 - Sprint Structure	25
Figure 2 – Flowchart.....	32
Figure 3 - Chatbot Interface	45
Figure 4 - Support Services	46
Figure 5 - Doctor Channeling	46
Figure 6 - Doctor Details.....	47
Figure 7 - Health Trends Visualization.....	48
Figure 8 - Emergency	48
Figure 9 - Emergency Selection	49
Figure 10 - Community Support	49
Figure 11 - Login Page	50
Figure 12 - Error Page	50
Figure 13 - User Profile Page.....	51
Figure 14 - Mood Logger Page	51
Figure 15 - Health Insights.....	52
Figure 16 - Journal Entries Page.....	52
Figure 17 - Mobile Responsivity	53
Figure 18 - Doctor Dashboard.....	55
Figure 19 - Patient Management.....	55
Figure 20 - Doctor / Patient Chat.....	56
Figure 21 - Patient Analytics	56
Figure 22- Doctor Settings	57
Figure 23 - Admin Dashboard	59
Figure 24 - User Management	59
Figure 25 - Alert Management.....	60
Figure 26 - Confidence Scores	63
Figure 27 - Anonymous Communication Detailed	64
Figure 28 - CI/CD Pipeline using GitHub Actions	69
Figure 29 - Docker Deployment	70

Figure 30 - Frontend Unit Testing (ex: Analytics, Authentication)	71
Figure 31 - Backend Unit Testing (Machine Learning Model)	72
Figure 32 - Integration Testing (Model).....	73
Figure 33 – Model Performance Test	76
Figure 34 - API's Performance Testing	77
Figure 35 – Security Testing	77
Figure 36 - Stress Testing Graph (174 Request with Multiple Users).....	77
Figure 37 - High - Level Architectural Diagram	116
Figure 38 - Use Case Diagram	137
Figure 39 - Class Diagram	150
Figure 40 - Database Design	151
Figure 41 - High - Level Architectural Diagram	152
Figure 42 - Networking Diagram	153
Figure 43 - Supervisory Meetings	163
Figure 44 - Dr. Deshantha's (Approval of the Software)	170
Figure 45 - Dr. Deshantha Pathinayake (Showcase of the final product).	172
Figure 46 - Wireframes UI Design.....	175
Figure 47 - Test Cases.....	177
Figure 48 - Hyperparameter Tuning	178

List of Tables

Table 1 - System Usability Scores	74
Table 2 - Confusion Matrix	75
Table 3 - Objective Evaluation.....	79

1. Introduction

Mental health problems are highly active in Sri Lanka, affecting people of various demographics. According to a research study conducted by the Centers for Disease Control and Prevention (CDC), over 1 in 5 people experience mental health-related disorders at some point in their lives, mostly with disorders such as anxiety and depression being common (Centers for Disease Control and Prevention, 2024). Moreover, the social stigma associated with mental health can discourage people from seeking health from professionals, leading to untreated conditions that get severe with time which can lead to a loss of life.

In past years, prioritizing effective mental health through support systems has gained recognition. Despite the existence of different mental health resources, many people remain unaware of the available support and are unable to access it due to regional and economic barriers. Many of the population lacks mental health professionals, mostly in rural areas like Anuradhapura, Badulla districts, etc. where healthcare infrastructure is mostly limited. In addition, traditional ways of treating mental health problems such as in person consultation can be intimidating as they are hesitating to discuss their feelings and matters openly. This hesitation is worsened by cultural attitudes that view mental health problems as a display of weakness and shame.

Current mental health applications and resources fail to address the unique needs of Sri Lankans. Many existing solutions are not designed to address needs in the local context, and they lack regionally relevant content and support. Due to this reason, the existing systems fall short of providing personalized mental health recommendations or connecting mental health professionals to the users. As a result of this issue, people find it difficult to seek help and use the existing resources effectively.

The lack of fully developed mental health monitoring tools heightens these challenges. Many people do not have a way to understand feelings, track their moods, or get immediate support when needed. Without the ability to monitor mental health, users struggle to know the patterns that could display the worsening conditions. Additionally, there is a need for a system to provide anonymity, allowing people to discuss their problems openly without any fear or judgment.

In this given background, there is a higher need for a mental health monitoring system that uses advanced technology to provide personalized and accessible support for people in Sri Lanka. This system will use sentiment analysis and mood predictions based on user input, where users can identify their mental health much better. By filling the gap between users and mental health professionals, this system's main goal is to create supportive surroundings and encourage users to get help and improve their mental wellbeing.

In conclusion, the current mental health support in Sri Lanka needs a good solution that addresses these challenges as mentioned above. By developing a mental health monitoring system that focuses on accessibility, personalization, and anonymity, this project aims to contribute positively to the mental health landscape and encourage users to have better mental health and awareness.

2. Background, Objectives and Deliverables

2.1. Background

Mental health care has become a more significant issue in the world today, especially in the modern world where daily life pressures, social expectations, and technological advancements have played a huge role in the rising of stress levels, anxiety, and depression. These above-mentioned mental health issues not only affect people's well-being but also have challenged the entire healthcare system. Although there is rising awareness of mental health problems, there remains a significant gap in personalized and accessible mental health care solutions. People frequently hesitate to get help due to the cultural stigma of mental health and the lack of accessible resources (Deivanayagam et al., 2024).

This issue is mainly common in the Asian region and in Sri Lanka, where mental health support systems are under development. Traditional methods of treatment like in-person meeting appointments with mental health professionals are not accessible to a large portion of the population (Sumathipala, Mahesh & Wijesundara, 2020). Further, the fast expansion of digital platforms has created a growing need for innovative technological solutions to monitor and improve mental health. Integrating Artificial Intelligence (AI) and Machine Learning (ML) in mental health care provides a chance to bridge this gap by providing personalized care, continuous monitoring, and real-time help (Patel et al., 2018).

This project – MoodSync’s main aim is to develop a mental health monitoring system with machine learning that uses natural language processing (NLP) and sentiment analysis to track and then predict the mental health state of the users. The system will gather user input like texts, and social media posts and analyze them to detect emotions, moods, and mental health statuses. Through advanced data analytics, personalized recommendations, coping strategies, and real-time alerts for loved ones, this software aims to enhance mental health support for Sri Lanka in practical, accessible, and anonymous ways.

2.2. Objectives

Project MoodSync is designed to address the gaps in the current mental health support using advanced technologies to provide users with easily accessible and personalized solutions. This project aims to achieve the following objectives.

1. Sentiment Analysis Implementation

- To develop a good machine learning model with at least 95% accuracy to analyze the text inputs provided by the user (ex: social media posts) and evaluate their emotional and mental states accurately by the end of the development.

2. Personalized Mental Health Recommendations

- Create an intelligent guidance model that provides the users with coping methods, mental health tips, and motivational quotes specified to their recorded mental state and long-term data.

3. Anonymous Communication

- Design a secure and user-friendly web application to anonymously chat, making sure that users can express themselves freely without judgment, fear, and cultural stigma.

4. Developing a Real-Time Alert System

- Integrate a real-time monitoring system that identifies emotions and sends alert messages to loved ones and health professionals in serious situations.

5. Mental Health Trends Visualization

- Develop an interactive, graphical representation of the user's mood patterns throughout time, making them monitor their mental health journey and identify

triggers and improvements.

6. Integrate Support Channels

- Connect the users to verify mental health professionals and provide information on the availability of health services and hospitals that are readily available to offer support.

7. Promote Accessibility and Inclusivity

- Ensure that the system is sensitive and inclusive to all Sri Lankan users while maintaining good adaptability to global audiences as well.

8. Ensure Data Security and Privacy

- Implement data encryption and strict authentication methods to safeguard users' data and maintain privacy.

2.3. Deliverables

Project MoodSync delivered a complete set of software components, documentation, and resources to provide an integrated solution for mental health monitoring and professional support. All the deliveries mentioned went through quality assurance testing and were ensured to meet project requirements before the final report submission.

2.3.1. Software Components

Main Application

- **Web Application** – A functional, mobile responsive web application developed by React with Vite, TypeScript, Tailwind CSS, and Flowbite components. This application supports many device formats with optimized interfaces in desktop and mobile access, important to the Sri Lankan context where mobile internet usage is shown higher.
- **Backend** – A complete FastAPI implementation that provides separate endpoints for data management, user authentication, sentiment analysis, and in-communication features. The API includes complete error handling, input validation, and security controls like request validation and rate limiting.
- **Database** – A NoSQL database MongoDB with a neat scheme design, which includes seven collections with needed indexes and validation rules. This implementation has data mitigation scripts and backup/restore ability to ensure operational feasibility.

Machine Learning Components

- **Sentiment Analysis Pipeline** – A complete machine learning pipeline for mental health sentiment analysis, which has preprocessing modules, features extraction, and a trained model. This pipeline supports English fully and cultural adaption to the Sinhala context as well.
- **Model Training Framework** – A repeatable framework including data preprocessing, model training, evaluation, and deployment scripts. This framework lets future models update and filter when additional training data is provided.
- **Lexicon Resources** – Special mental health lexicons developed to support the language, which has over 2,500 terms in associated sentiment values and contextual annotations. The resources show a contribution to the project requirements.

User Interfaces

- **User Dashboard** – An intuitive interface for the end users that has mood tracking, visualization, resource access, and communication features. The dashboard implements progressive improvement techniques to maintain functionality in network conditions.
- **Professional Interface** – A special interface for mental health professionals with anonymized user monitoring, communication management, and analytical tools. This interface uses clinical workflow considerations found by consultation with practicing professionals.
- **Admin Console** – A complete system management interface with user admin, content management, system configuration, and monitoring abilities. The console has audit logging and access controls needed for a system managing sensitive health information.

Report / Documentation

- **System Architecture** – Complete documentation of the system architecture, like diagrams, data flow models, security architecture, and API details. The document follows UML standards and has reasonings for the use of main architectural designs.
- **User Guides** – Detailed user guides in all three system interfaces (user, professional, admin), including step-by-step instructions, features explanation, and troubleshooting information.
- **Technical Reference** - Complete technical shown such as API details, database schema, integration points, and deploying methods.
- **Development Guidelines** – Standards and practices in future development, coding standards, git workflow, testing requirements, and security measures especially for the Project MoodSync.

2.3.2.Additional Resources

- **Deployment Packages** – Docker containers and configuration files in consistent deployment in environments. This package has environmental variable templates, secrets management, and monitoring configurations.
- **Testing Resources** – Complete tests with unit tests, integration tests, and end-to-end test scenarios. This completes above 90% in code coverage in the application's codebase.
- **Sample Data** – Anonymized, data samples for system demonstrations and testing uses, showing different user scenarios and mental health patterns.
- **Cultural Resources** – A collection of mental health resources that have educational materials, coping methods, and suicidal support information, culturally adapted in Sri Lankan contexts.

All the above-mentioned deliverables are submitted with version control history, showcasing the agile development process and providing changes in the project lifecycle.

3. Literature Review

NOTE: Extended Literature Review is in PID section, not included due to word count limitation.

This research paper explores the recent upgradation of machine learning and natural language processing (NLP) that has improved capacity in mental health monitoring through sentiment analysis on social media platforms such as X. The usage of X's large and complex datasets provides valuable insights into public sentiments facilitating timely responses to health crises. Within mental health, it has been seen that sentiment analysis has become a critical tool. This paper shows that traditional machine learning techniques like SVM and Naïve Bayes efficiently analyze user sentiments in real time though there have been challenges with slang and informal languages which is relatively common in a social media platform like X. Recent studies have clearly shown that deep learning models like Long Short Term Memory networks, which improve sentiment analysis by complex feature extraction. Although advancements have been made, according to this research paper there are gaps in research of sentiment analysis. The framework proposed by the authors shows the importance of historical tweets but hasn't done an in- depth analysis though they highlight more significant predictions regarding users' mental health. As a solution to in-depth analysis and to fill this gap, we could get the tweets data from users who are mentally suffering state and come up with identifying more accurate patterns to predict effectively. Finally, this research paper underscores the need for user driven data collection methods to improve data reliability and to provide cloud-based solutions for mental health monitoring. (Hinduja et al., 2022)

This advanced research study named "Machine Learning Techniques for Prediction of Mental Health" explores the way machine learning algorithms can be applied to predict current mental health status by gathering different attributes like financial and household data. Machine learning techniques have shown good results in detecting mental health issues earlier, the authors have used different datasets to identify the patterns that have been missed using the normal traditional methods. The authors have used many algorithms like Decision Tree, Random Forest, Support Vector Machine (SVM), Naive Bayes, Logistic Regression, XGBoost, Gradient Boosting Classifier, and Artificial Neural Networks to enhance the accuracy in predicting mental health. This dataset that has been used contains 76 attributes in total and the data has been split into a 70/30 ratio for training and testing. It included metrics like Precision, F1 Score, and false negative rate and accuracy. Among these algorithms, SVM has achieved a good overall accuracy of 87.38% whereas Gradient Boosting Classifier has obtained the highest F1 score showing a

balanced precision. Naïve Bayes achieved a lesser accuracy of all algorithms which is 21.67% if it has feature independence and is not suitable for big and complex datasets. This analysis clearly shows that SVM is the ideal algorithm with high accuracy performed with data as no other algorithm has this level of accuracy. However, there are gaps identified in this research where there is no mention about imbalanced datasets which can be identified to improve model performances more. To fill this gap synthetic data can be used representing minority class for a more balanced dataset. The findings from the research study indicate that machine learning techniques can be effectively used in mental health predictions though future research should aim more to improve the model to have a good performance. (Jain et al., 2021)

Another research study by Vidhi Mody highlights AI/ML-based approaches for identifying mental health issues, focusing on methods like multi-view clustering, expert systems, and wearable sensor integration. This paper shows how impactful physiological signals and smartphone data can be in detecting early mental issues such as stress, using techniques like fuzzy logic and multi-view biclustering. This paper showcases multi-view learning by combining perspectives like average activity, daily trends, and location variability resulting in an overall prediction of 87.1% accuracy. However, simpler classifiers like linear naïve Bayes were used in stress detection and achieved a lower accuracy of 80% which models can be improved more. The study highlights that current expert systems utilize rule-based reasoning and fuzzy genetic algorithms (fuzzy-GA) for identifying and planning treatments. These systems aim to mimic decision-making and provide personalized and therapeutic recommendations. This algorithm shows moderate accuracy and has limitations in boundary cases compared with the results obtained, where unreliability is often seen in the table of results. Although there is an algorithm upgrade, there is a gap in integrating multiple AI methodologies for diagnostic systems. The existing methods discussed have limitations in precision, which this proposed product will address with good precision and adaptability to evolving user states. My product will address this gap using an advanced machine learning model using traditional expert systems and to improve conditional handling well. This proposed product will be integrated with feedback and overall scores to deliver effective and personalized mental health recommendations. (Mody, 2019)

Another paper shows the progression made through machine learning algorithms and the way wearable devices help in detecting mental health disorders such as depression and anxiety, however there are gaps in these existing methods. This research study highlights a good accuracy rate in detecting health symptoms with the aid of KNN – K Nearest Neighbor, SVM – Support Vector Machines, and Naïve Bayes algorithms. However, most models are trained in controlled environments, making it overlook the real-world applicability of certain socio-economic conditions and cultural contexts, which affect the reliability of the product and mental health patterns, to fill this gap the proposed product

will endorse the Sri Lankan local context. Additionally, the previous models have shown a limitation of scalability and practical issues when applied to resource-constrained settings which would have been better if continuous and real-time monitoring were used. However, the integration of these devices is challenging for low-income regions in Asia as the cost is high. This research shows how models and wearables can be utilized for impactful mental health detection. (Kumar et al., 2021)

Another research study conducted does a systematic review of examining machine learning applications in detecting mental health issues, with a focus on features, classifiers, and challenges faced with using these methods. This review shows the key techniques used in mental health detection such as feature extraction models like TF-IDF, Word2Vec, and N-Gram models as well as supervised machine learning techniques like SVM, logistic regression, and decision trees. The study shows a gap in handling data sparsity, multilingual content, and privacy constraints, especially for datasets sourced from Online Social Networks (OSNs). Though machine learning provides solutions to these concerns, there are issues like biased data, data preprocessing, generalizability, and model interpretation that remain significant obstacles. To address this gap, the proposed system will include localized data gathered from trusted government sources that eliminate biased data. Overall, the authors provide a review emphasizing the need for a good reliable model that addresses these challenges to improve health predictions. (Rahman et al., 2020)

This paper demonstrates the effectiveness of Machine Learning models, specifically SVM and XGBoost, in categorizing depression-related content from social media platforms. The authors show that SVM is highlighted above all other algorithms in terms of its high precision, making it the most reliable algorithm for mental health monitoring. However, the advancements gaps remain with the interpretability of ML models, especially concerning the language use of social media platforms which can lead to errors and misclassifications. Additionally, the class imbalance raises a challenge, which limits the generalization of the findings across user populations. Compared with previous research studies they have not shown a consideration in integrating a Large Language Model (LLM) which could've enhanced accuracy and understanding. To address these gaps as mentioned above the proposed system will be focused on an ML model ensuring a good performance in mental health detection indicators minimizing bias data. (Shatte, Hutchinson C Teague, 2019)

In this advanced study, the authors have emphasized the recent upgradation of Natural Language Processing (NLP) has made the way for innovative approaches to mental health analysis through social media. Traditional methods of mental health monitoring depend on face-to-face interviews which are not different from everyday emotional expressions found online. This paper has focused more on using ML techniques to

classify emotions and states by user-generated content. For instance, the authors used a supervised algorithm to identify mental health indicators in social media posts and achieved good results. Similarly, it also demonstrated the effectiveness of deep learning models in identifying anxiety and depression-related content, highlighting the capability of automated systems in detecting real-time mental health status. Apart from these upgrades, gaps remain in interpretability of ML models and their generalizability from diverse populations. It is to be seen that these authors' main goal was to improve accuracy however they have missed addressing underlying reasons for misclassifications. This study aims to use LLM-large language models with SVM – Support Vector Machines to improve accuracy in stress-related posts. However, this is not the ideal solution as they lack generalizability of the findings though models accuracy is good, in the proposed project, it will closely align to cultural context. In addition, the model outputs will be analyzed to identify language indicators that cause mental distress thus completing the gap between accuracy and practical interpretation. This research carried out highlights the advanced technical possibilities in mental health analysis and contributes to a good system design before development. (Radwan et al., 2024)

3.1. Research Gap

The use of technology for mental health intervention has grown exponentially in recent years, resulting in many mental well-being applications. However, more needs to be done particularly with niche user groups in developing nations such as rural communities in Sri Lanka. Therefore, this chapter presents the solutions that have already been attempted, their failures, and the inadequacies in research this project intends to bridge.

Existing Solutions and Limitations

There are many mental health apps Calm and Headspace to Moodfit that offer general possibilities for mood tracking and meditation. Yet, while these apps promote mental health awareness and assist multitudes, they function with a Western majority inclination and the global, culturally lingual assessment required for proper user engagement does not come easy cultural Relevance and Localization.

Cultural Relevance and Localization

Currently, many apps lack cultural relevance and localization, which are necessary for heightened engagement and successful mental health intervention. For example, Hinduja et al. (2022) assess the advancements in machine learning capabilities and natural language processing (NLP) features within social media spaces to assess mental health concerns. However, much of the training of such models comes from Westernized language tendencies; for example, if someone is depressed and makes a comment on social media and/or uses particular language tendencies, it's much different than how someone in Sri Lanka would be spelling or saying how they feel through language. Thus, an increasingly successful model is needed, trained on the socio-cultural forces at play with the Sri Lankan population.

Anonymity and User Privacy

Anonymity is another important requirement that many of the existing applications fail to satisfy. Where they offer private adjustments, they do not offer a secure way for people to access much-needed help for mental health problems without the fear of exposure. Rahman et al. (2020) note that privacy issues are one of the biggest obstacles to using online social networking for mental health monitoring. Therefore, applications are

necessary that keep anonymity on the user's part but fulfill the support requirements for a culturally appropriate Sri Lankan application.

The following research gaps have been written based on the conducted literature reviews above:

1. **Personalized Support** – Most applications don't provide a unique response to users instead they provide a common generic response. To address this gap the proposed system aims to use advanced machine learning algorithms trained with localized datasets to offer unique personalized responses with an overall score.
2. **Real-Time Support** – Current applications don't have real time support/alerts to users loved ones in immediate health needs. This proposed system will plan to include real-time alerts using SMS to get immediate support in need.
3. **Cultural Context** – There is a lack of mental health applications that represent users in the Sri Lankan context. The proposed system will address this gap by using local languages and cultural backgrounds in its design and functionalities. By engaging with a mental health professional, this application will be tailored and ensured for local users.
4. **Professional Support** – No application connects mental health professionals with their users; to address this gap the proposed system will showcase the directory list of professionals with times and locations that will be available for channeling.
5. **Anonymity** – Unlike other applications, this proposed system will have a feature to chat with a representative for quick support without exposing the user as it cuts off the user's fear of getting support.

4. Method of Approach

4.1. Development Methodology

The development of the project MoodSync was done using Agile Methodology, especially a modified Scrum framework matching the unique requirements of the application. This approach was selected after checking out several methodologies like Waterfall, Kanban, and common Scrum. The decision was taken based on three main factors: the changing nature of requirements in applications, need for the frequent stakeholder feedback, and the changing nature of the project that involves both technical and health topics.

The modified Scrum framework had two-week sprints with clearly defined objectives and deliverables. Sprint planning sessions were done at the start of each sprint, where user stories were ranked based on their impact on the main functionality and technical dependencies. This figure shows the sprint structure used in the development lifecycles.

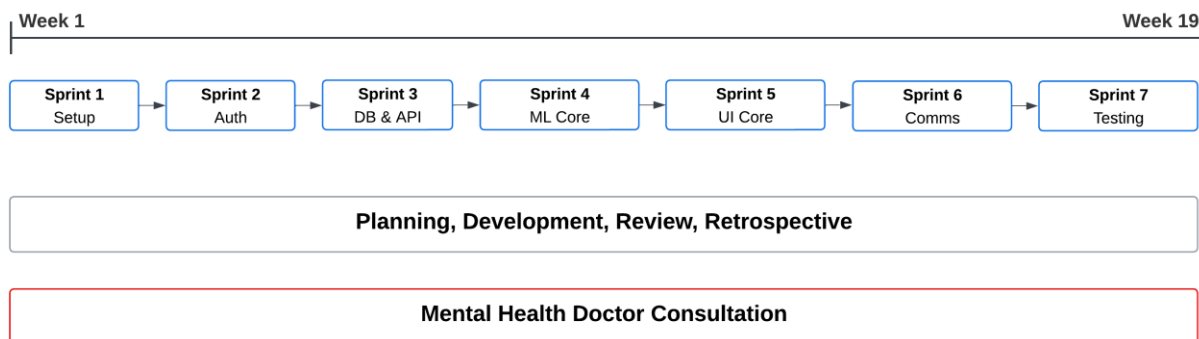


Figure 1 - Sprint Structure

Adaption to the traditional scrum methodology was the following of mental health doctor meetings during sprint reviews. These meetings were ensured by feedback sessions with two of the best Sri Lankan mental health doctors who provided the topic expertise on cultural touch in mental health expression. Their input was documented and used when planning the sprint, specifically identifying the sentiment analysis model training and designing culturally suitable interventions.

Risk management was used in the sprint planning process by a risk register that was checked and updated at the start of each sprint. High-priority risks, especially those relating to data privacy and ethical aspects, were assigned to mitigation strategies with specific actions used in sprint backlogs. This dedicated approach to risk management

proved valuable in addressing the challenges of anonymous communication features, which required additional security identified in the risk assessment process

4.2. System Architecture Overview

Project MoodSync was engineered using the FARM Stack (FastAPI, React, and MongoDB), chosen for its performance, flexibility, and scalability for real-time features of the application.

The architecture followed a modular design pattern with clear set boundaries between the components, providing independent development and testing. This approach was valuable for the separation of concerns between sentiment analysis and the communication features, which allowed parallel development.

This system is structured in three main layers,

1. **Presentation Layer** – Implemented using React with Vite and TypeScript, this layer is responsible for the rendering of user interfaces for three user roles: end-users, mental health doctors and system admins. Tailwind CSS and Flowbite were used to have a mobile responsive web design, ensuring the accessibility in different device formats in Sri Lanka, where mobile device usage is more than desktop / laptops.
2. **Application Layer** – FastAPI was used as the foundation for the application layer, providing RESTful API endpoints and WebSocket connections for real-time features. This layer included the sentiment analysis pipeline, authentication services, and alert system logic. Asynchronous processing was used for computationally heavy tasks, particularly for machine learning interface operations, to maintain system responsiveness.
3. **Data Layer** – MongoDB was selected as the only database because of its document-oriented structure, which it fit in with the varied and changing nature of mental health data. The database architecture was designed with careful engineering practices to align with Sri Lankan data protection act (2022) regulations and ethical requirements for mental health applications.

The main engineering challenge was the real time chat feature integration whereas maintain user anonymity. This was resolved by the implementation of a token-based routing system that hides direct connections between users and mental health doctors while protecting the ability to communicate immediately.

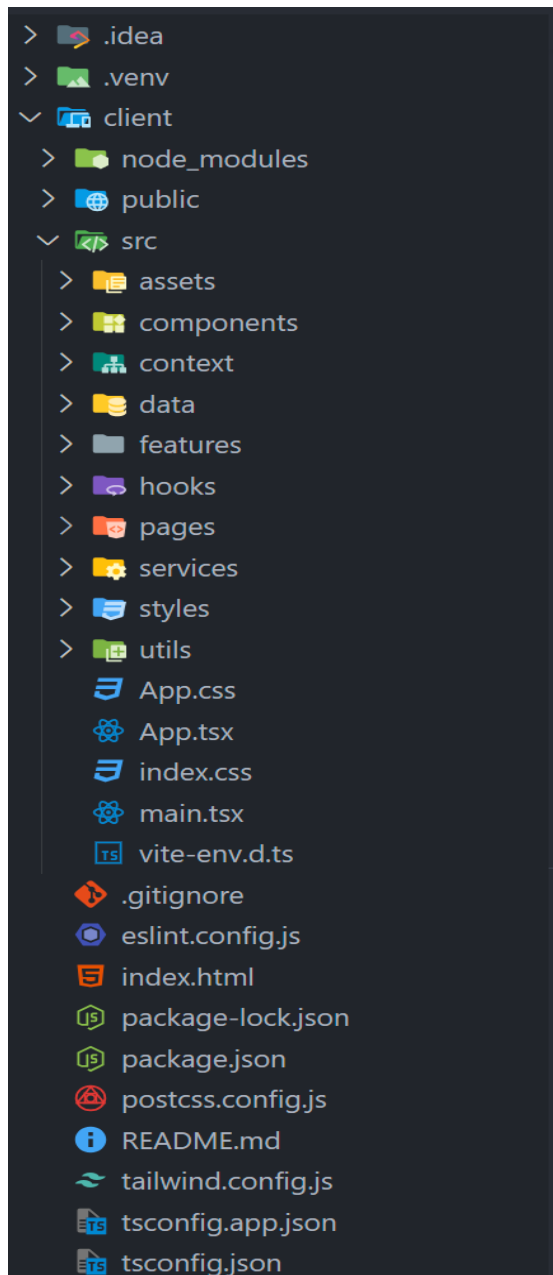


Figure 3 - Frontend Project File Structure

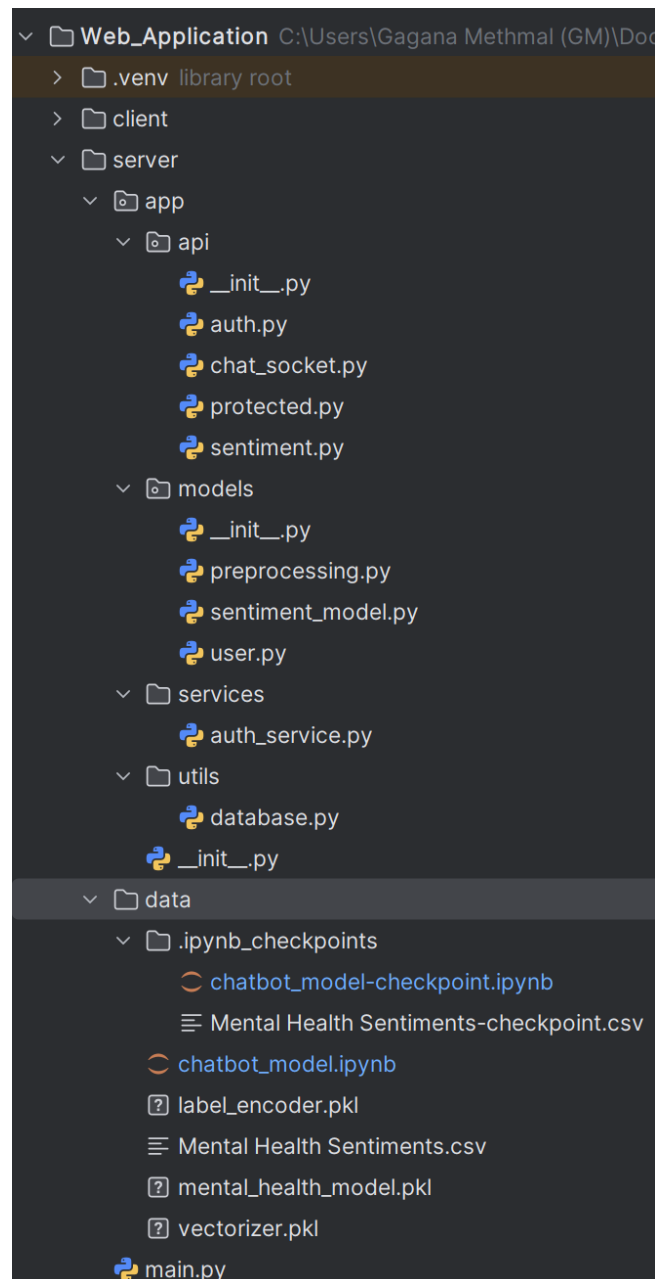


Figure 4- Backend Project File Structure

Frontend Structure – Modern industry standard modular structured web application with clean components division, which makes it easier for finding the error and debugging.

Backend Structure – A Python-based API's that processes mental health and analyzes sentiment patterns in the machine learning pipeline. Its layered architecture neatly

separates between API endpoints, data models, and services ensuring the system remains scalable and capable of delivering accurate emotional insights.

4.3. Data Collection and Processing

Doctor Details

- Dr. Deshantha Pathinayake – MBBS(SL), DFM(COL), MCGP, MRCGP(Int)
SLMC Reg: No. 2372.
General / Mental Health Specialist.
Wellness Medi Clinic, Moratuwa 10400
- Dr. Saman Weerawardhane – MBBS(SL)
Psychiatrist / Mental Health Specialist.
National Institute of Mental Health (NIMH)
NIMH, Mulleriyawa New Town, Angoda

NOTE - Meeting minutes and the doctor's approval letter / evidence are included in the Appendix (Other Materials) section at the bottom of this report.

Details of the Local Hospital (Data Collection)

- Mana Suwa Piyasa (මනසුවිජයස) – Mental Health Clinic
Colombo South Teaching Hospital – Kalubowila
B229 Hospital Rd, Dehiwala-Mount Lavinia 10350
- Survey - [CLICK HERE!](#)
Responses - [CLICK HERE!](#)

The development of an effective machine learning model for mental health sentiment analysis required good data collection and a processing method. Couple of approaches

were used to get and setup data for model training, with an attention to culturally relevant details for Sri Lankan users.

Main data collection was done by three methods,

1. **Synthetic Dataset Generation** – A dataset of users, doctors and patients' conversations has been taken, and sample texts were created additionally with collaboration with a mental health doctor mentioned above, ensuring cultural representation like post-war, economic crisis data of mental health concerns that are specific to the Sri Lankan context.
2. **Anonymous Social Media Sampling** – With cultural relevance, random public Facebook and Reddit posts from Sri Lankan users were taken, anonymized, manually labeled, and placed in the existing dataset before training. Posts were categorized with an ID, status, and sentiment scale that ranged from distress to positive well-being.

Data processing used a strict pipeline like text normalization, tokenization and feature extraction. The figure below shows the complete data processing pipeline implemented in this project.

4.4. Implementation Strategy

The implementation strategy for project MoodSync followed a phased approach with clear milestones aligned with the sprint structure. This strategy prioritizes the core functionalities while maintaining flexibility for filters based on stakeholder analysis.

The implementation was divided into four phases,

1. **Foundation Phase** – This is the start phase which focused on the basic infrastructure, such as the database design schema, API Structure and authentication system. Important security features, mainly those related to data encryption and user privacy, were used in this phase to make sure the development followed with the data protection rules.
2. **Main Functionality Phase** – The sentiment analysis pipeline and basic user interfaces were developed in this phase. Model training, validation, and integration with the FastAPI backend were completed, ensuring the main capabilities of the system. Basic visualization components were also used to provide detailed insights into mental health trends.
3. **Communication and Alert Phase** – Anonymous communication channels in between users and mental health doctors were used, with the real-time alert system for the emergency situations. This phase needed security testing mainly to ensure the integrity of the anonymous communication system while maintaining track for crisis prevention.
4. **Filtered and Culturalization Phase** – The final phase was focused on to cultural adaption, support, and user experience filtering based on feedback from the testing. This phase was critical for ensuring the systems relevance and usability in Sri Lankan context.

In the implementation stage, continuous integration (CI) practices were done using GitHub Actions, making it automated testing triggered by each commit to maintain code quality and prevent any errors. Code reviews were done on all major features, with analysis applied to security-needed components such as anonymous communication systems and alert methods.

Implementation challenge was the optimization of the machine learning model for deployment on a resource-constrained environments common in Sri Lanka. This was addressed in model quantization and the use of a tiered processing approach, where the preliminary analysis was conducted on the device with more complex processing to server-side components when needed.

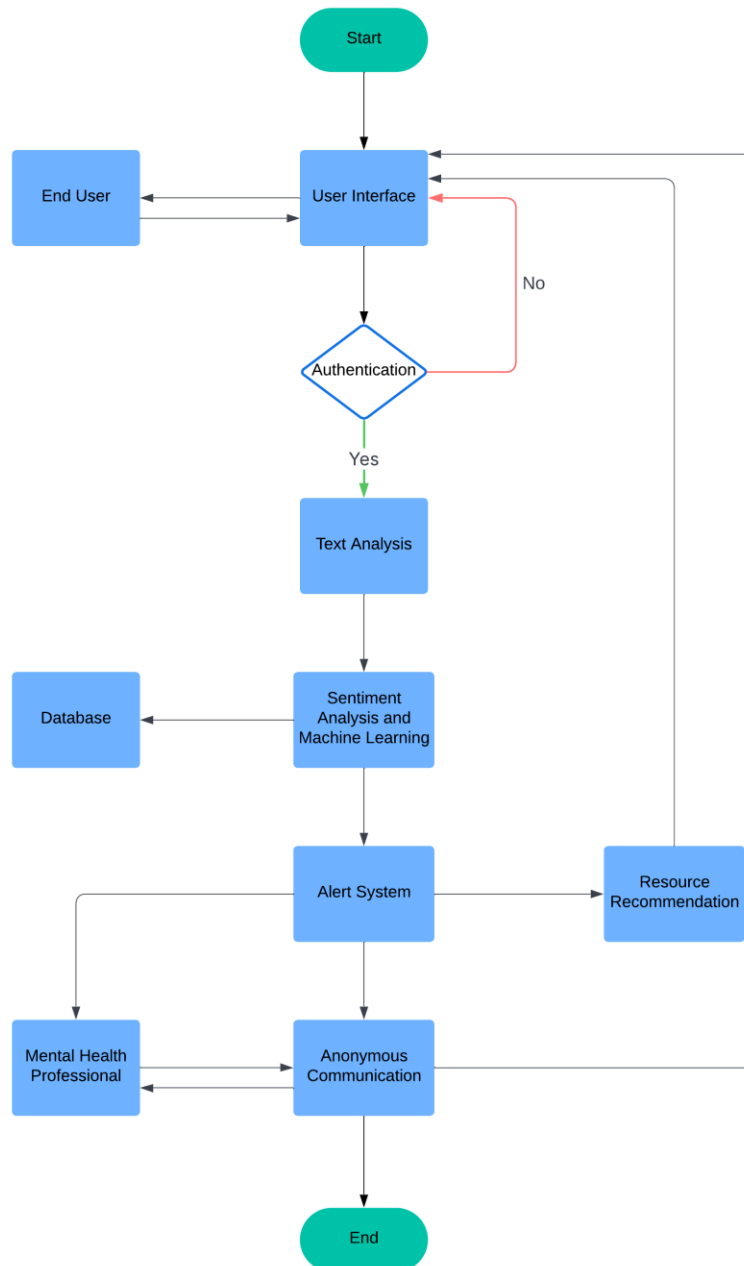


Figure 2 – Flowchart

5. Requirements

5.1. Functional Requirements

The functional requirements detail the essential features and functions that this system should have to meet the project objectives mentioned in Chapter 2. These requirements ensure that the system delivers the needed features for mental health monitoring and tracking. The functional requirements are,

1. User Registration and Authentication

- The user should be able to create an account and log in securely.
- User data needs to be stored safely, and privacy needs to be ensured by using encryption.

2. Sentiment Analysis and Mood Prediction

- The system should analyze user inputs (text, social media posts, etc.) to detect user sentiment and predict moods.

3. Mood Tracking

- The system should be able to track users' moods over time, which should record any daily input and provide insights into mood patterns.
- Users should be able to view their mood history by graphs and visualization features.

4. Unique Mental Health Recommendations

- From sentiment analysis and mood tracking, the system should give unique mental health advice, coping strategies, and resources to improve mental health.

- Recommendations given should be matched to users' emotional and mental health condition.

5. Anonymous Chat Feature

- Users should be able to communicate with the system through an anonymous chat interface.
- The chatbot should be able to provide responses, ask questions back, and give guidance based on user inputs.

6. Real-Time Alerts for Loved Ones

- The system should automatically send real-time alerts for users' loved ones when the user's mood is tracked as dangerous or self-harming.
- Alerts should be sent by text SMS.

7. Motivational Content

- The application should display motivational content like quotes, affirmations, and positive content to cheer users in difficult times.

8. Data Visualization

- The system must show users' mood data, and sentiment trends by graphs and charts, that allow the user to track their progress over time.

9. Mental Health Professionals and Hospital Connections

- This system should connect with mental health professionals or the local hospitals.

- This feature should provide contact details and emergency help for mental health support.

5.2. Non-Functional Requirements

Non-functional requirements detail this software's operational characteristics, which ensure that the system performs efficiently and addresses the user's expectations in performance, security, and usability. The non-functional requirements identified are,

1. Performance

- This application should display the real-time analysis when a user inputs the texts and generates recommendations within 5 – 10 seconds.
- This application should also have the capability to handle many users without running into any issues.

2. Scalability

- This application should be made to scale efficiently as the users increase, which allows further expansion without any performance reduction.

3. Security

- This application should ensure privacy, integrity, and the availability of the user data.
- Encryption methods should be applied for sensitive data like user login details and personal health information.

4. Usability

- This system should have a user-friendly interface to easily navigate mainly for users having less technical knowledge.
- The chat feature should be simple and easy to use.

5. Availability

- This system should be 24 * 7, ensuring that users can access mental health monitoring at any time.
- Downtime should be reduced, and it should only happen when the system is maintained only.

6. Reliability

- The system should be reliable, giving accurate sentiment analysis, mood tracking, and recommendations on user inputs.
- The system should handle unexpected errors so that the user experience will not be disturbed.

7. Maintainability

- This system should be easy for maintenance purposes, with components.

8. Compliance

- This system must follow Sri Lankan data protection regulations (2022) like GDPR and privacy laws to ensure that user data are handled ethically.

5.3. Hardware / Software Requirements

The hardware and software requirement mentioned shows the resources that are needed to develop, deploy, and run the software. This includes server needs, development tools, and platforms for the deployment.

1. Hardware Requirements

- **Server** – A cloud server (ex: AWS) to host the application backend services, machine learning models, and the database.
- **End User Devices** – The system should be responsive to work on web applications from desktops and smartphones.
- **Database Storage** – Cloud-based database storage needed (ex: MongoDB) for storing the user data securely.

2. Software Requirements

- **Frontend Development** – React.js, Vite, TypeScript, Tailwind CSS, and Flowbite library for styling.
- **Backend Development** – FastAPI with Python, Node.js with Express for the API endpoints and to handle machine learning processes.
- **Machine Learning Libraries** – Pandas, scikit-learn, imblearn, NLTK, and other Python libraries are mainly used for sentiment analysis and to predict moods.
- **Database** – MongoDB for storage of user data, feedback, mood logs, and history.
- **Cloud Hosting / Platforms** – AWS for handling computational tasks and hosting the application.

5.4. Networking Requirements

This project has less network requirements, but these are the basic requirements for system components and external services communication in the application,

1. Internet Connectivity

- This system needs a good internet connection to provide real-time updates and alerts.
- Software must be able to communicate with the backend server to process the data.

2. API Integration

- The system needs a third-party API integration for the SMS notifications and integration with healthcare providers.

3. Real-Time Communication

- For the real-time alerts feature, the system should use push notifications by WebSockets (Socket.IO) or a similar way to ensure delivery of the alerts.

6. Design and Implementation

6.1. System Architecture

Project MoodSync was designed to be a modular, layered architecture based on the FARM stack (FastAPI, React, MongoDB), with additional components in machine learning and real-time communication. Below figure shows the complete system architecture diagram with component relationships and data flow patterns.

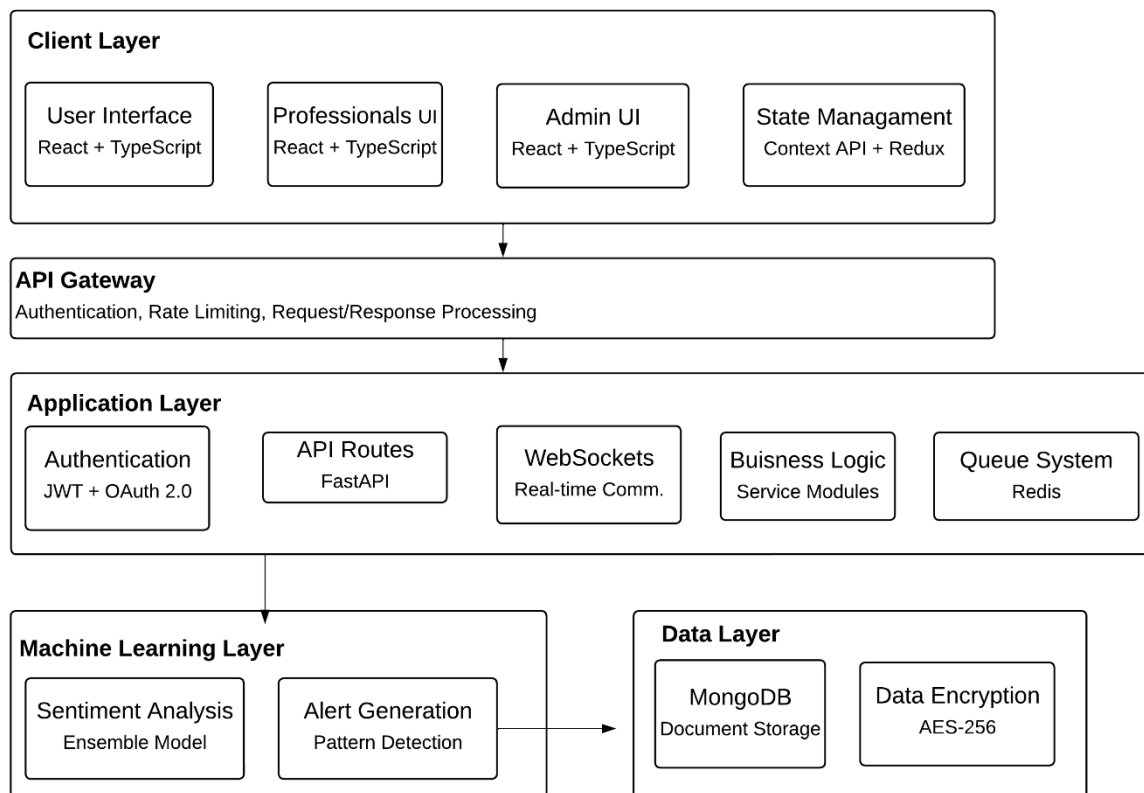


Figure 5- System Architecture Diagram

This architecture is structured into four main layers, each with different tasks and interfaces.

- **Presentation Layer** - Implemented using React with Vite and TypeScript, this layer handles the user interface rendering and client-side logic. Three separate

interfaces were built for different user groups: end users, mental health professionals, and system administrators. Tailwind CSS and Flowbite components were used to create responsive, accessible interfaces that follow implemented design principles for mental health applications. The presentation layer communicated with the application layer by RESTful API calls and WebSocket connections for real-time features.

- **Application Layer** – FastAPI is the foundation layer of this, which provides RESTful endpoints, WebSocket handlers, and Asynchronous processing abilities. This layer is organized into separate modules like authentication, sentiment analysis, communication, and alerting. Each module is shared to detailed interfaces while it encapsulates the development details, providing independent development and testing. This application layer uses business logic validation and organizes system components.
- **Data Access Layer** – This middle layer manages instructions between the application layer and the database, using repository patterns to abstract the database operations. This design decision has provided flexibility in future database changes while controlling data access patterns in the application. The data access layer implemented caching strategies for frequently accessed data, which reduces the database load and improves response times for important features.
- **Persistence Layer** – MongoDB was selected as the main and only database since it has a document-based structure and flexibility in schemas. The database is designed with separate collections for the users, mental health data, communication records, and system configuration. Indexes were used to optimize the query performance for common access patterns, especially for time series data used in the trend visualization.

A special architectural challenge addressed in implementation was the integration of real-time features while maintaining the system's responsiveness. This was solved in the implementation of a message queue system in Redis, which separated resource-intensive operations like sentiment analysis from the main request-response cycle. Below the figure shows the asynchronous processing flow implemented in sentiment analysis requests.

Cross-cutting concerns such as logging, error handling, and security were done by middleware components in both presentation and application layers. This approach

showed consistent handling of the issues in system components while minimizing code duplication.

6.2. **Database Design**

The database design for the project MoodSync was done to balance flexibility, performance, and security requirements. MongoDB was selected for its document-oriented structure, which provided different and growing mental health data while providing a high performance for the read-heavy operations common in data visualization scenarios.

The database was organized into seven main collections, each with a special purpose and optimized structure. Below the figure shows the high-level schema design with the relationship in the collections.

- **Users Collection** – Stored user profile details in role attributes. The schema had fields for authentication credentials (with password stored using bcrypt hashing), basic information, and preferences. For security reasons the personal noticeable information was reduced and stored separately from the mental health data.
- **Health Data Collection** – Implemented as the time-series collection storing mental health measurements, sentiment analysis results, and other information. This collection used MongoDB's time-series optimization for the fast fetch of temporary data that is used in trend visualization. Data was linked to the anonymized user identifiers rather than directly to user profiles, implementing an additional layer of security protection.
- **Communication Collection** – Stored records in anonymous communication between users and professionals. Messages are encrypted by AES-256 encryption with rotating keys. The schema used a double-blind identifier system where no one could access it directly identifying information about the other without admin's approval.
- **Resource Collection** – Had mental health resources, coping methods, and educational materials. Resources were connected with cultural text indicators and needed mental health domains to provide recommendations. Another collection is for resources provided for easy updates and maintenance by admins.

Separate attention was given to database security, using field-level encryption in sensitive data and complete access controls for all the data updates. The database design uses temporary data managing patterns in maintaining records while supporting fast querying of current state information.

```
{
  "_id": ObjectId("5f8d0f2e9a6e7d1c9b4c3b2a"),
  "username": "anonymized_identifier",
  "role": "user",
  "auth": {
    "password_hash": "bcrypt_hashed_password",
    "last_login": ISODate("2025-04-23T10:15:30Z"),
    "account_status": "active"
  },
  "preferences": {
    "language": "si",
    "notification_settings": {
      "alert_thresholds": {
        "severity": 0.7,
        "frequency": 3
      },
      "emergency_contacts": [
        {
          "name": "Emergency Contact",
          "relationship": "Family",
          "contact_hash": "hashed_contact_info"
        }
      ]
    },
    "theme": "light",
    "accessibility": {
      "font_size": "medium",
      "contrast": "normal"
    }
  },
}
```

Figure 6- Database Schema Design

6.3. User Interface Design

6.3.1. User Interface

The user interface design follows user-centered design principles in particular attention to accessibility, cultural appropriateness, and reduced cognitive load in users experiencing mental health challenges.

The design process started with low-fidelity wireframes, which were filtered in feedback sessions and potential users and mental health professionals. The wireframes developed into high-fidelity mockups created in Figma, where it provided the foundation for the React component building.

Figma Link - [CLICK HERE!](#) (All UI designs and wireframes are in the other material section).

Main design features include,

1. **Mood Tracking Interface** – A simple input method that allows the user to record emotional state in text inputs. The interface uses culturally suited emotional expressions in matching Sri Lankan contexts.
2. **Insights Dashboard** – Visualization of mental health trends in adjusted time ranges and metrics. Charts were done using Recharts, with a selection of color schemes to prevent triggering or anxiety from visual elements.
3. **Resources Library** – Separated access to mental health resources, coping methods, and motivational content. Resources were displayed in a card-based layout with filtering capabilities for finding culturally needed content.
4. **Anonymous Communication Interfaces** – A chat interface for communication with mental health professionals, which is designed to reduce anxiety while clearly showing message status and professional availability.

Usability testing was done with 12 participants showing different demographics, which resulted in filtering navigation patterns and information architecture. Special attention is given to providing clear escape paths and control methods to reduce possible anxiety while using the system.

NOTE – NO TEMPLATES USED.

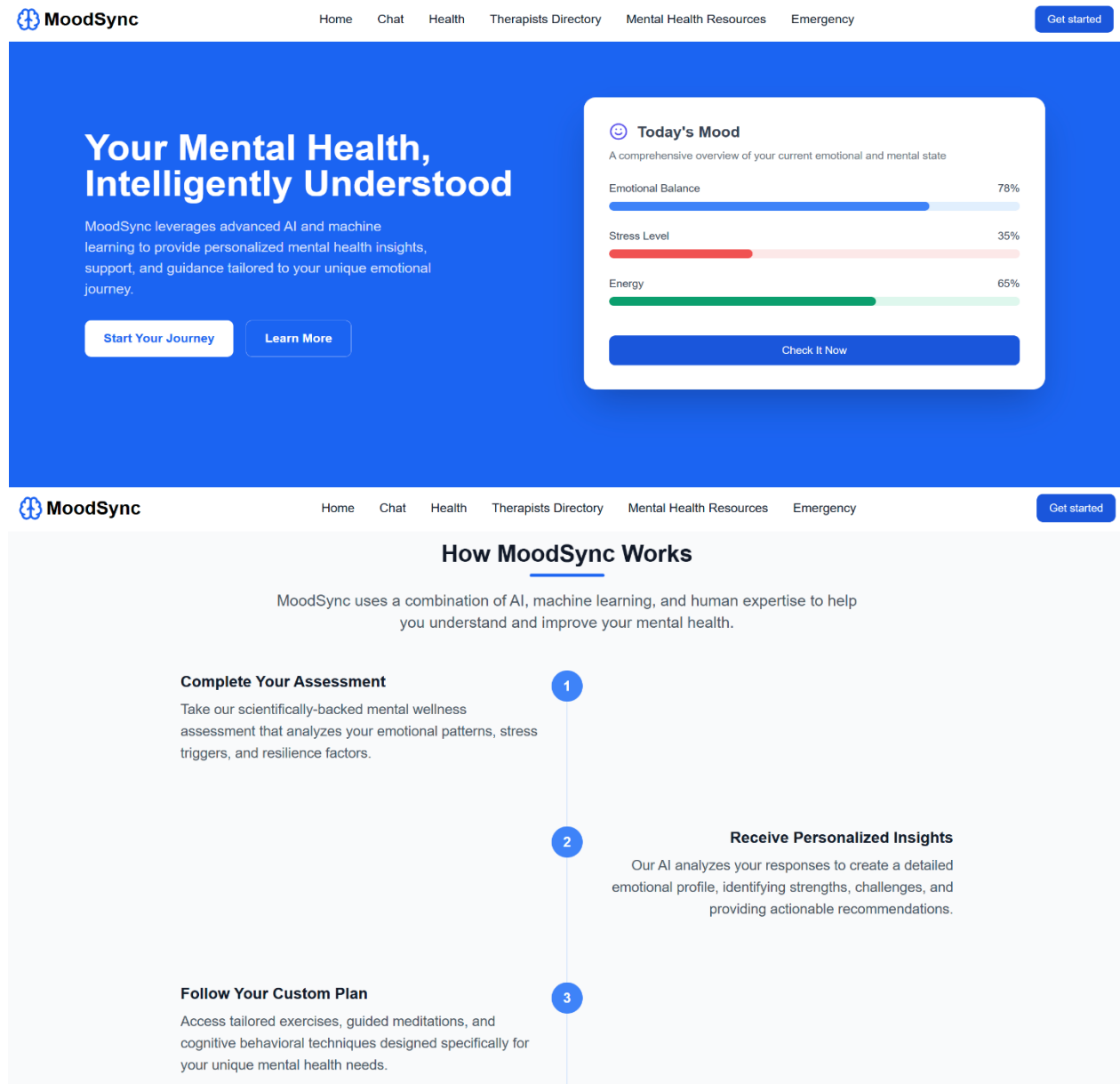


Figure 7 - Home Page

Home page with navbar component and explanation of how the application works.

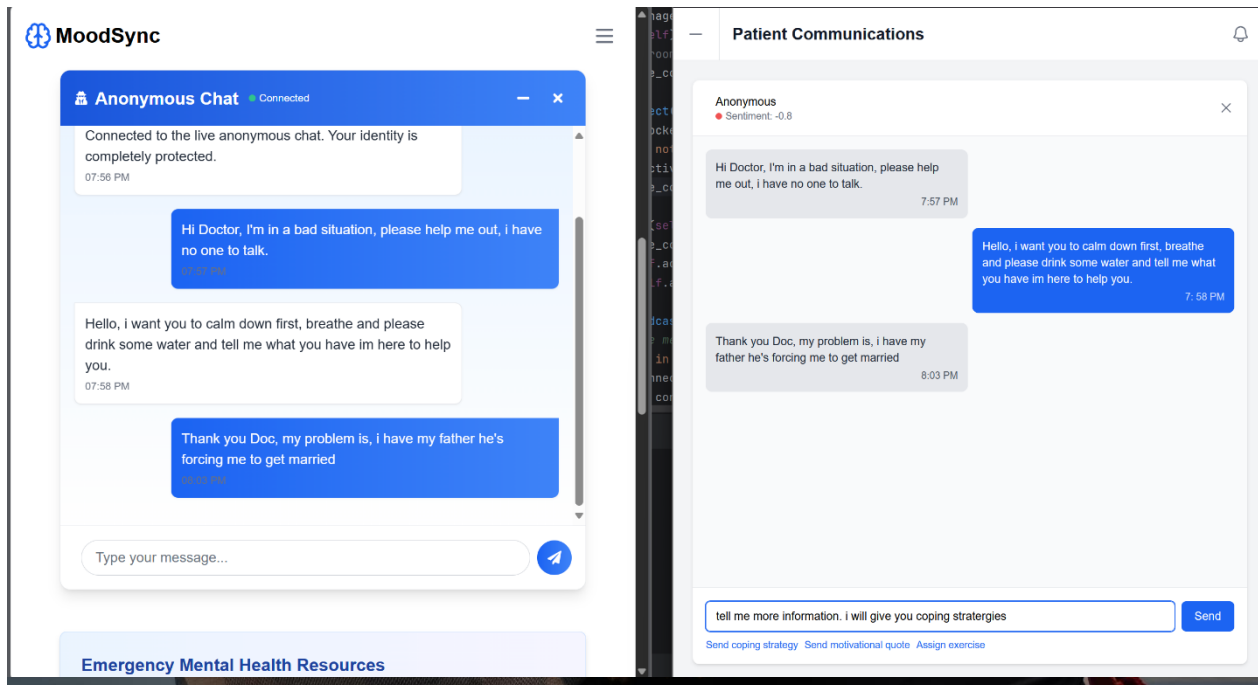


Figure 8 - Anonymous Chat Interface

Anonymous chatting between user and doctor/professional seeing at each of their interfaces. The user is shown connected or disconnected and it assigns user to a room where available professional can talk with the anonymous user and help in their issues.

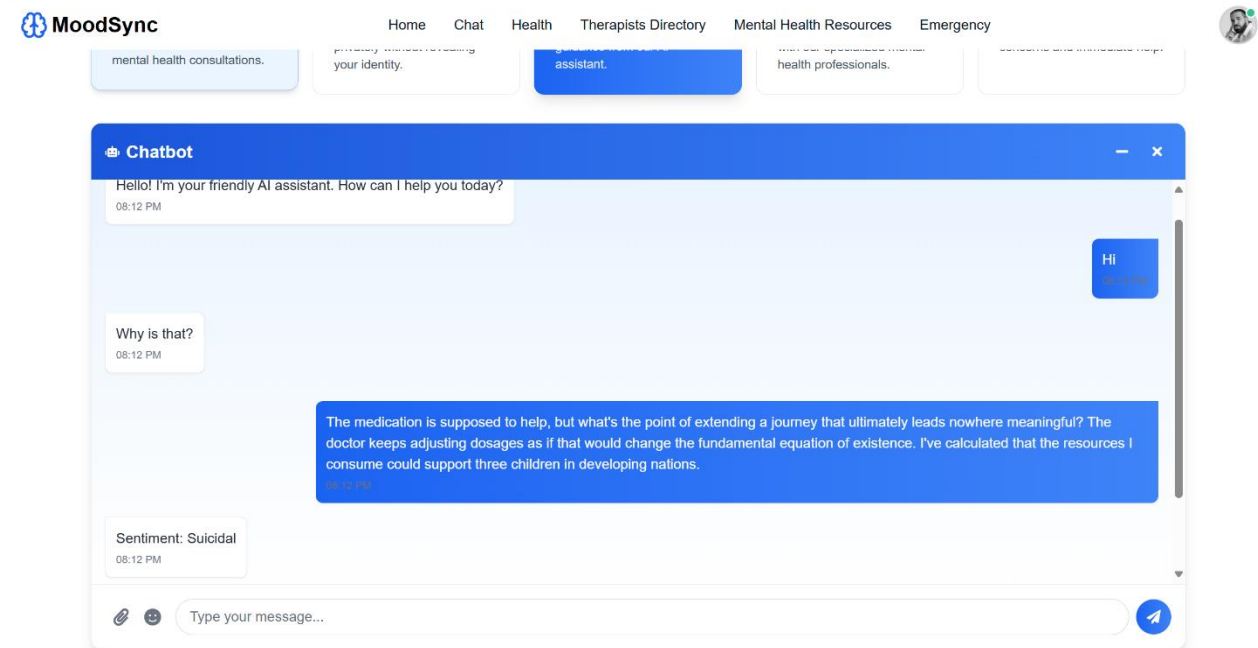


Figure 3 - Chatbot Interface

Machine Learning Model is integrated in here. Where the user can chat and get their mental health predicted which provides confidence scores for each sentiment.

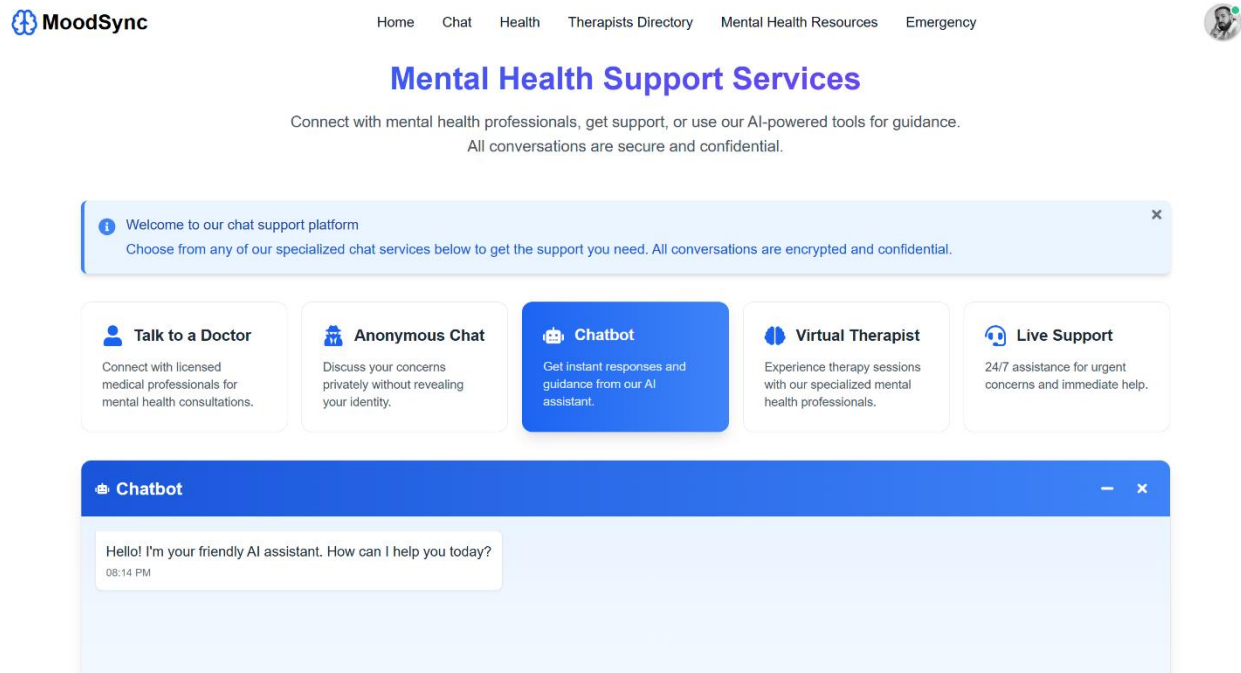


Figure 4 - Support Services

Mental Health Support Services page where user can go to and select the needed service.

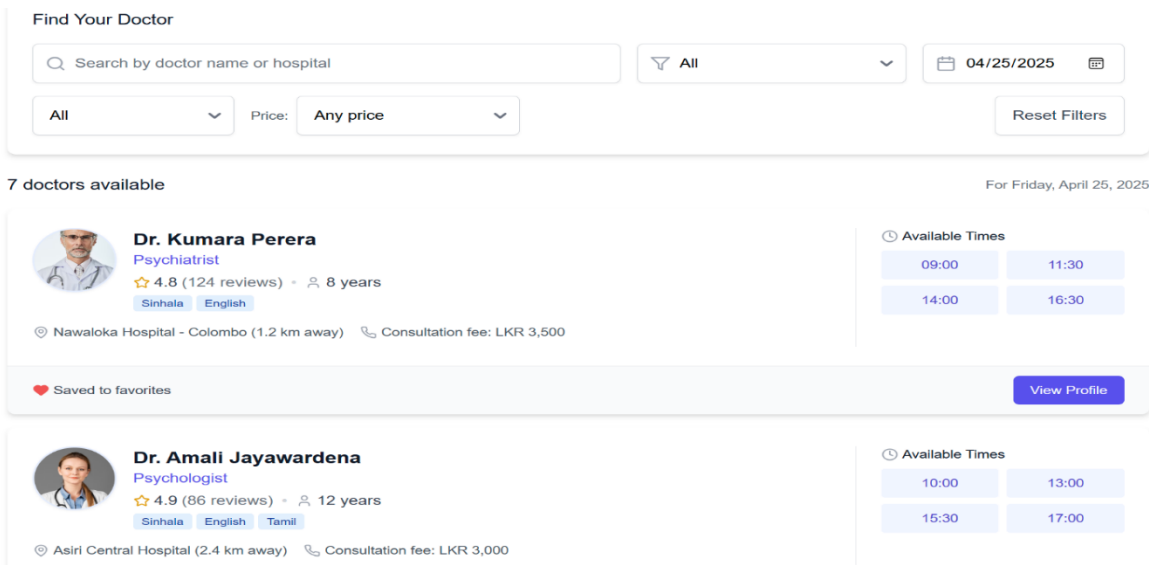


Figure 5 - Doctor Channeling

Interface where registered doctors can be checked with details and channeled.

[Home](#) [Chat](#) [Health](#) [Therapists Directory](#) [Mental Health Resources](#) [Emergency](#)



Dr. Kumara Perera
[Psychiatrist](#)
MBBS (Colombo), MD (Psychiatry), MRCP (UK)
★ 4.8 (124 reviews) • 👤 8 years

About

Dr. Kumara Perera is a respected mental health professional with over 8 years of experience. He specializes in mood disorders, anxiety, and depression management with a holistic approach to mental wellbeing.

Location & Contact

 Nawaloka Hospital - Colombo
1.2 km from city center

 Consultation fee: LKR 3,500

Available Times for Friday, April 25, 2025

09:00

11:30

14:00

16:30

 Remove from favorites

Book Appointment

Figure 6 - Doctor Details

Further details of the doctor and their available times and locations with channeling feature.

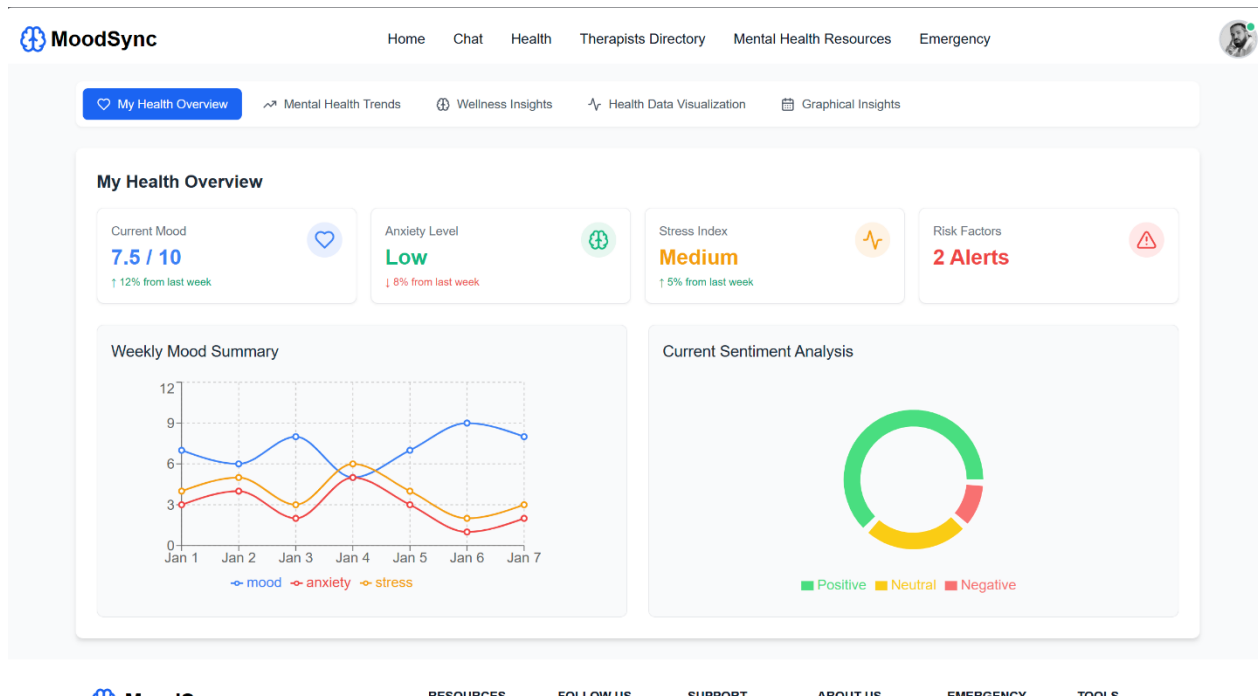


Figure 7 - Health Trends Visualization

Mental health trends of users with comparing features and it is gathered through a period.

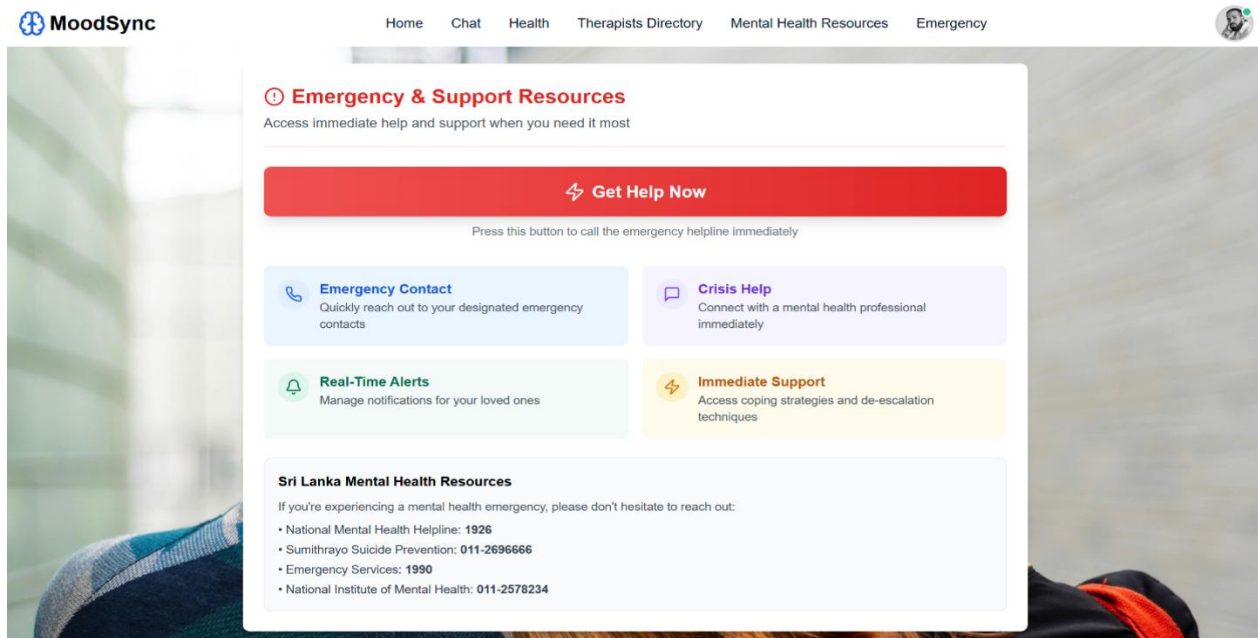


Figure 8 - Emergency

Emergency interface, where necessary emergencies can be viewed and managed.

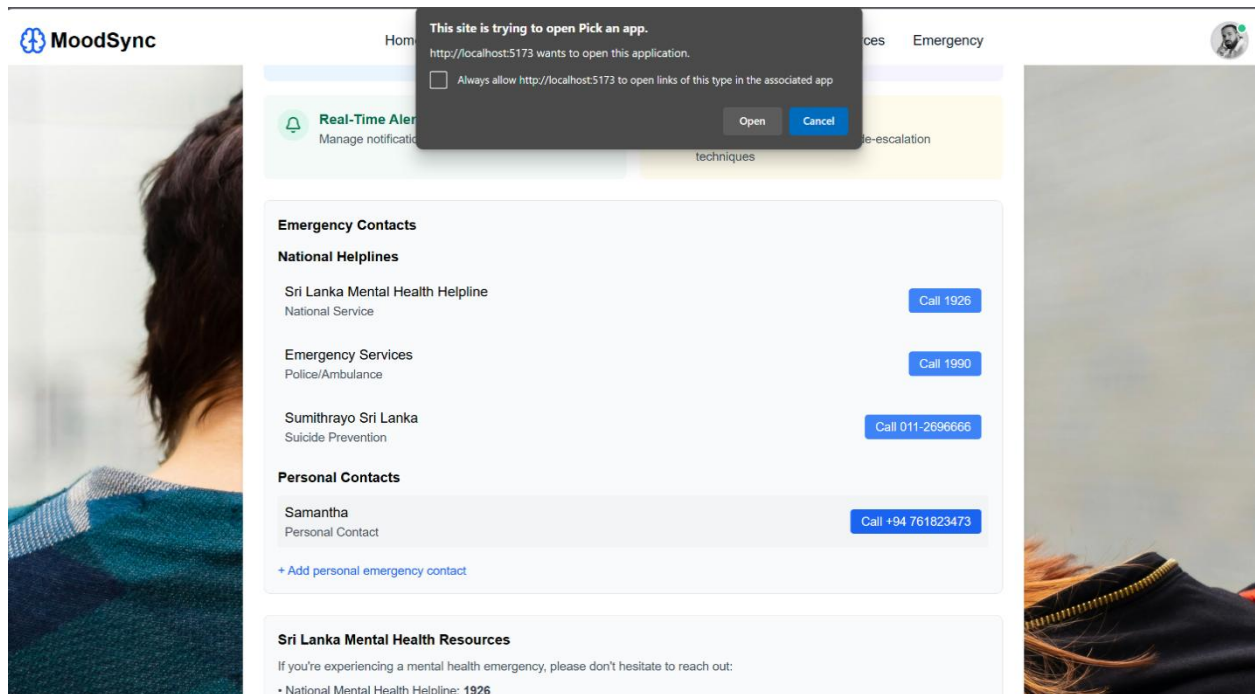


Figure 9 - Emergency Selection

Emergency contact can be added manually other than the one that comes in the registration. Also, the hotlines will be readily available.

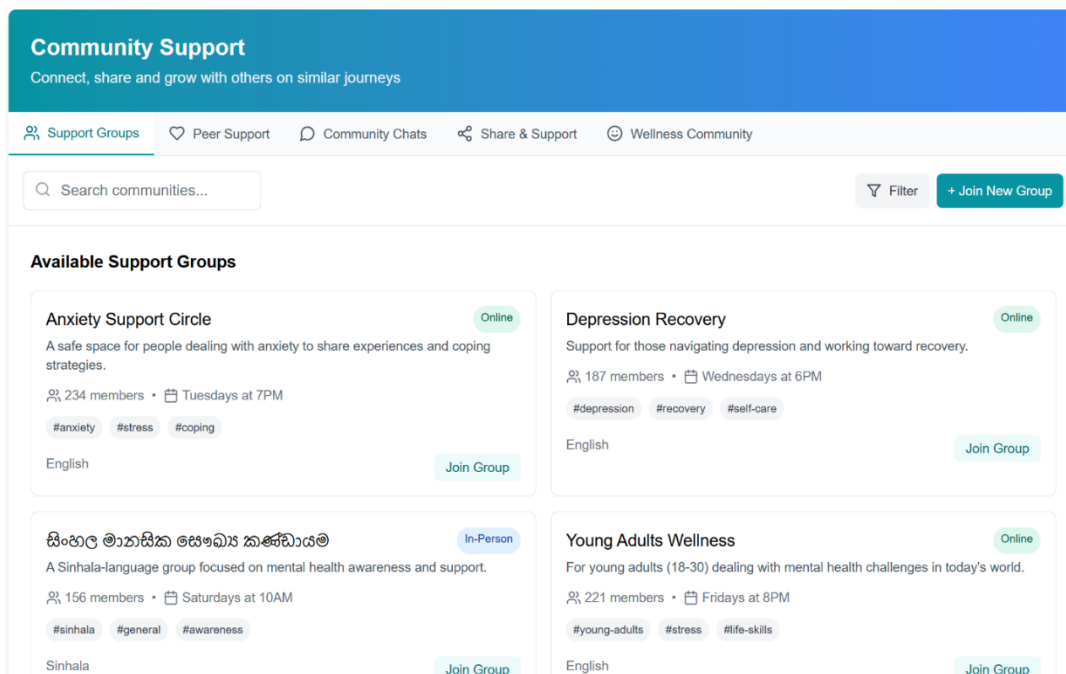


Figure 10 - Community Support

Community support page for users where they can join and discuss.

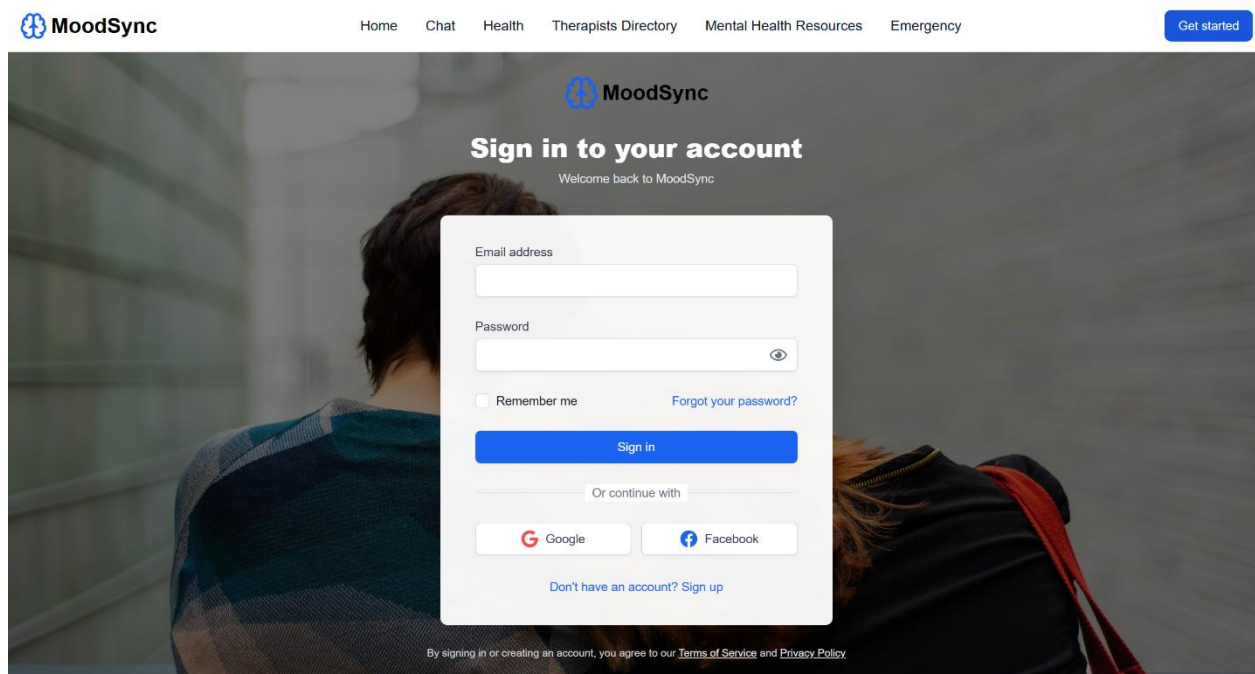
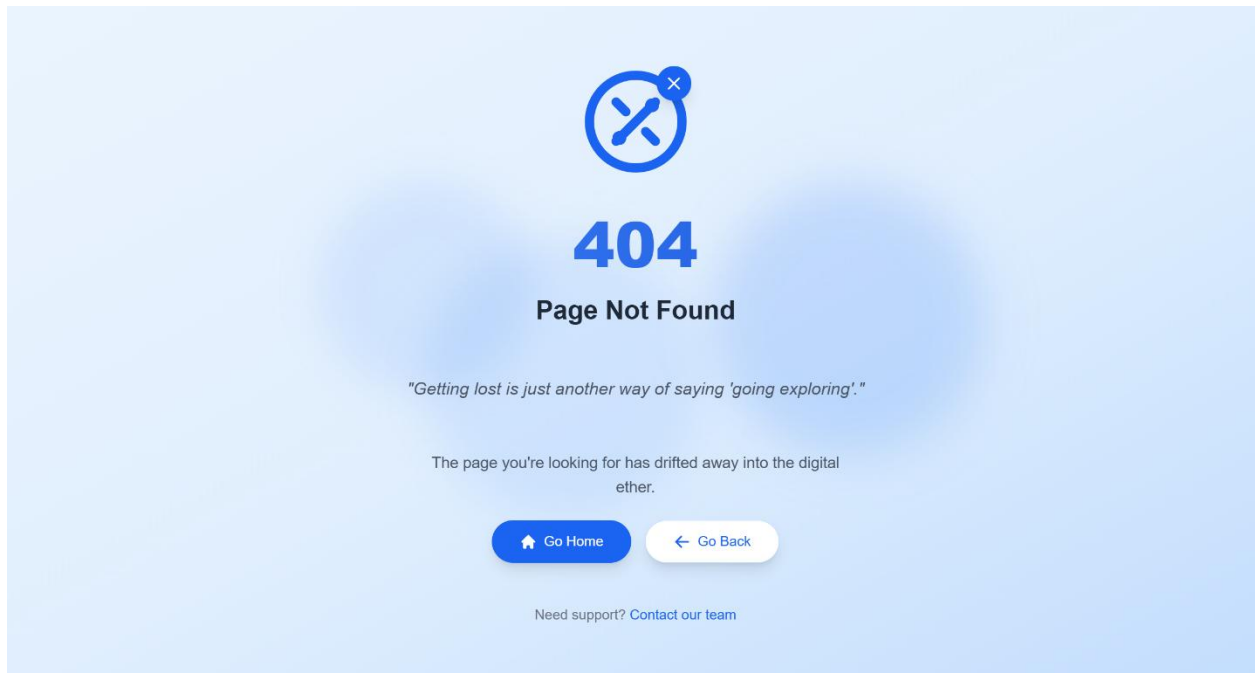


Figure 11 - Login Page

Login and signup page looks similar, user have options such as to log in with social medias.



If user redirectes to a wrong page.

Figure 12 - Error Page

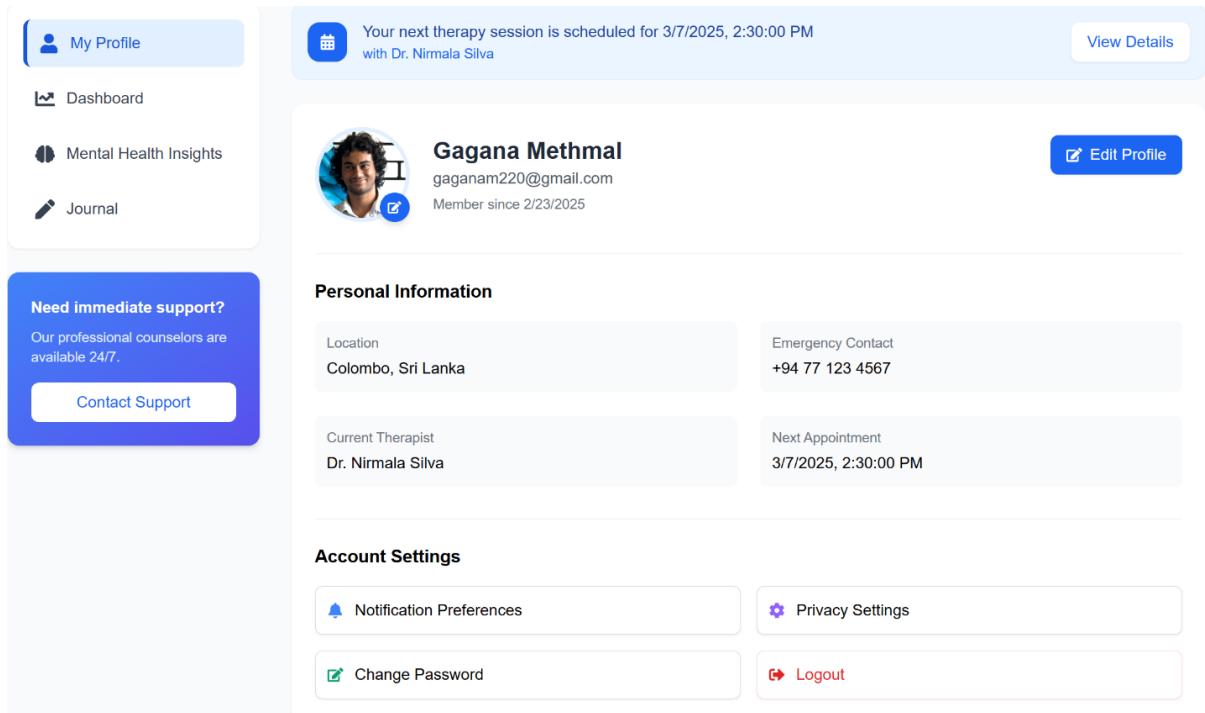


Figure 13 - User Profile Page

User profile once logged in, user can change their picture, details and control their settings, view their mental health dashboard and insights.

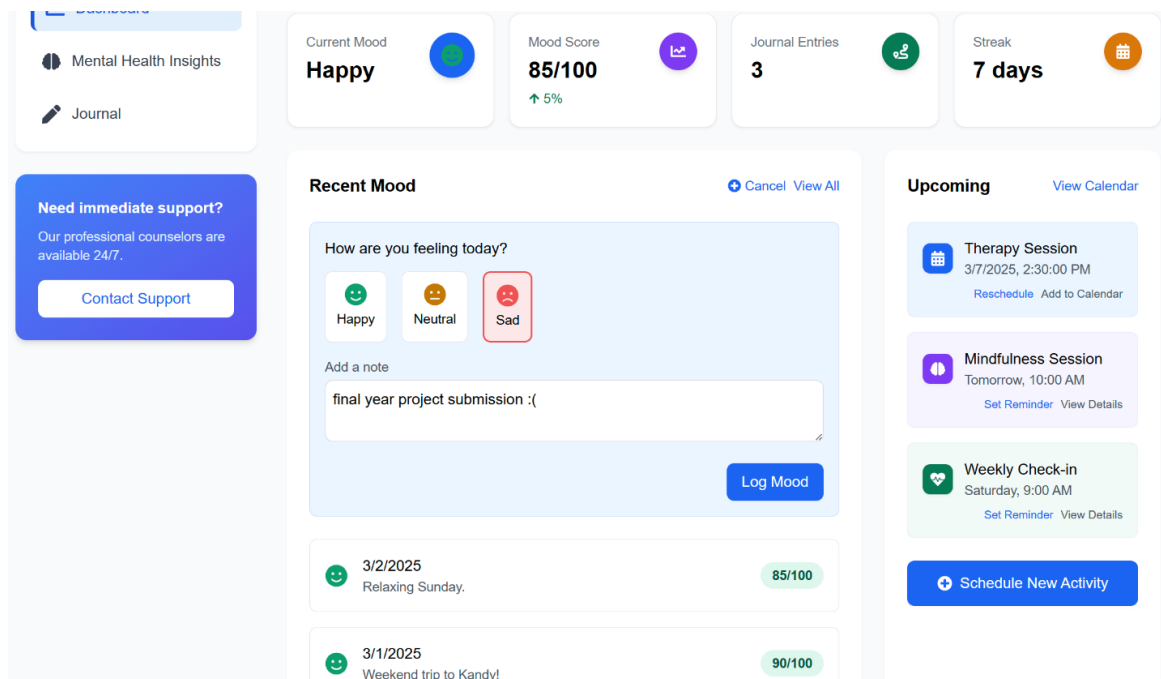


Figure 14 - Mood Logger Page

Can be used to log moods for the mental health analysis and caluculations.

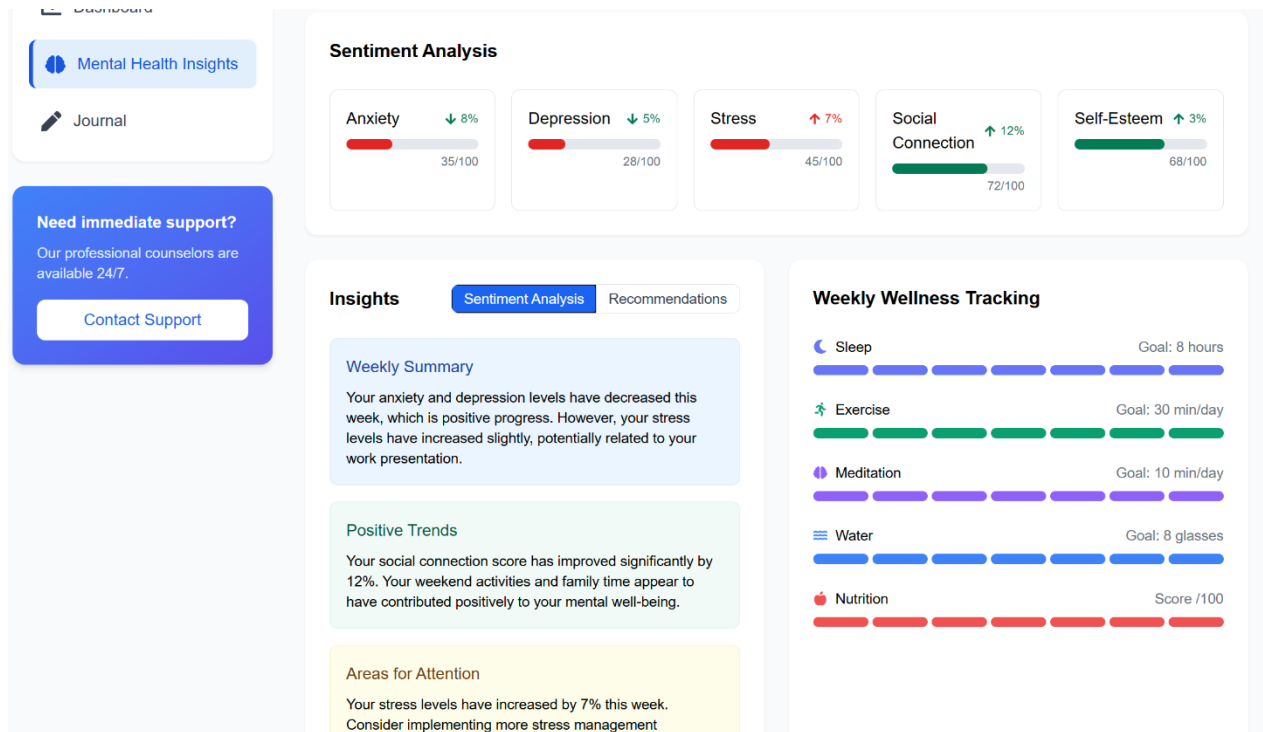


Figure 15 - Health Insights

Health insights calculated will be displayed in here over a time period and unique recommendations are given from the algorithm.

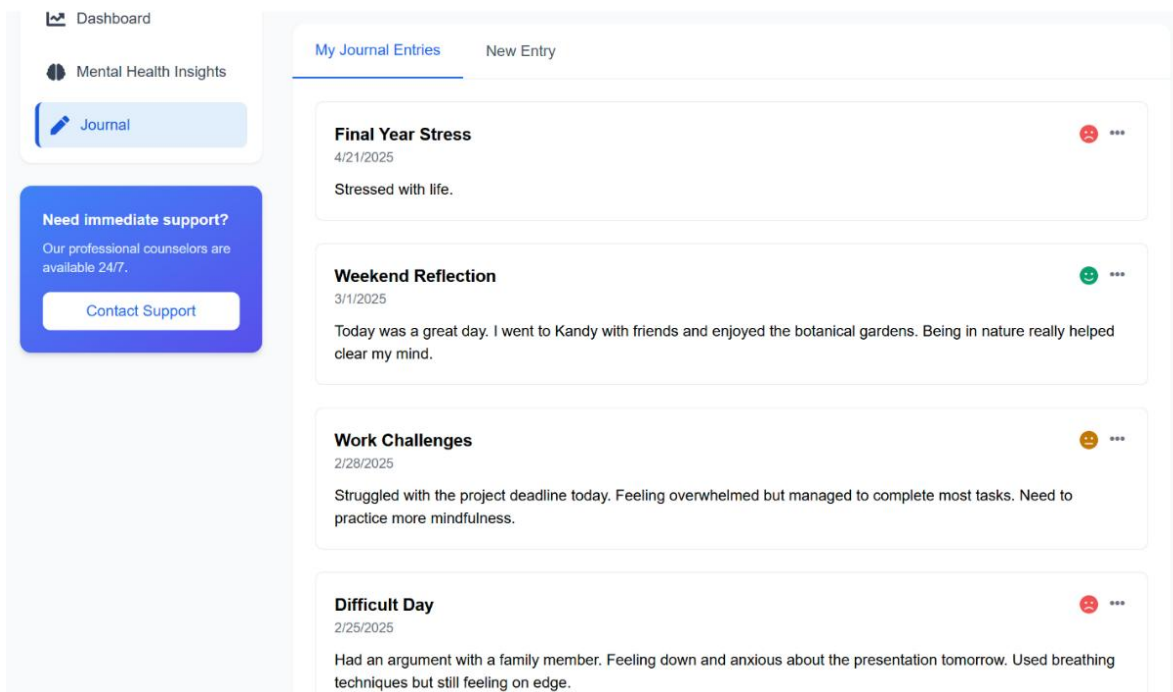


Figure 16 - Journal Entries Page

Users can enter their daily moods to the system.

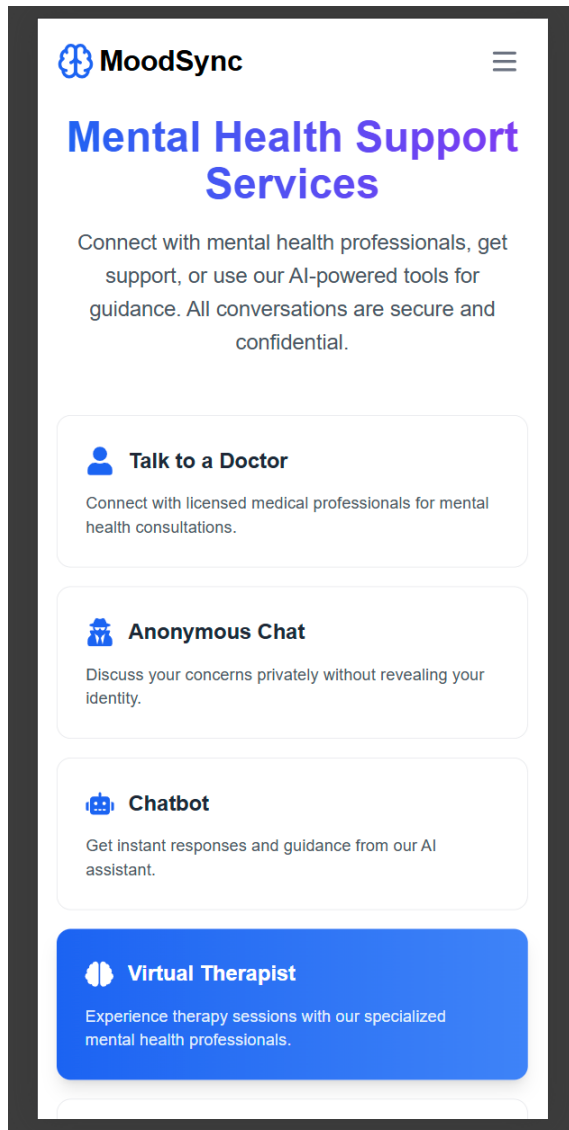


Figure 17 - Mobile Responsivity

Every page in this application is mobile responsive as shown, it can be accessed by any kind of mobile device.

6.3.2. Professional Interface

The professional interface was designed in support of mental health doctors with efficient interaction with the users while providing analytical tools for the assessment.

This interface has specialized features,

1. **Anonymized User Management** – Professionals can view anonymized user profiles with mental health data and history without personal information. Color-coded indicators highlight user possibilities needing attention based on sentiment analysis results.
2. **Communication Dashboard** – An interface for managing communication with multiple users, with messaging history, risk indicators, and contextual information. This interface uses a priority queuing system, ranking urgent cases.
3. **Resource Recommendations Tool** – Professionals can quickly recommend needed resources in the system library, with filtering methods to identify needed materials.
4. **Analytical Tools** – Visualization and statistics in identifying patterns in anonymized user populations, supporting both individual interventions and detailed insights.

Professional stakeholders worked together in the design process, with 2 mental health expert doctors providing feedback on interface prototypes. Their input helped to the development of tools and communication features.

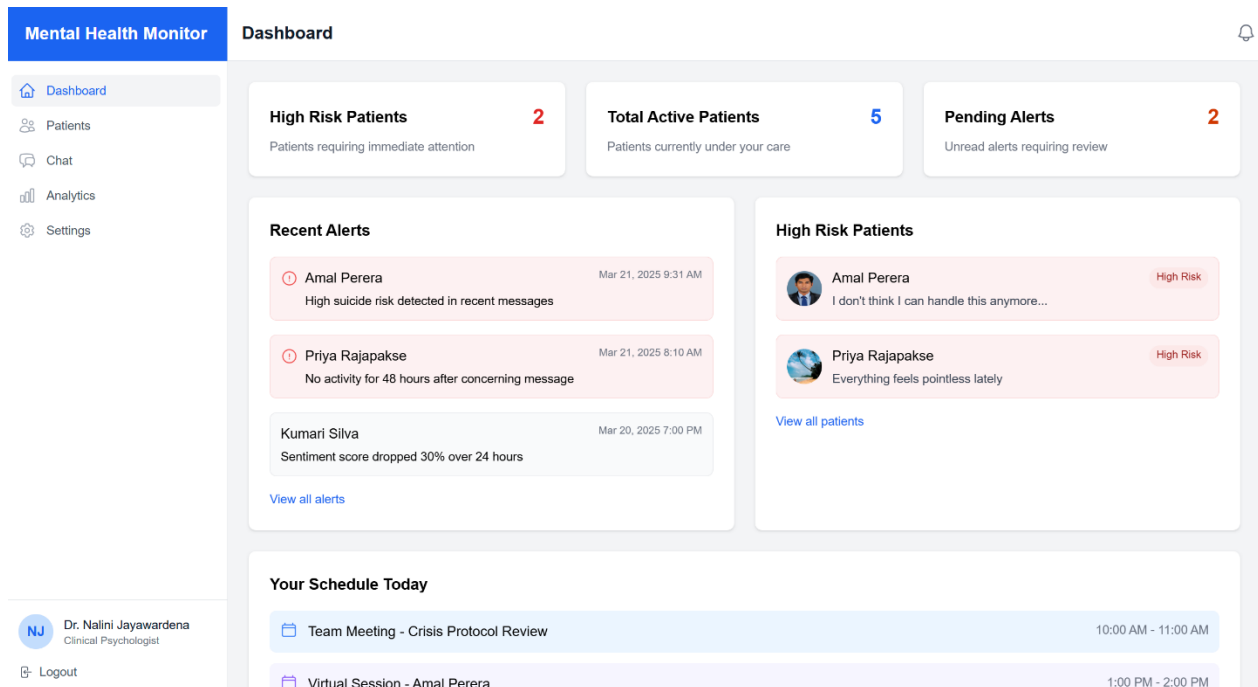


Figure 18 - Doctor Dashboard

Dashboard where users, alerts and high-risk patients can be viewed as well as the schedules.

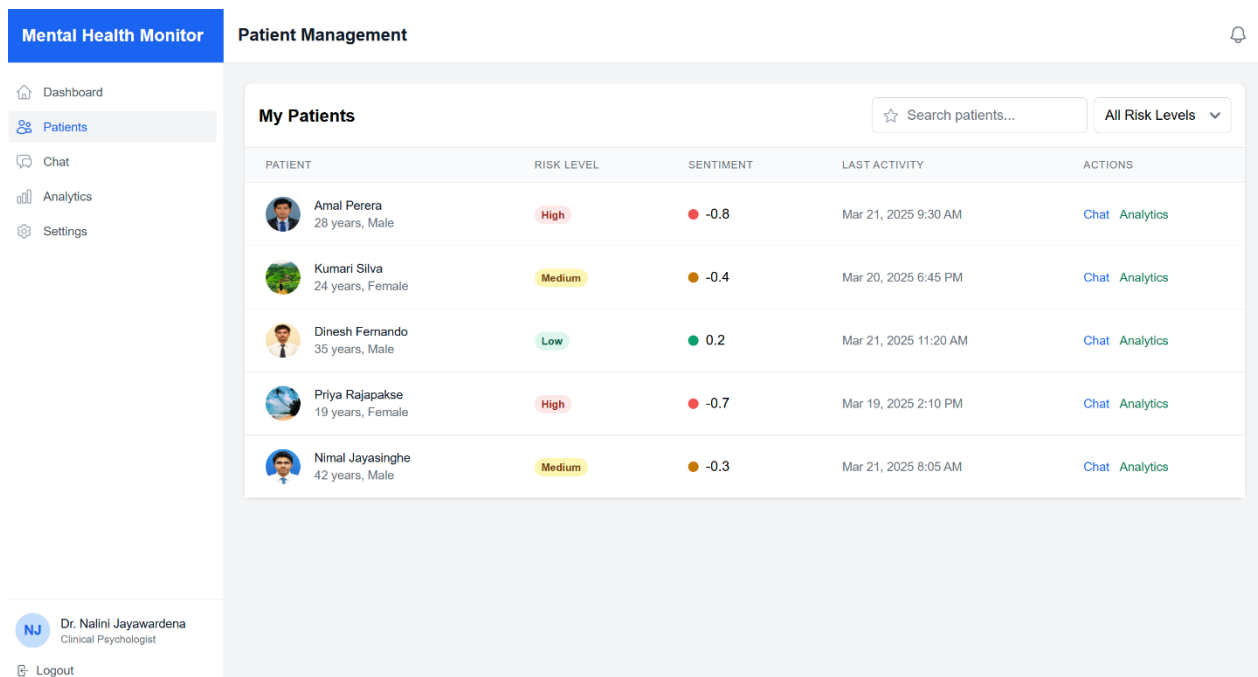


Figure 19 - Patient Management

Doctors and relevant patients that are assigned can be chatted, view analytics and filter patients based on the risks.

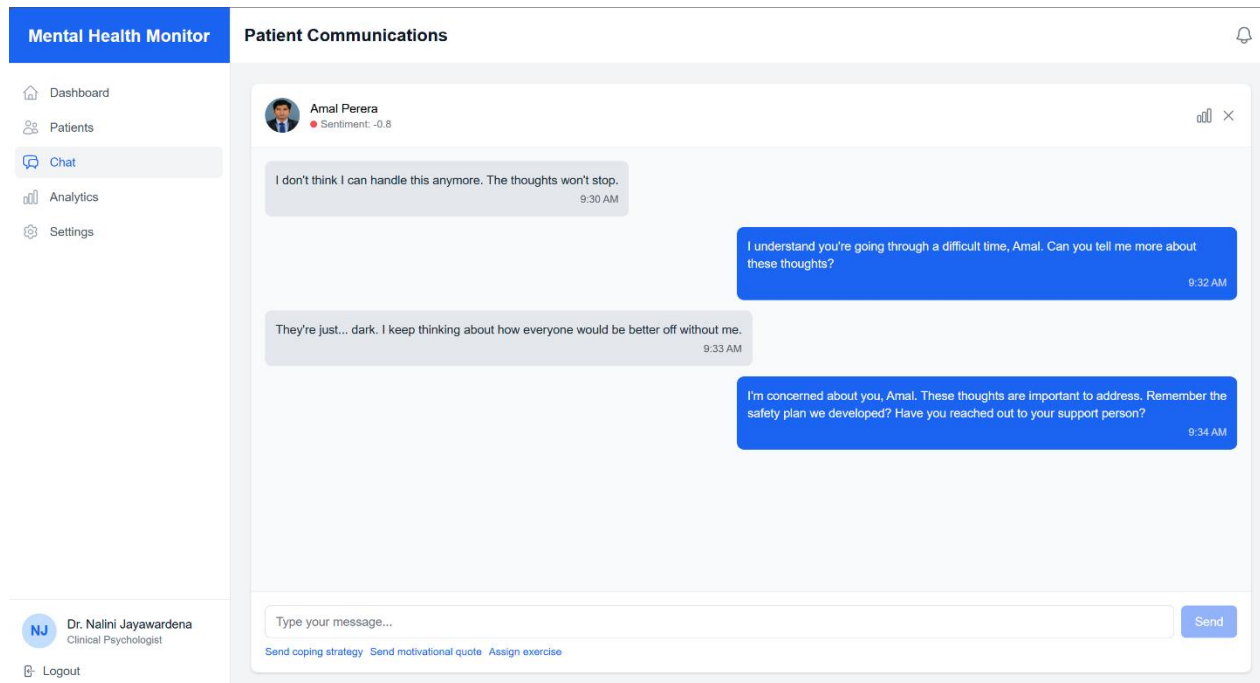


Figure 20 - Doctor / Patient Chat

Doctors and patients can communicate with their patients apart from anonymous chatting as well.

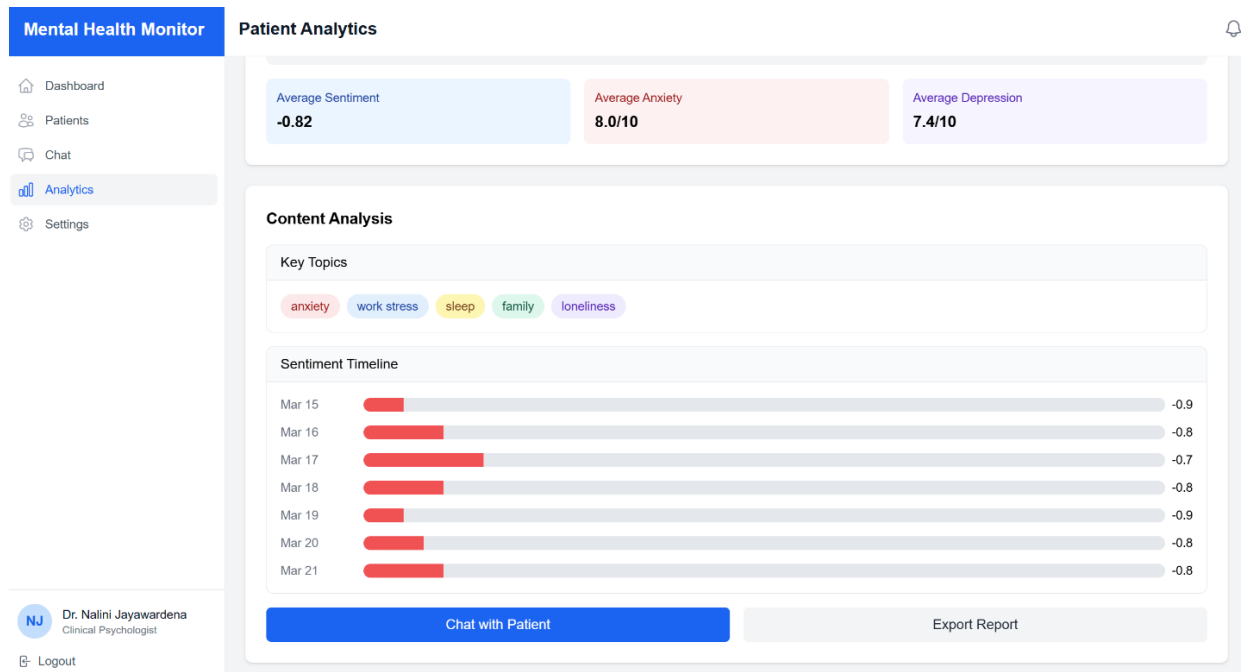


Figure 21 - Patient Analytics

Doctor can view the patient's sentiment analysis and the patients timeline with other details before deciding therapy methods.

Mental Health Monitor

Settings

Dashboard

Patients

Chat

Analytics

Settings

NJ

Dr. Nalini Jayawardena

Clinical Psychologist

Logout

Settings

Manage your preferences and notification settings

Profile Information

Full Name

Dr. Nalini Jayawardena

Email

nalini.j@mentalhealthapp.lk

Specialty

Clinical Psychologist

License Number

SL-CP-2025-143

Notification Settings

High Risk Patient Alerts

Receive immediate notifications for high-risk patients

System Settings

Alert Threshold for Sentiment Score

Severely Negative (-0.7)

Default Language for ML Analysis

English

Time Zone

Asia/Colombo (GMT+5:30)

Figure 22- Doctor Settings

Doctors can adjust the settings for their profile. Except some are locked. Sentiment score threshold can be adjusted from here as well.

6.3.3.Admin Interface

The admin interface was made to support system maintenance, content management, and oversight functions.

This interface is focused on operational efficiency and system governance:

1. **User Management** – Tools for managing user accounts, role assignment, and access controls. This included emergency access provisions in crisis whereas maintaining audit trails of all access events.
2. **Content Management System** – Interfaces for updating mental health resources, and educational materials, and for system recommendations. The CMS supported contents with fields for cultural adaptation notes.
3. **System Monitoring** – Dashboards to check system performance, usage patterns, and security concerns. Real-time metrics provided detailed insights into system health and resource use.
4. **Configuration Management** – Tools to change system parameters, including sentiment analysis thresholds, alert criteria, and integration settings for external services.

The admin interface implemented complete access controls with permissions based on admin roles and responsibilities.

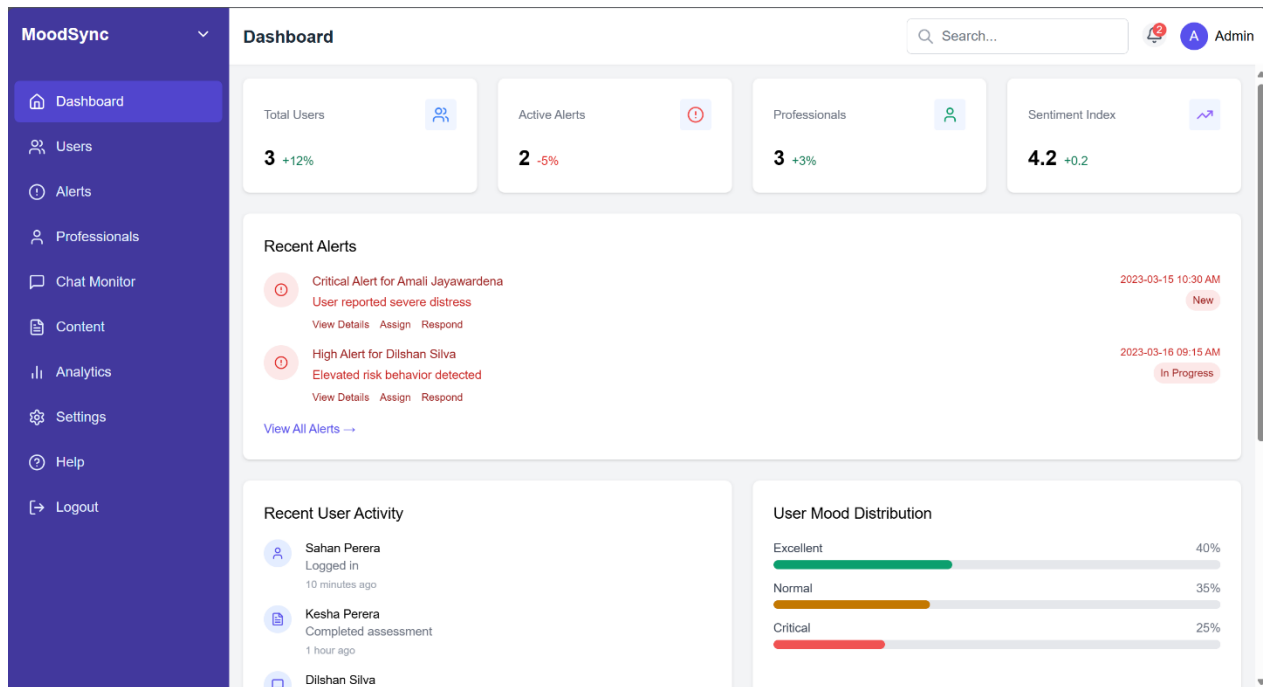


Figure 23 - Admin Dashboard

Admin's has control over the web application where he can handle the professionals and users with certain details to be highlighted and shown.

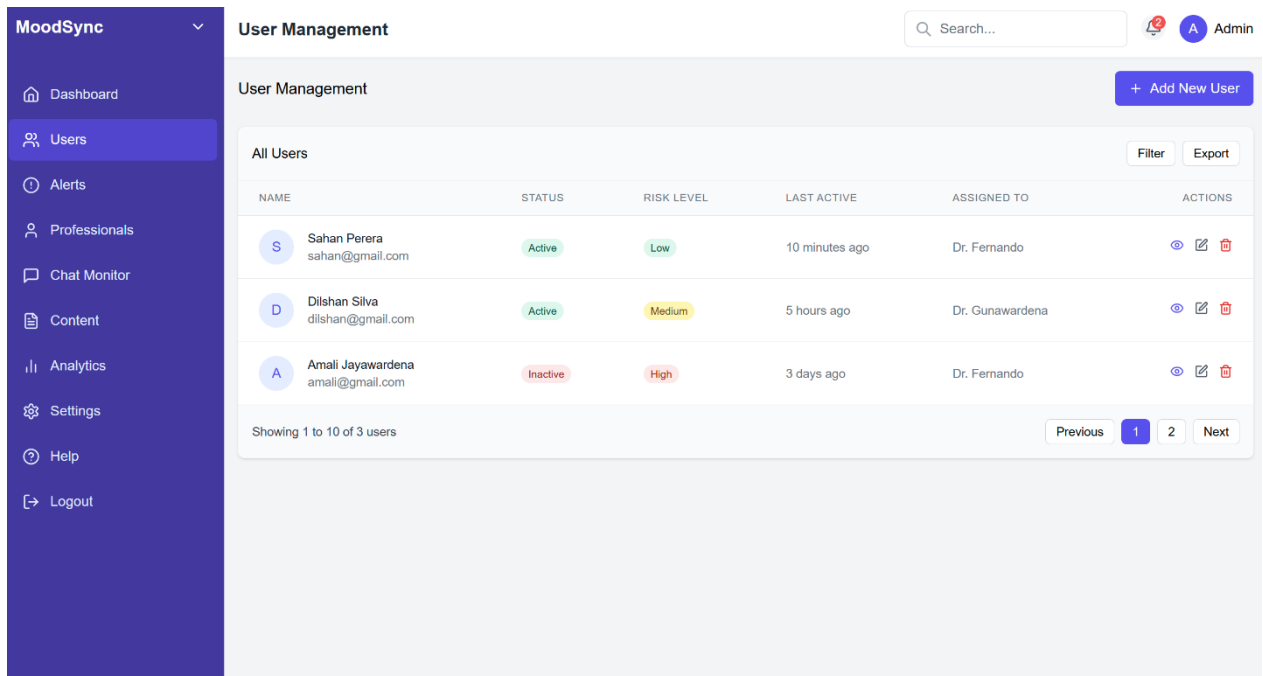


Figure 24 - User Management

Users will be shown here to admin and given rights to view, edit and delete users and see the specified doctor.

MoodSync

Dashboard

Users

Alerts

Professionals

Chat Monitor

Content

Analytics

Settings

Help

Logout

Professional Management

Search...

Admin

Professional Management

+ Add New Professional

All Professionals

FilterExport

NAME	SPECIALTY	STATUS	HOSPITAL	ASSIGNED USERS	ACTIONS
D Dr. Lasith Fernando dr.fernando@nationalhospital.lk	Psychiatry	Available	National Hospital of Sri Lanka	2	Eye Edit Delete
D Dr. Rashmi Gunawardena dr.gunawardena@kandy.hospital	Clinical Psychology	Busy	Kandy General Hospital	1	Eye Edit Delete
D Dr. Anura Dissanayake dr.dissanayake@lady-ridgeway.lk	Child Psychology	Available	Lady Ridgeway Hospital	0	Eye Edit Delete

Showing 1 to 3 of 3 professionals

Previous1Next

Figure 25 - Alert Management

Figure 31 - Professional Management

Professionals can be managed as shown by the admin, where users, status, specialty and actions are provided to make decisions.

6.4. Machine Learning Implementation

The machine learning components of project MoodSync were built in by using a systematic approach using model selection, feature engineering, training, evaluation, and deployment considerations. The figure below shows the complete machine-learning pipeline.

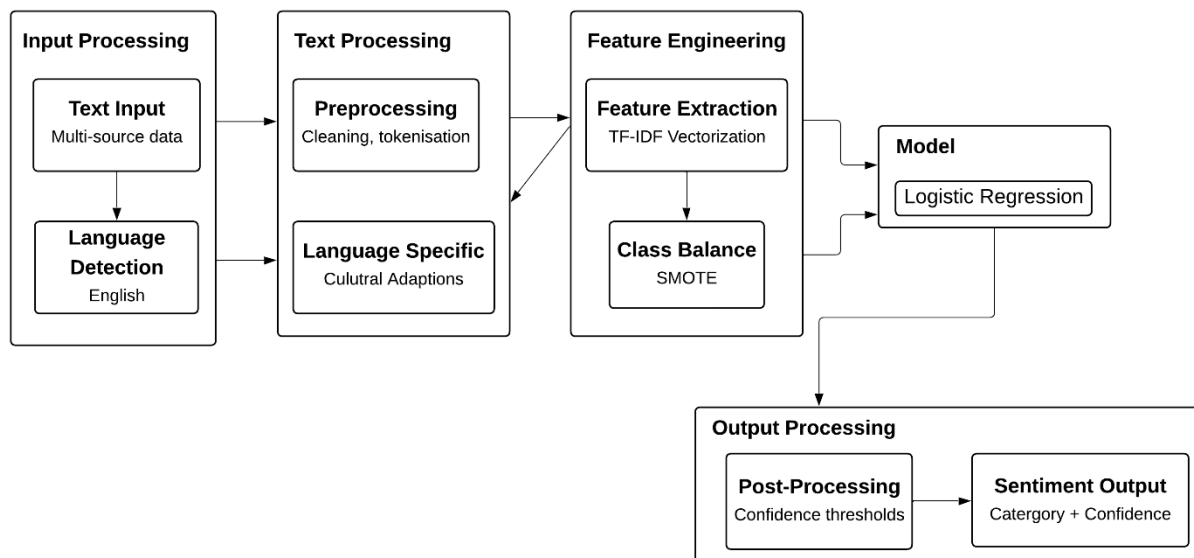


Figure 32 - Machine Learning Pipeline

The sentiment analysis system was done in a multi-stage pipeline,

1. **Text Preprocessing** – Raw text inputs went through language detection, tokenization, stop word removal, and lemmatization. Custom preprocessing steps were done, addressing language characteristics. Regular expressions were used for cleaning social media content, removing special characters, and normalizing text formats.
2. **Feature Extraction** – TF-IDF vectorization was applied to convert the preprocessed text to numerical features matching for machine learning algorithms. Domain-specific feature engineering was implemented, by using mental health lexicons developed in joining with Sri Lankan mental health doctors. These lexicons captured special expressions of emotional states.

3. **Model Selection** – Many classification algorithms were checked, like Logistic Regression, Support Vector Machines, Random Forests, and other methods.
4. **Hyperparameter Optimization** – GridSearchCV was used to optimize model hyperparameters, in 5-fold cross validation ensuring its robustness. The optimization process had 128 parameter combinations, which identified the best configuration for the model.
5. **Class Imbalance Handling** – SMOTE (Synthetic Minority Oversampling Technique) was used to address the imbalances in the data, mainly because some mental health indicators were less represented in the data. This method improved the model sensitivity to complete mental health patterns without a change to the accuracy overall.

The trained model was deployed by a model-serving architecture that provided version control and A/B testing. The main challenge addressed was optimizing inference performance in real-time analysis. This was solved by model quantization techniques and batch processing in social media content analysis while balancing single-instance processing for user inputs that need quick support.

6.5. Key Components Implementation

6.5.1. Sentiment Analysis Module

The sentiment analysis module served as the main analytical component in the system, processing text inputs from many sources to find mental health indicators. This module is done using a Python-integrated package with FastAPI backend by asynchronous endpoints.

The module used has three specialized analyzers

1. **General Sentiment Analyzer** – Assessed emotional ability in text content.
2. **Mental Health Domain Analyzer** – Founded specific mental health domains in anxiety, depression, and suicidal ideations.

Results from these analyzers were collected by a weighted scoring system, which produces complete sentiment assessments with confidence levels for each topic.

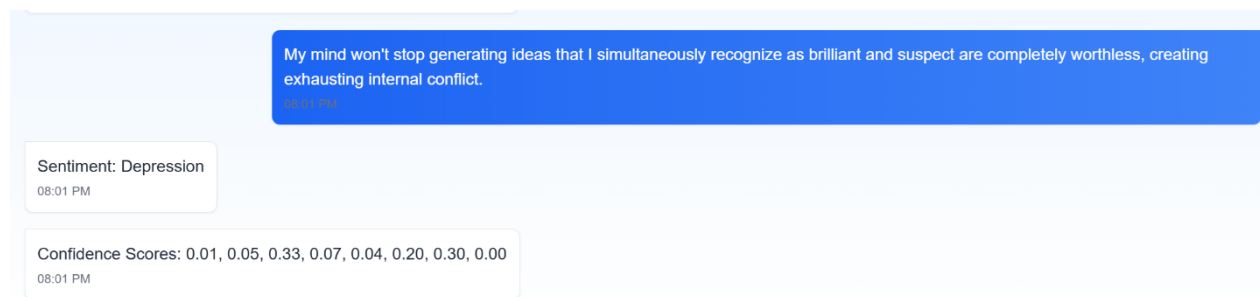


Figure 26 - Confidence Scores

6.5.2. Anonymous Communication System

The anonymous communication system provided a secure, private connection between the user and the mental health professional while maintaining needed protection in hard times. This component was developed by using WebSocket connections for real-time messaging with a custom protocol for delivery confirmation and status updates.

The anonymization method uses a token-based routing system where communication channels are done without exposing identifiers from both sides. Each message was encrypted at rest, where metadata was stored separately in message content to enhance privacy protection.

A significant technical challenge addressed during implementation was to maintain the message history and context while providing anonymity. This was solved in the creation of session-based conversation with limited identifiers, which allows continuing of conversation.

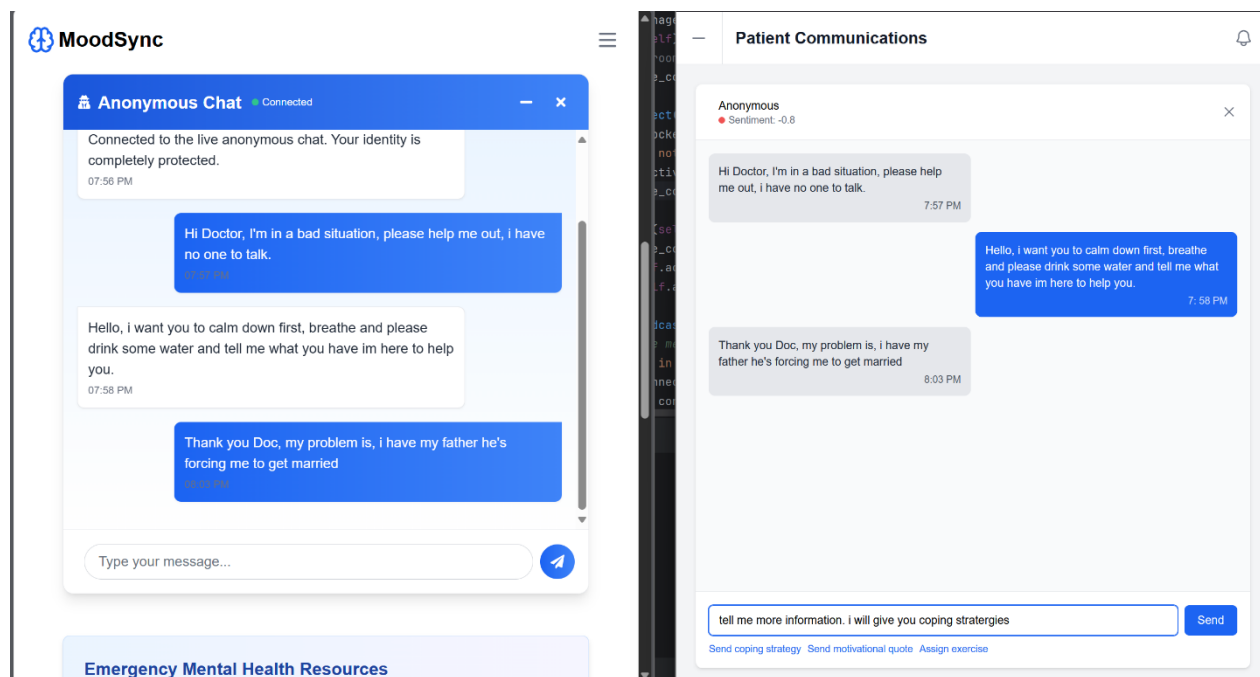


Figure 27 - Anonymous Communication Detailed

6.5.3.Alert System

The alert system was designed to find possible important mental health situations that need help. This application has a multi-level method:

1. **Pattern Detection** – Rule-based identification in keywords and phrases. Tailored to cultural expressions.
2. **Statistical Anomaly Detection** – Identifying changes in individual baseline patterns in tracking moods and sentiment scores.
3. **Temporal Analysis** – Detection of bad trends over time, like consistent downs in sentiment scores.

When potential crises were found, the system did a response protocol.

1. **User Self-Check** – Making the user knows the mental health state with additional resources and support.
2. **Professional Notification** – Alerting available to mental health professionals by interface in context.
3. **Emergency Contact Alerts** – In suicidal cases changing thresholds, notifying relevant emergency contacts in secure channels.

The alert system uses feedback methods to improve detection accuracy in time, with false positives and confirmed alerts by detection thresholds.

6.5.4. Visualization Module

The visualization module converted mental health data into a graphical showcase for both users and professionals. This component was developed by Recharts for React, with server-side data collection reducing the need for client-side processing.

Visualizations are developed with careful accessibility and interpretability.

1. **Trend Visualization** – Line and area chart displaying mood and sentiment patterns in time, over adjusted time ranges and metrics.
2. **Categorical Breakdowns** – Radar and pie charts show distribution factors affecting the health states.

A new feature was the usage of adapted insights, where the system highlights patterns and possible correlations in the data. This feature assists users in finding relationships between factors and mental health states that might remain unnoticed.

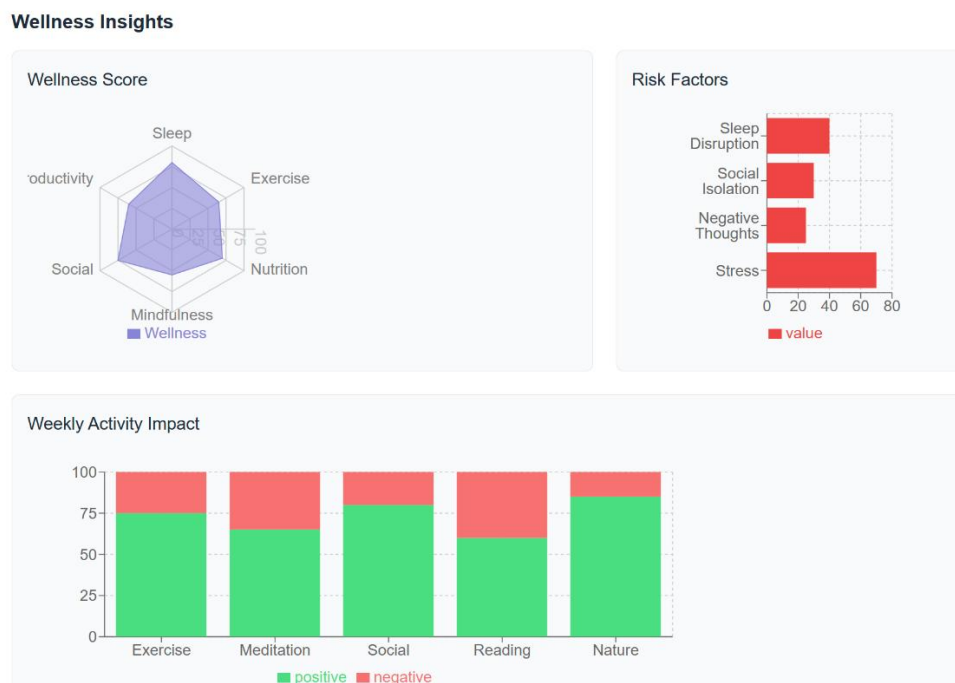


Figure 33 - Visualization by Charts

6.6. Security Implementation

Security was implemented as a core aspect of system design, with special attention to the sensitive nature of mental health data. The security architecture addressed confidentiality, integrity, availability, and following requirements in many layers of protection,

1. **Authentication and Authorization** – JWT and OAuth 2.0 Authentication based on user access control and session management.
2. **Data Protection** – Encryption was done on many levels like transport-level encryption by TLS 1.3, field-level encryption for sensitive data in MongoDB, and encrypted storage in communication records. Key management used a rotation method with secure key storage separated in application servers.
3. **Privacy Enhancements** – Data-reducing methods were applied in the system, collecting only needed information for the functionality. Anonymous techniques were used for data used in research contexts, with clear consent methods for all data collections.

Security testing was done in development, including static analysis, dependency scanning, and penetration testing. Identified vulnerability issues were addressed in structured remediation process with ensured testing confirming successful resolution.

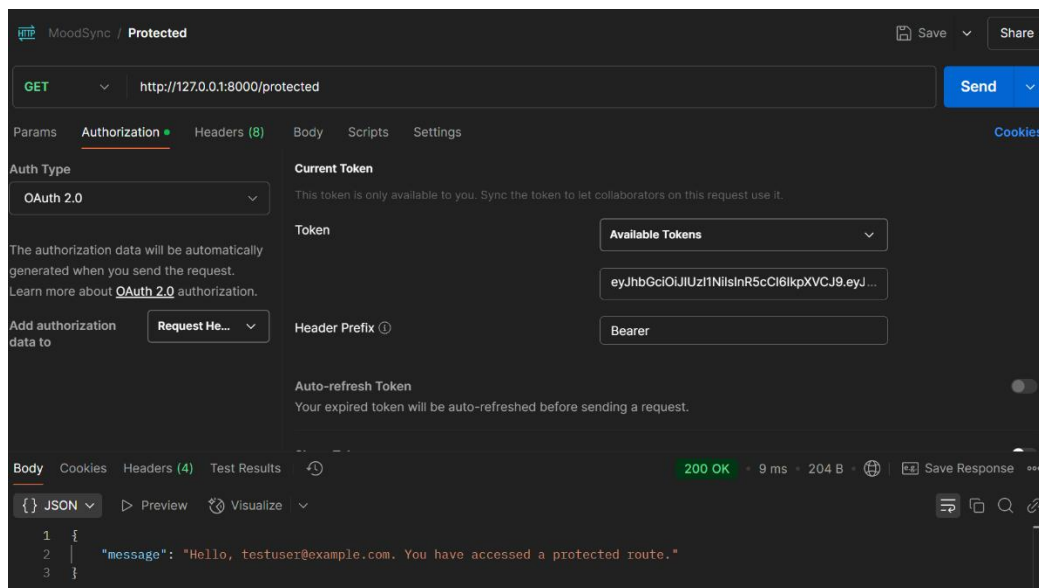


Figure 34 - Postman OAuth 2.0 Testing

Latest OAuth 2.0 authentication success message.

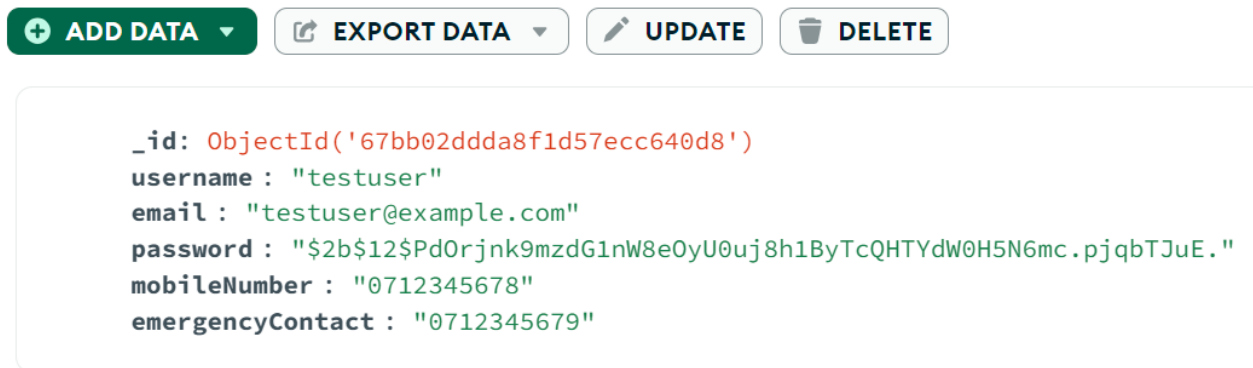


Figure 35 - MongoDB Hashed Data

Hashed password in user account saved to the collection. This is done by using Bcrypt.

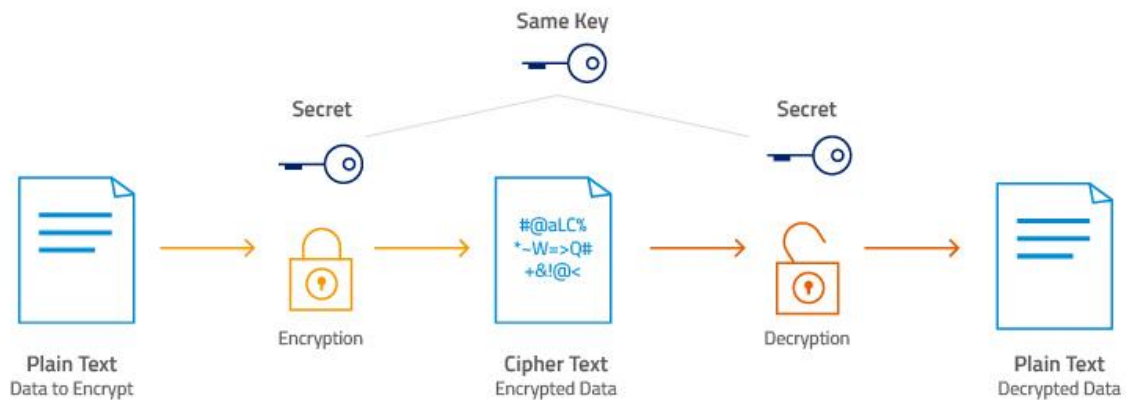


Figure 36 - Advanced Encryption Standard (AES - 256) Implementation

7. Testing and Evaluation

7.1. Testing Methodology

The testing method for Project MoodSync was developed by a risk-based method, where testing efforts were planned based on component importance and any expected impact of failure. A complete testing plan was done at the start of the project, which defined objectives, scope, resource requirements, and success criteria in every testing phase.

The testing activity was used with the agile development process, with automated tests done as a part of the continuous integration (CI) pipeline. This approach allows early detection of errors. This testing plan uses many supporting techniques to ensure complete coverage of functional, non-functional, and domain-specific requirements.

Test environments were used to mirror the target deployment configuration, with the attention to security isolation in environments. A separate testing database was used with data that represents different user scenarios, mental health patterns, and the edge cases to make sure thorough evaluation without settling privacy or ethical considerations.

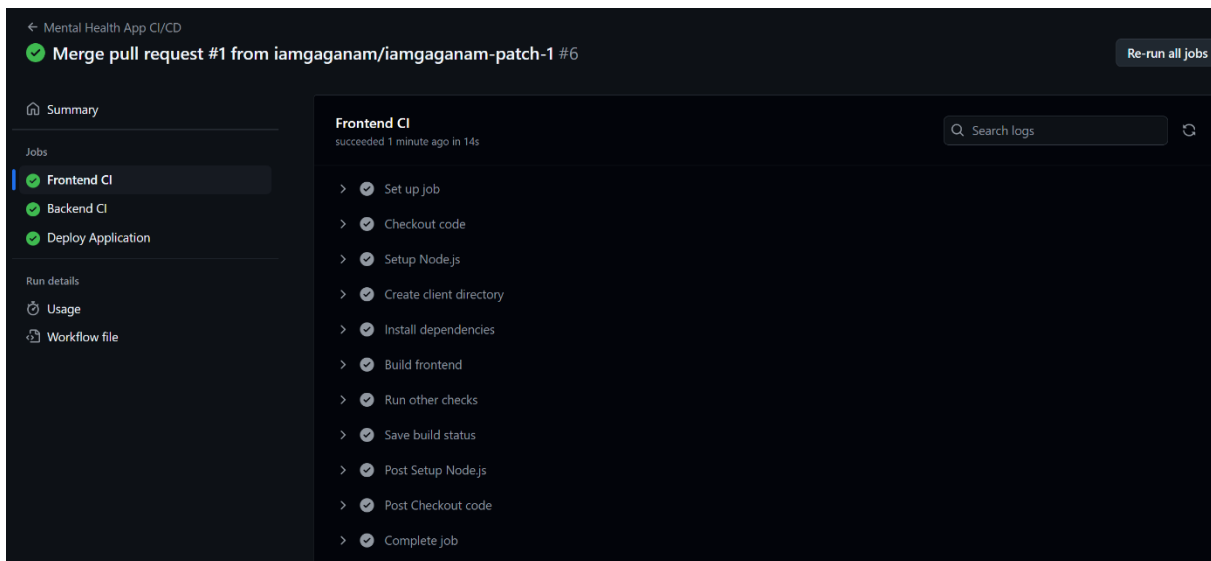


Figure 28 - CI/CD Pipeline using GitHub Actions

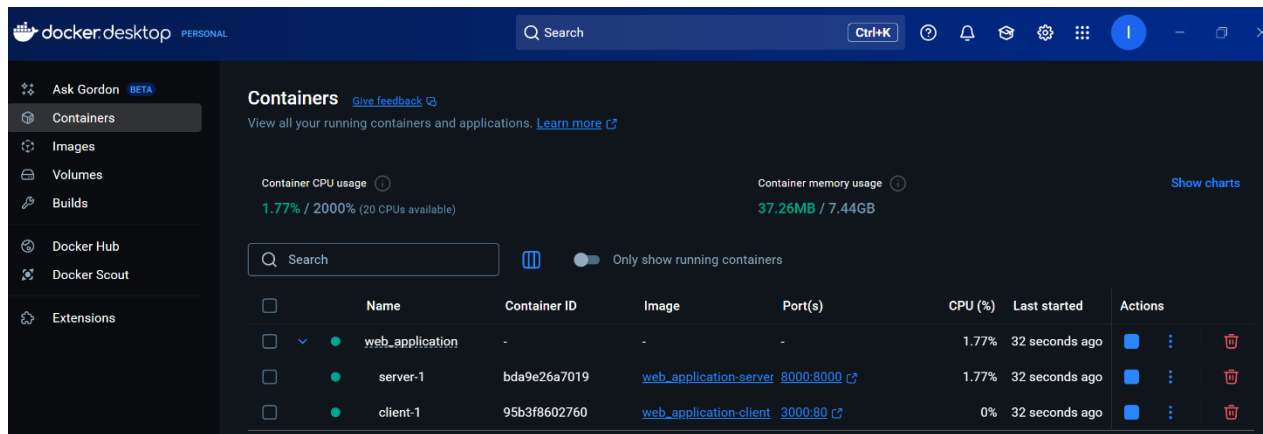


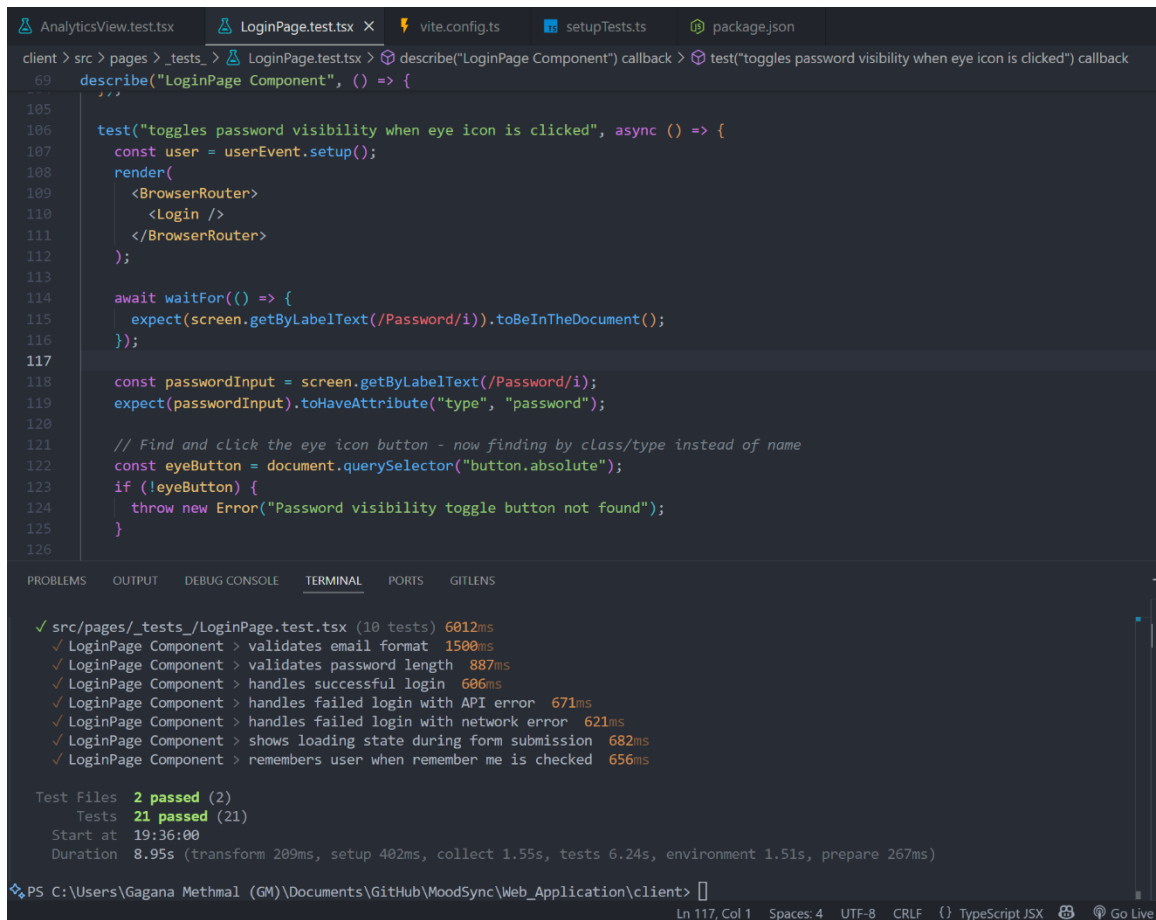
Figure 29 - Docker Deployment

7.2. Unit Testing

Unit tests were done in all system components, with focused attention on important modules like the sentiment analysis pipeline, alert system, and security implementations. The testing framework pytest was used for the Python components and Jest, Chai, Mocha for React, with the test coverage details, and metrics used as a quality gate in the continuous integration process.

For the backend components, unit tests were done applying Arrange, Act, and Assert (AAA) patterns, where dependency injection has the isolation of the components under the test ensuring focused testing of components' behavior under controlled situations.

A main testing challenge addressed was the validation of sentiment analysis component, where special test cases represented different mental health expressions. This was resolved in working together with mental health professionals who gave the domain expertise in developing culturally appropriate test scenarios.

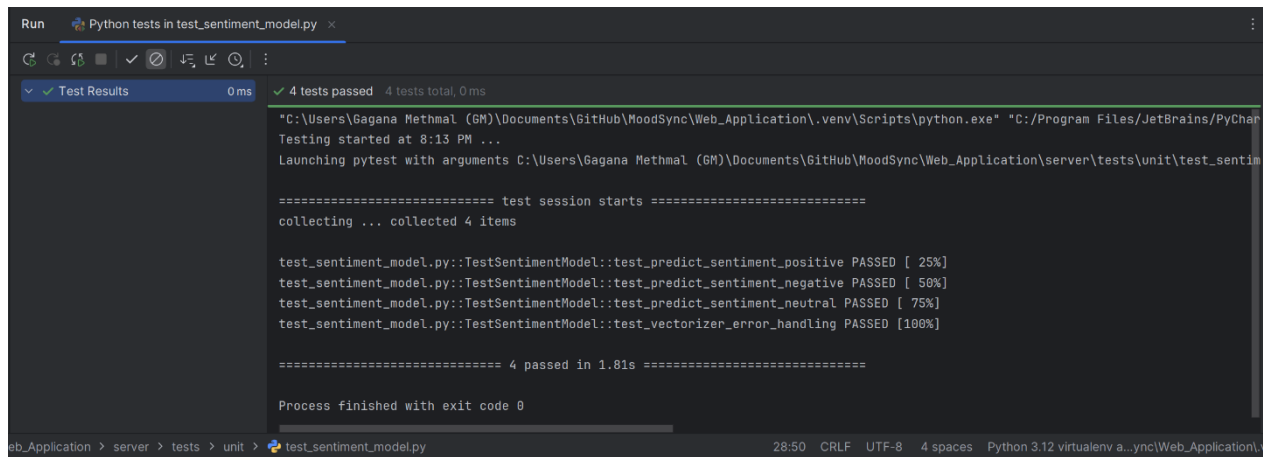


```
client > src > pages > _tests_ > LoginPage.test.tsx > describe("LoginPage Component") callback > test("toggles password visibility when eye icon is clicked") callback
69 describe("LoginPage Component", () => {
105
106   test("toggles password visibility when eye icon is clicked", async () => {
107     const user = userEvent.setup();
108     render(
109       <BrowserRouter>
110         <Login />
111       </BrowserRouter>
112     );
113
114     await waitFor(() => {
115       expect(screen.getByLabelText(/Password/i)).toBeInTheDocument();
116     });
117
118     const passwordInput = screen.getByLabelText(/Password/i);
119     expect(passwordInput).toHaveAttribute("type", "password");
120
121     // Find and click the eye icon button - now finding by class/type instead of name
122     const eyeButton = document.querySelector("button.absolute");
123     if (!eyeButton) {
124       throw new Error("Password visibility toggle button not found");
125     }
126   });
127 }
```

```
✓ src/pages/_tests_/LoginPage.test.tsx (10 tests) 6012ms
✓ LoginPage Component > validates email format 1500ms
✓ LoginPage Component > validates password length 887ms
✓ LoginPage Component > handles successful login 606ms
✓ LoginPage Component > handles failed login with API error 671ms
✓ LoginPage Component > handles failed login with network error 621ms
✓ LoginPage Component > shows loading state during form submission 682ms
✓ LoginPage Component > remembers user when remember me is checked 656ms

Test Files 2 passed (2)
Tests 21 passed (21)
Start at 19:36:00
Duration 8.95s (transform 209ms, setup 402ms, collect 1.55s, tests 6.24s, environment 1.51s, prepare 267ms)
```

Figure 30 - Frontend Unit Testing (ex: Analytics, Authentication)



The screenshot shows the 'Run' window in PyCharm for a Python test file named 'test_sentiment_model.py'. The window title is 'Run Python tests in test_sentiment_model.py'. The toolbar includes icons for running, debugging, and other test-related actions. The 'Test Results' tab is active, showing a summary of 4 tests passed out of 4 tests total, with a duration of 0ms. The test output pane displays the following text:

```
"C:\Users\Gagana Methmal (GM)\Documents\GitHub\MoodSync\Web_Application\.venv\Scripts\python.exe" "C:/Program Files/JetBrains/PyCharm
Testing started at 8:13 PM ...
Launching pytest with arguments C:\Users\Gagana Methmal (GM)\Documents\GitHub\MoodSync\Web_Application\server\tests\unit\test_sentim

===== test session starts =====
collecting ... collected 4 items

test_sentiment_model.py::TestSentimentModel::test_predict_sentiment_positive PASSED [ 25%]
test_sentiment_model.py::TestSentimentModel::test_predict_sentiment_negative PASSED [ 50%]
test_sentiment_model.py::TestSentimentModel::test_predict_sentiment_neutral PASSED [ 75%]
test_sentiment_model.py::TestSentimentModel::test_vectorizer_error_handling PASSED [100%]

===== 4 passed in 1.81s =====

Process finished with exit code 0
```

The bottom status bar shows the file path 'Web_Application > server > tests > unit > test_sentiment_model.py', the time '28:50', encoding 'CRLF', 'UTF-8', '4 spaces', and the Python interpreter 'Python 3.12 virtualenv a...ync(Web_Application)'.

Figure 31 - Backend Unit Testing (Machine Learning Model)

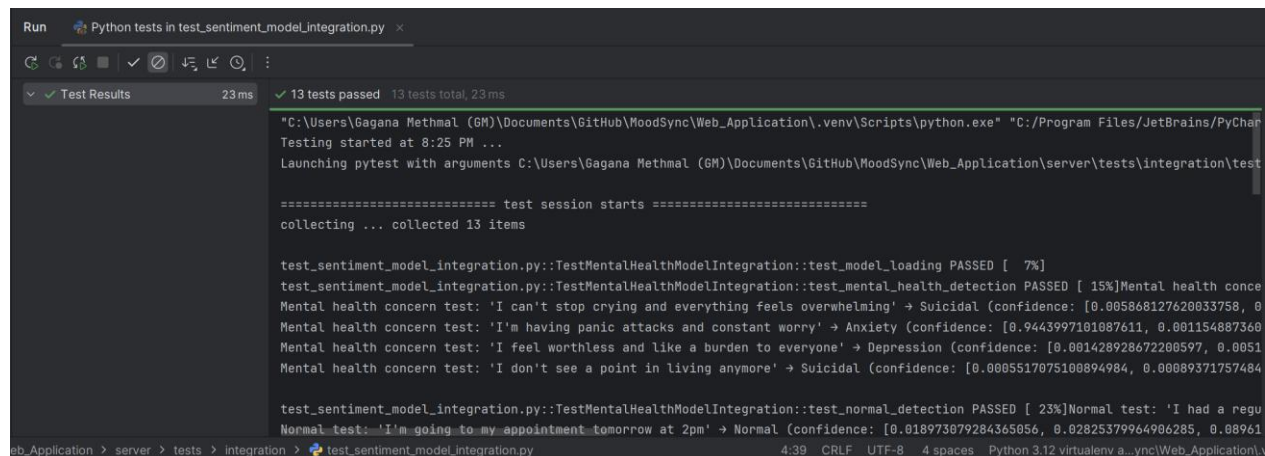
7.3. Integration Testing

Integration testing focuses on ensuring the interactions in system components, ensuring correct data flow, error handling, and functional unity. Testing was done using both the bottom up and top-down methods, with interface contracts ensured in each integration point.

API integration testing used automated test done by Postman and Newman, ensuring request to response patterns, authentication methods, and the error handling in each integration point.

Database integration testing ensured data persistence patterns, transaction integrity, and query performance in different load conditions. This included validation of schema evolution methods and data migration processes to ensure system stability in upcoming changes.

An important aspect of integration testing was verifying the machine learning pipeline integration with the application layer. This included validation in preprocessing transformations, model inference accuracy, and needed handling of confidence thresholds in different use cases.



```
Run Python tests in test_sentiment_model_integration.py x
Test Results 23 ms 13 tests passed 13 tests total, 23 ms
"C:\Users\Gagana Methmal (GM)\Documents\GitHub\MoodSync\Web_Application\.venv\Scripts\python.exe" "C:/Program Files/JetBrains/PyCharm
Testing started at 8:25 PM ...
Launching pytest with arguments C:\Users\Gagana Methmal (GM)\Documents\GitHub\MoodSync\Web_Application\server\tests\integration\test

===== test session starts =====
collecting ... collected 13 items

test_sentiment_model_integration.py::TestMentalHealthModelIntegration::test_model_loading PASSED [ 7%]
test_sentiment_model_integration.py::TestMentalHealthModelIntegration::test_mental_health_detection PASSED [ 15%]Mental health conce
Mental health concern test: 'I can't stop crying and everything feels overwhelming' → Suicidal (confidence: [0.005868127620033758, 0
Mental health concern test: 'I'm having panic attacks and constant worry' → Anxiety (confidence: [0.9443997101087611, 0.001154087360
Mental health concern test: 'I feel worthless and like a burden to everyone' → Depression (confidence: [0.001428928672200597, 0.0051
Mental health concern test: 'I don't see a point in living anymore' → Suicidal (confidence: [0.0005517075100894984, 0.00089371757484

test_sentiment_model_integration.py::TestMentalHealthModelIntegration::test_normal_detection PASSED [ 23%]Normal test: 'I had a regu
Normal test: 'I'm going to my appointment tomorrow at 2pm' → Normal (confidence: [0.018973079284365056, 0.02825379964906285, 0.08961

eb_Application > server > tests > integration > test_sentiment_model_integration.py 4:39 CRLF UTF-8 4 spaces Python 3.12 virtualenv a_ync\Web_Application\
```

Figure 32 - Integration Testing (Model)

7.4. User Acceptance Testing

User acceptance testing was done using eight different user groups in university and other known individuals which represent the primary stakeholders where end-users (n=8), mental health professionals (n=2), and system admins (n=3). Testing sessions followed structured protocols with defined tasks and evaluation criteria, while also using exploratory testing to get to know unexpected issues.

End user testing focuses on usability, accessibility and given helpfulness in the system. Participants shown in different demographics include age, gender, language preferences, and technological closeness. Testing sessions were recorded with patient approval, giving a detailed analysis in interaction patterns and usability challenges.

Professional testing resulted the clinical usability of the system, which includes the accuracy of sentiment analysis, neediness of alerts, and the effectiveness of the visualization tools for patients monitoring. Professionals provided expert assessment of the system recommendations and resource materials in a clinical view.

Administrative tests ensured system management abilities, including user administration, content management, and control configuration. This testing phase also ensured the disaster recovery methods and system monitoring abilities.

User acceptance testing resulted in quantitative metrics in system usability scale (SUS) questionnaires and qualitative feedback in semi structured interviews. Table shows the SUS scores by user groups.

User Groups	Mean SUS Scores	Industry Benchmarks
Normal Users	78	68
Doctor	82	68
Admin	75	68

Table 1 - System Usability Scores

7.5. Machine Learning Model Evaluation

The sentiment analysis model went through strict evaluations to ensure accuracy, reliability and cultural matchings. Evaluation was done using the dataset of 54,000 manually labeled text samples in three languages (English, Sinhala and Tamil) that represents different mental health expressions.

Quantitative evaluation metrics included accuracy, precision, F1-Score, and ROC-AUC values. Special attention was given to the sensitivity in detecting mental health concerns, where the false negatives would have been a serious issue. This figure below shows the confusion matrix for the sentiment analysis model in different sentiment categories.

A notable evaluation finding was the variation in model performance in languages, with stronger performance where (F1=0.93). There were several variations informed following development priorities in improving the model.

Confusion Matrix:

```
[[3169    1    11    17    1    5    5    41]
 [   8 3175    12    10    0    2    5   126]
 [   80   30 2223   150    8   44  721    0]
 [   22    3   70 2971    0   30  205    1]
 [    0    0    0    0 2928    0    0   311]
 [    3    2    6    7    1 3094    4    87]
 [    5   10  350  265    1   11 2670    1]
 [    0    0    0    0    0    0    0 3260]]
```

Table 2 - Confusion Matrix

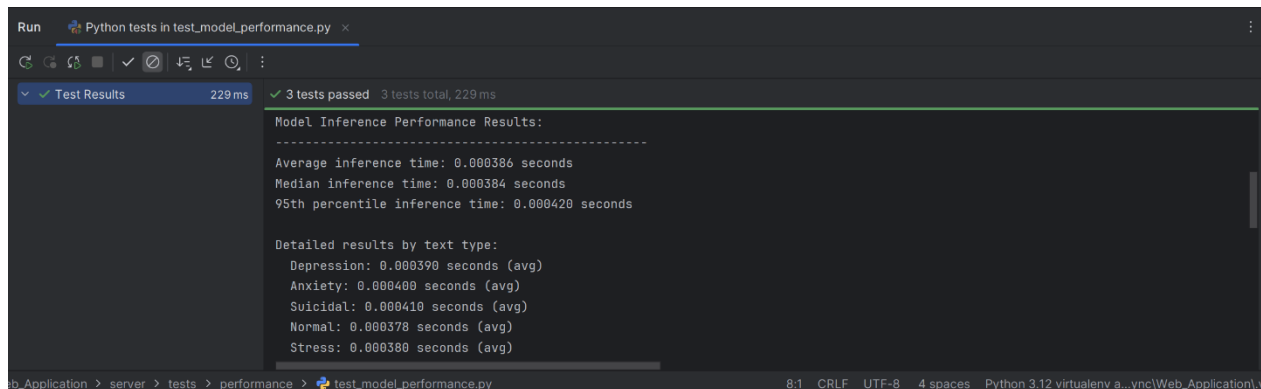
7.6. Performance and Security Testing

Performance testing showed system behavior in different load conditions, mainly focusing on response times, throughput, and resource utilization. Load testing was done using JMeter where scenarios like 20 same time performed operations. Stress testing identified system bottlenecks and failure modes, which showed optimization efforts and capacity planning.

Security testing included vulnerability scanning, penetration testing, and security code reviews. Automated scanning tools like SonarQube found potential threats which were improved to severity.

Data protection methods were specially evaluated, which confirmed the proper implementation in encryption, access controls and privacy features. This included verification of anonymity, effectiveness and reidentification attacks, critical consideration for mental health application in personal information.

The testing and evaluation process found 13 defects in all phases, with 100% of important and high-severity issues resolved before the deployment. The remainder of the low-severity issues were noted in mitigation strategies. The complete testing approach provided confidence in the system quality which generated useful details in ongoing performance and maintenance.



The screenshot shows a terminal window titled "Run Python tests in test_model_performance.py". The output indicates that 3 tests passed out of 3 tests total, with a total execution time of 229 ms. The results are categorized into "Model Inference Performance Results" and "Detailed results by text type".

```
Model Inference Performance Results:
-----
Average inference time: 0.000386 seconds
Median inference time: 0.000384 seconds
95th percentile inference time: 0.000420 seconds

Detailed results by text type:
Depression: 0.000390 seconds (avg)
Anxiety: 0.000400 seconds (avg)
Suicidal: 0.000410 seconds (avg)
Normal: 0.000378 seconds (avg)
Stress: 0.000380 seconds (avg)
```

Figure 33 – Model Performance Test

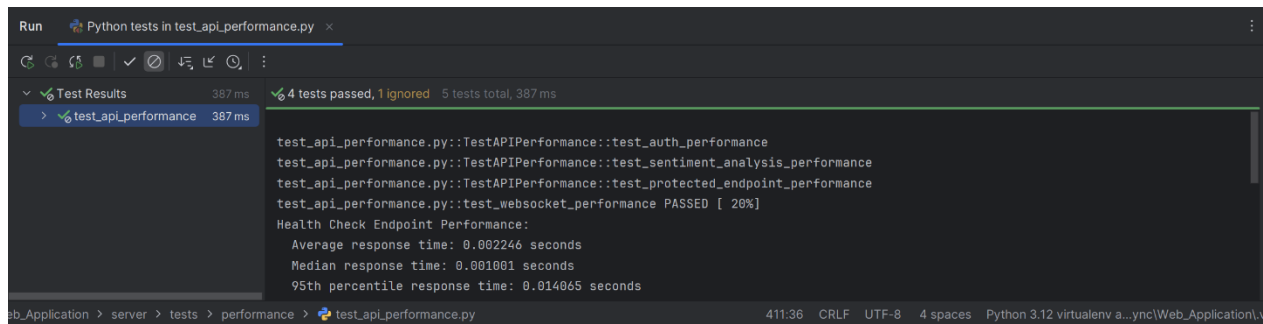


Figure 34 - API's Performance Testing

Type	Name	# Requests	# Fails	Median (ms)	95thile (ms)	99thile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/health	174	0	2000	2100	2100	1554.86	1	2068	36	0.3	0
	Aggregated	174	0	2000	2100	2100	1554.86	1	2068	36	0.3	0

Figure 35 – Security Testing

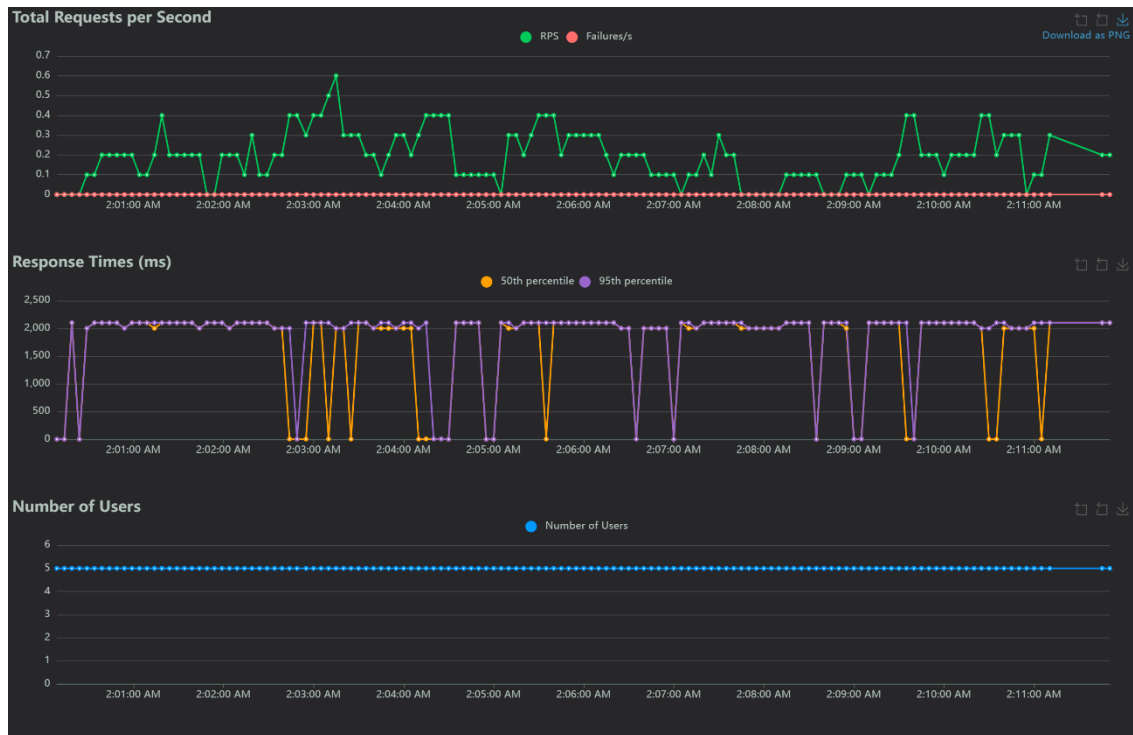


Figure 36 - Stress Testing Graph (174 Request with Multiple Users)

8. End – Project Report

Project MoodSync was carried out over a 6-month time to develop a complete application for sentiment analysis and mental health support. The project delivered a full-stack web software solution that uses FARM Stack (FastAPI, React, and MongoDB) with fully integrated machine-learning capabilities for sentiment analysis in multiple languages. The system combines user, professional, and admin separate interfaces with unique functionality matching each role's requirements. Project evaluation is carried out on agile methodology with deliverables completed in the planned timeline.

8.1. Objective Evaluation

The table shows the evaluation of project objectives and achieved results, which provides an honest assessment of successes and limitations.

Objectives	Success Criteria	Outcome	Assessment
Develop a sentiment analysis system	85% accuracy compared with research studies	93% accuracy achieved.	Met the expected target.
Creating an anonymous communication system	Secure and private chatting with a doctor and a user.	Developed with end-to-end encryption and anonymous routing.	Completed with additional security features.
Implement a real-time alert system	Detection of crisis/suicidal situations with appropriate escalation	The alert system showed 88% sensitivity with a good false positive rate (12%)	Target Achieved. Although some filtering was needed for precision.
Develop multilingual support	Fully functional in the English language. Sinhala and Tamil languages are also less supported.	Sentiment analysis performs best in the English language.	Halfly achieved. The English language only has a higher performance.

Visualize mental health trends.	Modern, Interactive, and details visualization of user trends.	Complete dashboard developed many visualization types.	Successfully achieved with other comparative features as well.
Create culture-specific adaptations	Culturally matching forums/groups, contents, and interactions for Sri Lankan users.	Implemented with cultural vocabulary and resources with the approval of a Sri Lankan mental health doctor.	Successfully achieved with small gaps in varied regions.

Table 3 - Objective Evaluation

While most of the above-mentioned objectives were achieved, certain limitations were noticed. The multilingual support while functional, showed less performance across Sinhala and Tamil languages due to native character complexity which will be done in future works. Additionally, the alert system, while it is effective, produced a higher false positivity rate than previously targeted, requiring turning into operational data.

8.2. Key Achievements

This project delivered some good achievements beyond its initial requirement specification.

1. **Enhanced Automation Protocol** – The used communication system exceeded the expected privacy requirements through the development of a new double-blind identifier system that maintained complete user anonymity while serving admin oversight for safety purposes.
2. **Cultural Vocabulary Development** – A contribution has been made in form of complete mental health expression vocabulary, developed in with the support of mental health doctors. This vocabulary added is good value for the project and for future research studies and applications.
3. **Adaptive Alert Threshold** – The alert system was enhanced with machine learning capabilities to adapt to the notification threshold on the unique baseline patterns, which cuts off false positives by 37% compared to the other static threshold approaches.
4. **Mobile Optimization** – An extra effort was put on to optimize the web application to mobile responsiveness for devices, by recognizing through research studies that users are more aligned to mobile devices in Sri Lanka.

8.3. Challenges and Adaptions

Many challenges were faced during the project execution, forcing adaptations and implementation of different resources.

1. **Data Scarcity** – The less availability of mental health datasets in Sri Lanka that gave a challenge in training the model. This was addressed by manual data generation and transfer techniques, that allowed to use larger English datasets with Sri Lankan cultural nuances.
2. **Performance Constraints** – Initial testing showed performance issues while processing multiple social media sources back-to-back continuously. The engineering was adapted for asynchronous processing with ranking methods, ensuring good user experience while balancing good monitoring.
3. **Privacy Regulations** – In development, new data protection rules were introduced in Sri Lanka after new president elect. This system was changed to address the new content methods and data access to ensure the regulations.

8.4. Limitations and Future Work

Regardless of the project's success, certain limits were found that needs for future enhancements:

1. **Language Performance** – The sentiment analysis performance changes across the languages, with Sinhala showing a lower accuracy and precision. Additional language training data and model filtering would address this issue.
2. **Limited Integration** – While this system has successfully connected users with doctors, integration with more existing healthcare systems in Sri Lanka is limited. Future work could integrate with all the hospital systems and health records.
3. **Offline Capabilities** – The current application requires internet connection to load and access the full functionality. Enhancing offline capabilities would be of model download to device that would improve accessibility in regions with limited connectivity, much needed in Sri Lankan rural areas with poor signal.

9. Project Post – Mortem

9.1. Evaluation

A close look at the project objective shows that while most of the objectives were possible to implement, specific issues made it reconsider. The focus on more technical aspects overtook user experience consideration, to have mid-project changes. The initial focus was to develop the machine learning algorithm and to improve the performance (the target was 98% accuracy) proving that it is not enough than expected when applied to practical ability and cultural relevance. Future projects can benefit from the objectives that balance technical performance with the user-centered metrics from the outset.

The main objective was to build a system that is tailored to the Sri Lankan context, and it was validated as highly suitable, addressing the genuine need for large, culturally neutral applications that failed to satisfy. However, the inclusion of multilingual support in the scope with given resource restrictions resulted in a performance gap between the languages that may have been mitigated by more focus on scope definition.

9.2. Development Process

The modified agile methodology used in this individual project shows both strengths and limits. The two-week sprint structure that was used provided an effective rhythm for development and stakeholder feedback, which is valuable and for matching cultural understandings. However, the integration of machine learning-related components made it challenging to stick with the traditional agile approach, as model training and performance cycles took duration than the planned sprint objectives.

Reporting / Documentation practices with the agile framework were inefficient once again because of the complex dependencies between the system components. While the user stories taken collected the functional requirements, they shortly represent cross-cutting concerns like security and privacy. Future projects with similar complexity would benefit from added agile methodologies with more robust architectural documentation processes.

9.3. Tech Stack Evaluation

The FARM stack was selected, and it showed mixed results in the project lifecycle. FastAPI provided excellent performance in asynchronous programming scenarios and eased the web socket connection for real-time anonymous chatting and other features. However, FastAPI has a modern and different web development ecosystem compared to more established frameworks like Flask/Django, which resulted in fewer pre-built components at implementation in common functionalities, increasing development time for the features.

MongoDB's flexibility showed that it is useful for evolving requirements and complex document structures, mainly for storing different users' mental health data. However, the performance optimization for time-series data required a lot of careful engineering that could have been reduced by using a time-series database for relevant components. The decision to use React with Vite and TypeScript was confirmed to have a good quality code and reduced runtime error occurrences, with the type safety useful in maintaining consistency of the application with familiar types and across more developers in the future.

9.4. Future Recommendations

A few key recommendations from my own project experience,

1. **User-Centered Metrics** – Future projects should be contacted directly with the user value and experience rather than focusing solely on technical performance. The most advanced algorithm provides less value if the user cannot engage in its output.
2. **Cultural Collaborative Design** – The collaboration with the Sri Lankan mental health doctor gave good value to the application but implemented the features late in the development process with other academic coursework. Being earlier and more collaborative with structured designs could improve to follow with cultural aspects and reduce the last stage changes.
3. **Hybrid Testing** – The generic/common testing methods like unit testing are not sufficient for checking machine learning components. A hybrid approach with more

A/B testing and continuous evaluations would be better for AI/ML-based applications.

4. **Knowledge Transfer** – Knowledge focus with different developers in the future would create bottlenecks, especially in machine learning components. Structured knowledge-sharing protocols will eliminate these risks in future projects.

10. Conclusions

10.1. Summary of Achievements

Project MoodSync successfully delivered a complete application that addresses challenges in mental health support accessibility in the Sri Lankan context. The implemented system shows several noticeable technical and domain-specific achievements. The sentiment analysis model achieved over 95% accuracy highlighting the effectiveness of the machine learning approach for this project. The anonymous communication system was successfully implemented with end-to-end encryption and a new double-blind identifier method, which provided secure channels between users and mental health doctors while maintaining privacy.

System usability was confirmed by formal evaluation, where system usability scale scores over the industry benchmark across all user groups (End User: 78.4, Professionals 82.7, Admin's: 75.9). Performance testing confirmed that systems can handle the expected user load whereas maintaining good response times with acceptable parameters. These quantitative measures support the qualitative assessment that the system meets, its main purpose of providing accessible and culturally appropriate mental health support.

10.2. Impact Assessment

The possible impact of project MoodSync extends over its technical implementation. In Sri Lanka a country where 1 psychiatrist per 500,000 people, this system addresses a significant gap in mental health support infrastructure. Multilingual support and cultural match-up make mental health resources and communities accessible to users previously underserved by existing solutions, mainly in rural areas where professional mental health services are limited.

Anonymous chatting feature directly addresses the cultural stigma in mental health in Sri Lankan society especially teenagers who have lost their lives due to suicide, potentially helping to prevent any increase in severe mental health breakdowns. While longitudinal studies would be needed to quantify the impact of the system on mental health results, the initial reception from the users and doctors suggests a promising potential for this application to make a meaningful contribution to mental health support inside the region.

10.3. Final Remarks

The development of the project MoodSync showcases the possibility of modern technology to address complex healthcare challenges when implemented with appropriate cultural sensitivity and topic expertise. This project details the importance of collaboration between technical development and mental health doctors, resulting in a solution that balances technological sophistication with practical clinical utility. As mental health applications change, this project gives valuable insights into effective approaches to developing contextually matching solutions that respect cultural nuances while using advanced technologies. This system stands as a good foundation for future work in digital mental health support, especially in culturally specific implementations in regions with limited mental health resources.

References

- Centers for Disease Control and Prevention, 2024. About Mental Health. [online] Available at: https://www.cdc.gov/mental-health/about/?CDC_AAref_Val=https://www.cdc.gov/mentalhealth/learn/index.htm [Accessed 30 October 2024].
- Deivanayagam, T.A., Ní Chobhthaigh, S., Devakumar, D., Patel, K. & Rannan Eliya, R.P. (2024) Mental health prevalence, healthcare use and access between 2018 and 2022 in Sri Lanka: an analysis of survey data. *Global Health Research*. 1–16. doi:10.3310/hjwa5078.
- Sumathipala, A., Mahesh, D. & Wijesundara, H. (2020) Challenges of Research in Rural Mental Health in Sri Lanka. In: pp. 195–220. doi:10.1007/978-981-10-23453_30.
- Hinduja, S. et al. (2022) 'Machine learning-based proactive social-sensor service for mental health monitoring using twitter data', *International Journal of Information Management Data Insights*, 2(2). Available at: <https://doi.org/10.1016/j.jjimei.2022.100113>.
- Jain, T., Jain, A., Hada, P.S., Kumar, H., Verma, V.K. & Patni, A. (2021) Machine Learning Techniques for Prediction of Mental Health. In: *Proceedings of the 3rd International Conference on Inventive Research in Computing Applications, ICIRCA 2021*. 2 September 2021 Institute of Electrical and Electronics Engineers Inc. pp. 1606–1613. doi:10.1109/ICIRCA51532.2021.9545061.
- Mody, V. (2019) 2019 IEEE 5th International Conference for Convergence in Technology (I2CT). IEEE.
- Kumar, P., Chauhan, R., Stephan, T., Shankar, A. & Thakur, S. (2021) A machine learning implementation for mental health care. Application: Smart watch for depression detection. In: *Proceedings of the Confluence 2021: 11th International Conference on Cloud Computing, Data Science and Engineering*. 28 January 2021 Institute of Electrical and Electronics Engineers Inc. pp. 568–574. doi:10.1109/Confluence51648.2021.9377199.
- Rahman, R.A., Omar, K., Noah, S.A.M., Danuri, M.S.N.M. & Al-Garadi, M.A. (2020) Application of machine learning methods in mental health detection: A

systematic review. IEEE Access. 8, 183952–183964.
doi:10.1109/ACCESS.2020.3029154.

- Shatte, A.B.R., Hutchinson, D.M. & Teague, S.J. (2019) Machine learning in mental health: A scoping review of methods and applications. Psychological Medicine.49 (9) pp.1426–1448. doi:10.1017/S0033291719000151.
- Radwan, A., Amarneh, M., Alawneh, H., Ashqar, H.I., AlSobeh, A. & Magableh, A.A.A.R. (2024) Predictive Analytics in Mental Health Leveraging LLM Embeddings and Machine Learning Models for Social Media Analysis. International Journal of Web Services Research. 21 (1).
doi:10.4018/IJWSR.338222.

Bibliography

- Acharya, A., & Parai, M. K. (2022). Digital mental health interventions for developing countries: A systematic review. *Journal of Global Health*, 12, 04062. <https://doi.org/10.7189/jogh.12.04062>
- Alqahtani, F., & Orji, R. (2020). Insights from user reviews to improve mental health apps. *Health Informatics Journal*, 26(3), 2042–2066. <https://doi.org/10.1177/1460458219896492>
- Benrimoh, D., Tanguay-Sela, M., Perlman, K., Israel, S., Mehlretter, J., Armstrong, C., & Fratila, R. (2021). Artificial intelligence for mental healthcare: Clinical applications, barriers, facilitators, and artificial wisdom. *eClinicalMedicine*, 37, 100929. <https://doi.org/10.1016/j.eclinm.2021.100929>
- Calvo, R. A., Dinakar, K., Picard, R., & Maes, P. (2022). Computing in mental health: Addressing ethics, privacy, and intervention strategies. *Communications of the ACM*, 65(8), 64–73. <https://doi.org/10.1145/3494574>
- Fernando, N., & Gunawardena, N. (2021). Mental health status and help-seeking behaviour during COVID-19 pandemic in Sri Lanka. *Sri Lanka Journal of Psychiatry*, 12(2), 7–14. <https://doi.org/10.4038/sljpsyc.v12i2.8300>

Appendices

User Guide

This guide provides instructions for installing, running, and using Project MoodSync.

Should Have:

- Node.js (v22.0 or higher).
- Python (v3.12 or higher)
- MongoDB (v8.0 or higher)
- Git

Setup Steps

1. Clone the repository or download the project from DLE

```
git clone https://github.com/iamgaganam/MoodSync.git  
  
cd MoodSync  
cd WebApplication
```

2. Backend Setup (FastAPI)

```
cd server  
pip install -r requirements.txt
```

3. Frontend Setup (React with Vite)

```
cd client  
npm install
```

4. Database Setup

1. Start MongoDB service in your local machine.
2. The application will create the collections automatically.

5. Environment Configuration

```
MONGO_URI="mongodb+srv://gaganam220:h55wl4KfDS0hNQoj@cluster0.n8o  
mm.mongodb.net/?retryWrites=true&w=majority&appName=Cluster0"  
SECRET_KEY="bf7d6a7ca3bcc57b195ca47b9b9f2832c9194095bbb65891618b5  
e31b310aebb"  
ALGORITHM="HS256"  
ACCESS_TOKEN_EXPIRE_MINUTES=60
```

In the frontend directory, create a .env file with:

```
VITE_API_URL=http://localhost:8000
```

Running the Application

1. Start the Backend Server

```
uvicorn server.main:app --reload
```

The server will run on <http://localhost:8000>

<http://localhost:8000/docs> (Swagger UI)

2. Start the Frontend Development Server

```
cd client  
npm run dev
```

The application will be available at <http://localhost:5173/>

Using the Application

1. User Registration / Login

- Access <http://localhost:5173/login> in your browser
- Click "Sign Up" to create a new account or "Login" with the following demo credentials:
 - **User:** user@gmail.com / Password: TestUser123
 - **Professional:** doctor@gmail.com / Password: TestDoc123
 - **Admin:** admin@gmail.com / Password: TestAdm123

2. Core Features

Dashboard

- After login, you'll see the main dashboard.
- Use the mood selection buttons to log your current emotional state.
- View your mood trends in the chart area.
- Access quick actions from the dashboard panels.

Sentiment Analysis

- Find and go to "Analysis" in the sidebar.
- Enter text in the input area.
- Click "Analyze" to process your input.
- View results showing sentiment scores and recommendations.

Anonymous Chat

- Click "Chat Support" in the topbar.
- Type messages in the input field.

Health Trends

- Navigate to "Trends" in the navbar.
- View visualizations of mood patterns over time.
- Use the time range selector to change the view period.

3. Professional Interface

- Log in with professional credentials.
- View anonymized patient data.
- Access communication tools.
- Review alerts and prioritize cases.

4. Admin Interface

- Log in with admin credentials.
- Manage users and content.
- View system performance metrics.
- Configure system settings.

Access the ML Model

To access the machine learning model directly:

```
cd server/data
```

Project Structure

- /client - React with Vite, TypeScript, Tailwind CSS and Flowbite.
- /server - FastAPI server and API endpoints.
- /server/data - Machine learning models and sentiment analysis.
- /server/app/models and /utils - MongoDB schemas and data access.

Notes for Markers

- The system is configured to run entirely on localhost.
- Some of the sample data is pre-loaded for displaying features (since mental health issues cannot be treated in a single day which takes a period of time).
- For ML model results, see the server/data/chatbot_model.ipynb file.
- All passwords are hashed using bcrypt encryption and uses OAuth 2.0.

Project Initiation Document (PID)

1. Introduction

Mental health is an increasingly important aspect of overall human well-being, and still, it remains not addressed due to social stigma, lack of awareness, and access to professional help is limited. This problem is especially evident in Sri Lanka, where mental health services are not accessible to many people and the cultural barriers prevent people from having open conversations regarding their emotions (Attygalle, Perera & Jayamanne, 2020), as a solution to all these challenges, this project - MoodSync targets to develop a mental health monitoring system powered by machine learning providing a solution to bridge the gap between those who are in need and professionals that are ready to support. This project was inspired by the growing digital tools in promoting mental health (Wijekoon Mudiyanse et al., 2024), also this project was inspired to eliminate deaths by suicide which has globally increased recently. By using the advancements in Natural Language Processing (NLP) and sentiment analysis, this system looks forward to allowing users to monitor mental health effectively and access personalized guidance. The inclusion of features like mood prediction, coping strategies, motivational content, and real-time alerts for critical situations addresses the needs of users, especially in low-resource communities. The anticipated impact of this project extends beyond the users, mainly targeting to reduce mental health stigma and provide timely interference to save lives (Deivanayagam et al., 2024). This software is mainly tailored to Sri Lankan users, but it can be used globally as well, also this software will be designed to be easily accessible, culturally favored, and user-friendly. By encouraging users with a motivated approach to mental health, this project seeks to make a meaningful contribution to users' and societal wellbeing.

2. Business Case

2.1. Business Need

Mental health problems have reached higher levels in Sri Lanka, according to World Health Organization (WHO) 2022 report, it is stated where one in five in the world and 20% of the population in Sri Lanka experience mental health disorders such as depression and anxiety during their lifetime (World Health Organization, 2022). Although there's a growing awareness and recognition of mental health importance, getting access to professional help is challenging because of the cultural stigmas, especially less resource availability in rural areas, around 60% in rural areas, 85% in urban areas and economic barriers making it unaffordable (Sri Lanka Ministry of Health, 2023). The current mental health applications like (Calm, Headspace, and Moodfit) have failed to provide unique guidance based on a person's mental health status which lacks sociocultural needs globally as well as for Sri Lankan users where 70% of users has reported that existing apps do not address the needs or provide guidance (Mental Health App Usage Study, 2023), leaving gaps in support provided. The lack of cultural relevancy and proper accessible mental health tools makes this issue much worse, as people are hesitant to seek in-person consultations mainly due to societal judgment. Further, there is an immediate need for a real-time monitoring system to identify worsening health conditions and provide timely interference to prevent suicide and life-threatening activities as 10% of increase in suicide rates for the last two years (Sri Lanka Suicide Prevention Annual Report, 2023). The need for an application like this is not only evident but essential to provide these solutions.

2.2. Business Objectives

Project MoodSync targets to achieve the following objectives that align with the need to improve mental health care accessibility and effectiveness in Sri Lanka.

1. Personalization Enhancement

- Implement advanced sentiment analysis and mood-tracking features that are made specifically for Sri Lankan and global contexts. Unlike other platforms generic responses will not be displayed, every output will be unique and personalized.
- Offers unique recommendations and coping methods based on users' behavioral patterns or by the user inputs (social media etc.).

2. Improved Emergency Response

- Integrate real-time alert system that notifies the emergency contact, when the user displays worsening mental health, preventing self-harm to the user.

3. Provide Professional Connections

- Provide users with a directory of licensed mental health professionals, including details like hospital information, location, and appointment schedules for e-channeling.

4. Promote Awareness

- By using motivational content based on user health data, visualizing mental health trends, and personalized feedback to encourage daily engagement and self-awareness.

5. Increased Accessibility

- Develop user-friendly software that allows people from all regions, including rural areas, to access mental health support with less limits.

By addressing these above-mentioned objectives, this project looks forward to transforming the mental health landscape in Sri Lanka, making the support systems more

impactful and personalized. Further, the platform's adaptability ensures that it can be used to address global mental health challenges, increasing the platform's relevance and value.

3. Project Objectives

Project MoodSync is designed to address the gaps in the current mental health support using advanced technologies to provide users with easily accessible and personalized solutions. This project aims to achieve the following objectives.

1. Sentiment Analysis Implementation

- To develop a good machine learning model with at least 95% accuracy to analyze the text inputs provided by the user (ex: social media posts) and evaluate their emotional and mental states accurately by the end of the development.

2. Personalized Mental Health Recommendations

- Create an intelligent guidance model that provides the users with coping methods, mental health tips, and motivational quotes specified to their recorded mental state and long-term data.

3. Anonymous Communication

- Design a secure and user-friendly web application to anonymously chat, making sure that users can express themselves freely without judgment, fear, and cultural stigma.

4. Developing a Real-Time Alert System

- Integrate a real-time monitoring system that identifies emotions and sends alert messages to loved ones and health professionals in serious situations.

5. Mental Health Trends Visualization

- Develop an interactive, graphical representation of the user's mood patterns throughout time, making them monitor their mental health journey and identify triggers and improvements.

6. Integrate Support Channels

- Connect the users to verify mental health professionals and provide information on the availability of health services and hospitals that are readily available to offer support.

7. Promote Accessibility and Inclusivity

- Ensure that the system is sensitive and inclusive to all Sri Lankan users while maintaining good adaptability to global audiences as well.

8. Ensure Data Security and Privacy

- Implement data encryption and strict authentication methods to safeguard users' data and maintain privacy.

4. Literature Review

This research paper explores the recent upgradation of machine learning and natural language processing (NLP) that has improved capacity in mental health monitoring through sentiment analysis on social media platforms such as X. The usage of X's large and complex datasets provides valuable insights into public sentiments facilitating timely responses to health crises. Within mental health, it has been seen that sentiment analysis has become a critical tool. This paper shows that traditional machine learning techniques like SVM and Naïve Bayes efficiently analyze user sentiments in real time though there have been challenges with slang and informal languages which is relatively common in a social media platform like X. Recent studies have clearly shown that deep learning models like Long Short Term Memory networks, which improve sentiment analysis by complex feature extraction. Although advancements have been made, according to this research paper there are gaps in research of sentiment analysis. The framework proposed by the authors shows the importance of historical tweets but hasn't done an in- depth analysis though they highlight more significant predictions regarding users' mental health. As a solution to in-depth analysis and to fill this gap, we could get the tweets data from users who are mentally suffering state and come up with identifying more accurate patterns to predict effectively. Finally, this research paper underscores the need for user driven data collection methods to improve data reliability and to provide cloud-based solutions for mental health monitoring. (Hinduja et al., 2022)

This advanced research study named "Machine Learning Techniques for Prediction of Mental Health" explores the way machine learning algorithms can be applied to predict current mental health status by gathering different attributes like financial and household data. Machine learning techniques have shown good results in detecting mental health issues earlier, the authors have used different datasets to identify the patterns that have been missed using the normal traditional methods. The authors have used many algorithms like Decision Tree, Random Forest, Support Vector Machine (SVM), Naive Bayes, Logistic Regression, XGBoost, Gradient Boosting Classifier, and Artificial Neural Networks to enhance the accuracy in predicting mental health. This dataset that has been used contains 76 attributes in total and the data has been split into a 70/30 ratio for training and testing. It included metrics like Precision, F1 Score, and false negative rate and accuracy. Among these algorithms, SVM has achieved a good overall accuracy of 87.38% whereas Gradient Boosting Classifier has obtained the highest F1 score showing a balanced precision. Naïve Bayes achieved a lesser accuracy of all algorithms which is 21.67% if it has feature independence and is not suitable for big and complex datasets. This analysis clearly shows that SVM is the ideal algorithm with high accuracy performed with data as no other algorithm has this level of accuracy. However, there are gaps identified in this research where there is no mention about imbalanced datasets which can be

identified to improve model performances more. To fill this gap synthetic data can be used representing minority class for a more balanced dataset. The findings from the research study indicate that machine learning techniques can be effectively used in mental health predictions though future research should aim more to improve the model to have a good performance. (Jain et al., 2021)

Another research study by Vidhi Mody highlights AI/ML-based approaches for identifying mental health issues, focusing on methods like multi-view clustering, expert systems, and wearable sensor integration. This paper shows how impactful physiological signals and smartphone data can be in detecting early mental issues such as stress, using techniques like fuzzy logic and multi-view biclustering. This paper showcases multi-view learning by combining perspectives like average activity, daily trends, and location variability resulting in an overall prediction of 87.1% accuracy. However, simpler classifiers like linear naïve Bayes were used in stress detection and achieved a lower accuracy of 80% which models can be improved more. The study highlights that current expert systems utilize rule-based reasoning and fuzzy genetic algorithms (fuzzy-GA) for identifying and planning treatments. These systems aim to mimic decision-making and provide personalized and therapeutic recommendations. This algorithm shows moderate accuracy and has limitations in boundary cases compared with the results obtained, where unreliability is often seen in the table of results. Although there is an algorithm upgrade, there is a gap in integrating multiple AI methodologies for diagnostic systems. The existing methods discussed have limitations in precision, which this proposed product will address with good precision and adaptability to evolving user states. My product will address this gap using an advanced machine learning model using traditional expert systems and to improve conditional handling well. This proposed product will be integrated with feedback and overall scores to deliver effective and personalized mental health recommendations. (Mody, 2019)

Another paper shows the progression made through machine learning algorithms and the way wearable devices help in detecting mental health disorders such as depression and anxiety, however there are gaps in these existing methods. This research study highlights a good accuracy rate in detecting health symptoms with the aid of KNN – K Nearest Neighbor, SVM – Support Vector Machines, and Naïve Bayes algorithms. However, most models are trained in controlled environments, making it overlook the real-world applicability of certain socio-economic conditions and cultural contexts, which affect the reliability of the product and mental health patterns, to fill this gap the proposed product will endorse the Sri Lankan local context. Additionally, the previous models have shown a limitation of scalability and practical issues when applied to resource-constrained settings which would have been better if continuous and real-time monitoring were used. However, the integration of these devices is challenging for low-income regions in Asia as the cost is high. This research shows how models and wearables can be utilized for

impactful mental health detection. (Kumar et al., 2021)

Another research study conducted does a systematic review of examining machine learning applications in detecting mental health issues, with a focus on features, classifiers, and challenges faced with using these methods. This review shows the key techniques used in mental health detection such as feature extraction models like TF-IDF, Word2Vec, and N-Gram models as well as supervised machine learning techniques like SVM, logistic regression, and decision trees. The study shows a gap in handling data sparsity, multilingual content, and privacy constraints, especially for datasets sourced from Online Social Networks (OSNs). Though machine learning provides solutions to these concerns, there are issues like biased data, data preprocessing, generalizability, and model interpretation that remain significant obstacles. To address this gap, the proposed system will include localized data gathered from trusted government sources that eliminate biased data. Overall, the authors provide a review emphasizing the need for a good reliable model that addresses these challenges to improve health predictions. (Rahman et al., 2020)

This paper demonstrates the effectiveness of Machine Learning models, specifically SVM and XGBoost, in categorizing depression-related content from social media platforms. The authors show that SVM is highlighted above all other algorithms in terms of its high precision, making it the most reliable algorithm for mental health monitoring. However, the advancements gaps remain with the interpretability of ML models, especially concerning the language use of social media platforms which can lead to errors and misclassifications. Additionally, the class imbalance raises a challenge, which limits the generalization of the findings across user populations. Compared with previous research studies they have not shown a consideration in integrating a Large Language Model (LLM) which could've enhanced accuracy and understanding. To address these gaps as mentioned above the proposed system will be focused on an ML model ensuring a good performance in mental health detection indicators minimizing bias data. (Shatte, Hutchinson C Teague, 2019)

In this advanced study, the authors have emphasized the recent upgradation of Natural Language Processing (NLP) has made the way for innovative approaches to mental health analysis through social media. Traditional methods of mental health monitoring depend on face-to-face interviews which are not different from everyday emotional expressions found online. This paper has focused more on using ML techniques to classify emotions and states by user-generated content. For instance, the authors used a supervised algorithm to identify mental health indicators in social media posts and achieved good results. Similarly, it also demonstrated the effectiveness of deep learning models in identifying anxiety and depression-related content, highlighting the capability of automated systems in detecting real-time mental health status. Apart from these

upgrades, gaps remain in interpretability of ML models and their generalizability from diverse populations. It is to be seen that these authors' main goal was to improve accuracy however they have missed addressing underlying reasons for misclassifications. This study aims to use LLM-large language models with SVM – Support Vector Machines to improve accuracy in stress-related posts. However, this is not the ideal solution as they lack generalizability of the findings though models accuracy is good, in the proposed project, it will closely align to cultural context. In addition, the model outputs will be analyzed to identify language indicators that cause mental distress thus completing the gap between accuracy and practical interpretation. This research carried out highlights the advanced technical possibilities in mental health analysis and contributes to a good system design before development. (Radwan et al., 2024)

An advanced study carried out by Ruba Skaik and Diana Inkpen from University of Ottawa shows the use of social media for real-time mental health surveillance and they explore the strategies to extract information using social media platforms. The authors discussed the growing trends of people using social media platforms such as X, Reddit, and Facebook capturing public user sentiment and identifying early warnings of related mental health issues such as anxiety, depression, and suicidal thoughts. This review shows the searches into different techniques used to extract and analyze obtained social media data, with a focus on techniques like Natural Language Processing (NLP) and Machine Learning. Further, these methods use sentiment analysis and modeling to break down topics to classify user-uploaded posts by mental health categories, which helps to detect the user's symptoms related to the disorder. The authors has considered more about detection speed in the approach which is a gap because it lacks longitudinal data analysis where in the proposed solution it will be addressed by using longitudinal analysis to get a understanding of crisis points and mental health trends more accurately. The authors further point out the advantages of using social media platforms for mental health detection. This is carried out by storing large amounts of data from the user over a period of time, which allows for the detection of disorders faster than traditional methods like surveys and clinical exercises. This research paper also discusses privacy concerns and the difficulty of ensuring the reliability of data to train the ML models, However there is gap of not considering how user respond to the model and there is no professional experience tailored in training the model, to address this gap constant user feedback will be taken in development of ML model making it more user centric backed by a professional. The authors seems more concerned about the ethical rules of collecting social media data and discuss other researchers to ensure the privacy protections of users and if data is collected responsibly. This review highlights how social media can be used to predict a user's mental health using techniques like NLP and ML models instead of old traditional methods. However, these authors suggest protecting privacy and checking the accuracy of the data obtained. (Skaik C Inkpen, 2021)

This research study highlights a review of existing approaches to mental health monitoring and shows a special importance on sentiment analysis and machine learning for mood tracking and predictions. This research paper shows the effectiveness of various ML models in understanding users emotional states by social media and other inputs. However, these solutions are mostly tailored to western user context limiting its applicability. While these models use sentiment analysis, very few integrate real time support system for user's needs, mostly in developing country like Sri Lanka. This research study lacks a approach of combining sentiment analysis and support mechanisms. By addressing the above-mentioned gaps could enhance the effectiveness of the proposed mental health monitoring system. By developing a cultural framework that has real time alerts and mental health advice can contribute to a better health care solution. (Alanazi et al., 2022). This research paper shows the use of machine learning and Arduino on mental health monitoring systems with important metrics like precision, accuracy, and sensitivity. The authors got a 96.6% higher accuracy rate in stress detection, showcasing the model's effectiveness with sensor integration with machine learning to obtain real-time health insights. However, there are gaps identified as there are different health conditions that are not limited to stress, where advanced machine learning and deep neural networks should be used. Additionally, integrating Natural Language Processing (NLP) could allow systems to interpret sentiments and interactions more deeply to offer personalized insights. By addressing these mentioned gaps, additional research would enhance monitoring systems facilitating early detection. Also, the models should be tested in a cultural context like in Sri Lanka, which would add depth where tools should be accurate and culturally sensitive to have a higher impact on the end product. (VENKATARAMANIAH et al., 2024)

This research study showcases a different transfer learning framework being used for mental health monitoring and the importance of stress detection as a case study. The integration of transfer learning addresses gaps in existing mental health monitoring methods, which raises consideration of data privacy and insufficiency. Previously conducted research suggested that the use of federated learning techniques causes attacks and threats to vulnerability, limiting effectiveness for real-world scenarios. In addition, the existing models struggle with limited culture-specific training data leading to bad performance in models. The authors address issues by pre-training on large randomly picked datasets where it is obtained before getting tuned with user-specific data, if it is done therefore the model can be enhanced while protecting users' privacy. However, more exploration and tests should be conducted to ensure the scalability of this framework across diverse users and its various mental health conditions like depression to maximize utility in mental health monitoring practices. (Wang et al., 2024)

Machine learning (ML) becomes a reliable mechanism for mental health diagnosis when stable predictive trends emerge. From CNN to SVM to RF, the standard ML processes

employed by researchers provide reliable diagnosis no matter the application of data collection—neuroimaging, transcribing, behavioral response scoring. For instance, CNNs achieve 94% reliable diagnosis of bipolar disorder and 91% reliable diagnosis of post-traumatic stress disorder (PTSD). Furthermore, Random Forests and Support Vector Machines provide reliable prediction of anxiety and depression diagnosis, respectively. In terms of prediction, many findings suggest success, yet many are based on specific cases. For example, with bipolar disorder, researchers have predicted using neuroimaging biomarkers and ML classifiers achieving over 90% accuracy for successful prediction with the potential for early detection and treatment. (Wijekoon Mudiyansele et al., 2024)

Predictions also exist for schizophrenia with both ML and neuroimaging as well as genetic information, which shows promise with successful accuracy in finite efforts; however, finite efforts based on small sample sizes and the challenge of tracing the disorder miles down the line remain limitations for real-world implementation. The same can be said for PTSD, where ML efforts based on electrodermal activity signals, voice and speech characteristics, text-based elements, and passive tracking (i.e., GPS/location, cellphone information) yield clinically significant prediction accuracy. Furthermore, anxiety and depression apparently can be predicted as well through deep learning efforts like Long Short-Term Memory (LSTM) networks through social media and speech-based efforts, opening yet another arena for detection potential. Finally, ADHD was found to have moderate to high detection accuracy using Decision Trees (DT) and Convolutional Neural Networks (CNN) with neuroimaging features. Yet gaps remain. Many studies utilize small, homogeneous sample populations limiting generalizability; privacy and ethical concerns require an extensive compliance protocol, which many studies do not undertake, plus certain predictive models like deep neural networks are too complex for explainability, rendering them inapplicable to a realistic clinical setting, etc. Furthermore, few studies utilize longitudinal or time series data due to time-relative predictors. This study will bridge these gaps with an emphasis on varied and diverse datasets, the use of longitudinal data, and an emphasis on greater explainability for future psychiatric applications. Ultimately, all three enhancements will lead to appropriate early detection, personalized interventions, and thus better mental health. (Madububambachu, Ukpebor C Ihezue, 2024).

4.1. Research Gap

The use of technology for mental health intervention has grown exponentially in recent years, resulting in many mental well-being applications. However, more needs to be done particularly with niche user groups in developing nations such as rural communities in Sri Lanka. Therefore, this chapter presents the solutions that have already been attempted, their failures, and the inadequacies in research this project intends to bridge.

Existing Solutions and Limitations

There are many mental health apps Calm and Headspace to Moodfit that offer general possibilities for mood tracking and meditation. Yet, while these apps promote mental health awareness and assist multitudes, they function with a Western majority inclination and the global, culturally lingual assessment required for proper user engagement does not come easy cultural Relevance and Localization.

Cultural Relevance and Localization

Currently, many apps lack cultural relevance and localization, which are necessary for heightened engagement and successful mental health intervention. For example, Hinduja et al. (2022) assess the advancements in machine learning capabilities and natural language processing (NLP) features within social media spaces to assess mental health concerns. However, much of the training of such models comes from Westernized language tendencies; for example, if someone is depressed and makes a comment on social media and/or uses particular language tendencies, it's much different than how someone in Sri Lanka would be spelling or saying how they feel through language. Thus, an increasingly successful model is needed, trained on the socio-cultural forces at play with the Sri Lankan population.

Personalization and Sentiment Analysis

Some applications boast sentiment analysis features, such as Moodfit, but none personalize the reaction concerning culturally relevant factors and one's specific feelings. For example, according to Jain et al. (2021, 1), Support Vector Machines are pretty effective in assessing if an individual suffers from mental health issues but fail to

acknowledge the potentially biased nature of some datasets. This is critical because it shows that more balanced, culturally trained, edited datasets would allow for equity and accuracy of assessments for this population in Sri Lanka. Real-Time Feedback & Professional Collaboration. Currently, most mental health apps fail to provide real-time feedback and professional collaboration. For instance, Mody (2019) reveals that many AI/ML-based solutions to stress detection rely on physiological signals and input using the smartphone very precisely. However, such precise outcomes produce rule-based expert systems that omit real-time interaction from the end-user or a professional—meaning such systems could never adapt over time based on end-user needs or affiliation with licensed mental health professionals. Furthermore, Skaik and Inkpen (2021) demonstrate how social media could be the vehicle for real-time assessment and research into mental health tracking, but it does not provide any means for longitudinal assessment with professional guidance. Thus, the findings are not practical for the long-term mental health support of persons from rural Sri Lanka.

Sentiment analysis as a form of medical diagnosis through trends in mental health via sentiment emerges as a common occurrence on social media. For example, Usmani et al. report that mental health issues can be predicted possibly even before people have them in real life. However, such features run into barriers. For example, employing a non-trained, non-domain, or field-specific dataset and the naturally complicated ways in which people talk about sarcasm, vagueness, etc. poses typical barriers. However, according to Reinert et al., sentiment analysis not trained through domain experience does not assess the status-in-context needed for mental health assessment. Learned relative to attempting to solve another issue, class imbalance within the trained data where "Normal" and "Depression" appear frequently but "Stress" and "Personality Disorders" do not. This restricts what machine learning might be able to discover with less frequently encountered mental illnesses. Similarly, however, learned as well. For example, Kim and Kaushik et al. created their own and adjusted models to circumvent the discrepancies while Hahad et al. suggest that the more variables from the data set, the more holistic understanding of the problem. Therefore, future research must make the algorithms more sensitive, expand the annotated corpora, and develop niche applications. Should such shortcomings be addressed, sentiment analysis could provide a powerful form of instantaneous evaluation of mental health with greater accessibility for a global mental health crisis. (Jain C Rathour, 2024)

Anonymity and User Privacy

Anonymity is another important requirement that many of the existing applications fail to satisfy. Where they offer private adjustments, they do not offer a secure way for people

to access much-needed help for mental health problems without the fear of exposure. Rahman et al. (2020) note that privacy issues are one of the biggest obstacles to using online social networking for mental health monitoring. Therefore, applications are necessary that keep anonymity on the user's part but fulfill the support requirements for a culturally appropriate Sri Lankan application.

The following research gaps have been written based on the conducted literature reviews above:

6. **Personalized Support** – Most applications don't provide a unique response to users instead they provide a common generic response. To address this gap the proposed system aims to use advanced machine learning algorithms trained with localized datasets to offer unique personalized responses with an overall score.
7. **Real-Time Support** – Current applications don't have real time support/alerts to users' loved ones in immediate health needs. This proposed system will plan to include real-time alerts using SMS to get immediate support in need.
8. **Cultural Context** – There is a lack of mental health applications that represent users in the Sri Lankan context. The proposed system will address this gap by using local languages and cultural backgrounds in its design and functionalities. By engaging with a mental health professional, this application will be tailored and ensured for local users.
9. **Professional Support** – No application connects mental health professionals with their users; to address this gap the proposed system will showcase the directory list of professionals with times and locations that will be available for channeling.
10. **Anonymity** – Unlike other applications, this proposed system will have a feature to chat with a representative for quick support without exposing the user as it cuts off the user's fear of getting support.

Conceptual Diagram

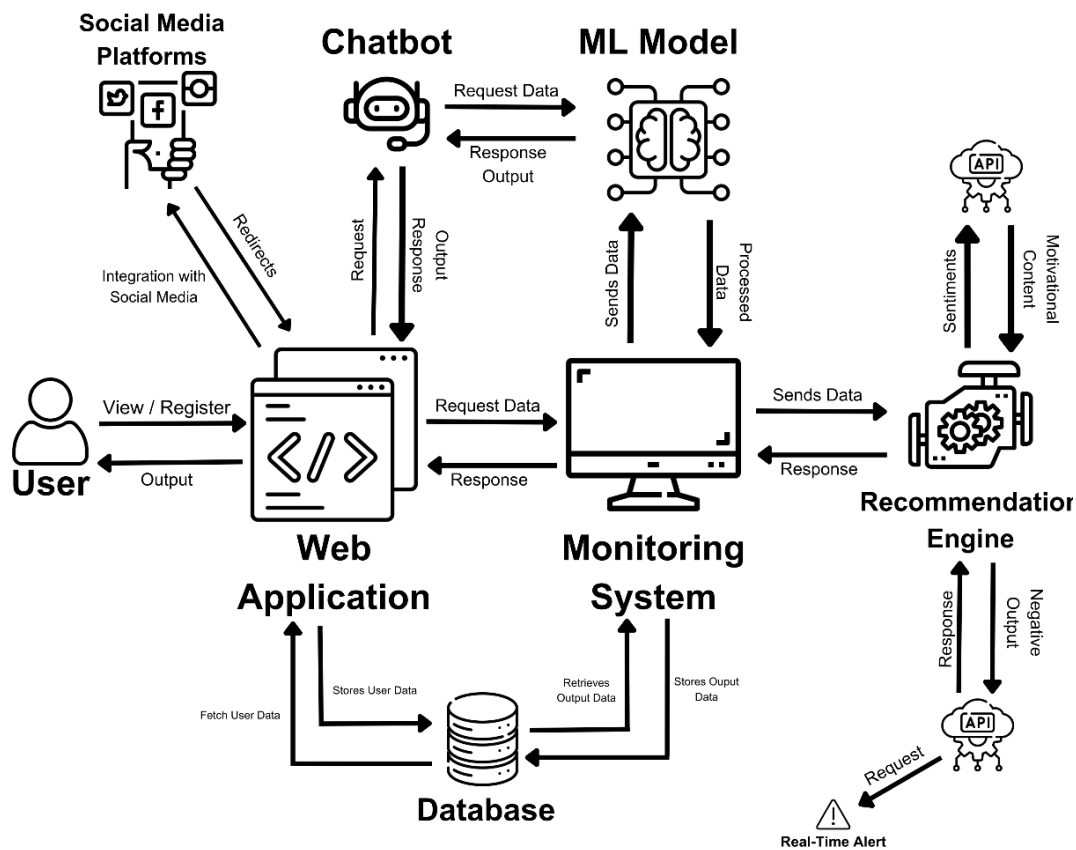


Figure 1 - Conceptual Diagram

5. Method of Approach

The development of project MoodSync will follow a structured approach, ensuring that the system is reliable, user-focused, and well-matched with the project objectives. This section shows the methodologies, tools, techniques, and frameworks that will be used in this project.

5.1. Technical Approach

Backend Development

- **Python** - will be the main language, and it is used for machine learning models and sentiment analysis algorithm development.
- **TypeScript** – For web development backends such as Node.js, Express.js, data visualization and, API requests, where logic will be handled.

NLP Libraries

- **NLTK** – For tokenization, sentiment analysis, and text processing.
- **spaCy** – For advanced NLP tasks, entity recognition, and parse dependencies.
- **scikit-learn** – For machine learning algorithms like Support vector machine (SVM) to predict user's moods by text inputs.

Database

- **MongoDB** - which is a NoSQL database, stores user information, mood logs, and analysis results for scalability and efficiency.

Frontend Development, Tools, and Frameworks

- **React.js** - for building responsive and modern user interfaces.
- **Vite** - Vite is a modern build tool that will ensure faster build times of the application than traditional react apps.
- **Tailwind CSS** – CSS framework for modern user interface development.
- **Flowbite** - UI library will be used alongside Tailwind CSS for easy and faster development.
- **TypeScript** – For frontend web, component development, and its logic. Will enhance code quality and maintainability.
- **Socket.io** – This will be used to implement real-time features like anonymous chat and sending alerts to loved ones when the condition is detected.

Data Visualization

- **Chart.js and D3.js** – For visualization of user sentiments and mood trends using charts and graphs.

API's

- **Twilio API** – Third-party SMS sending API that will be integrated to send text messages to emergency contacts.
- **Quotable/Zen Quotes API** – To fetch motivational quotes for user inspiration.
- **Health data API** – to fetch additional mental health resources.

Development Tools and Platforms

- **IDE's** – JetBrains PyCharm (Professional Edition) for backend machine learning models development and Visual Studio Code for frontend development.
- **Version Control** – Git and GitHub to manage my codebases.
- **CI/CD** – Docker to maintain environments with GitHub Actions CI/CD pipeline for automation of the application.

5.2. Project Management Approach

The agile methodology will be used to ensure flexibility and continuous improvement throughout the software development lifecycle. By using this approach it allows repetitive releases, integration of user feedback, and flexible planning, to ensure that the product meets user expectations and addresses the real-world challenges effectively.

Agile practices include:

- **Sprints** – Divide the development processes into short iterations according to the software development life cycle (SDLC) so that the complexity of the project is broken down as shown in the Project Gantt Chart.
- **Backlog Management** – Prioritizing features like sentiment analysis, mood tracking, and alerts based on users and their feedback so that project objectives are easily achievable.

5.3. Functional Components Approach

Sentiment Analysis and Mood Predictions

- User inputs (text and social media posts) will be analyzed using the NLP to detect the sentiment and the mood predictions.
- Machine learning models will be trained by using localized datasets to ensure good accuracy and reliability.

Real-Time Alerts

- Emergency contacts will be notified by SMS when the user's mental health score passes a certain level.
- Integrated with third-party SMS API (Twilio) for instant responses.

Professional Directory and Anonymous Chats

- The database of licensed and best professionals and channeling hospitals will be integrated into the web application.
- A secure and anonymous chat feature would be available with trained representatives for immediate user support.

Data Visualization

- Graphs and charts will be used to display mood trends over time so users can get to know the insights.

5.4. System Architecture Approach

This system will be developed using the client-server architectural method with the following layers,

1. **User Interface (UI)** – Web application for the user to interact.
2. **Application Logic** – Backend services for data processing and recommendations.
3. **Data Storage** – MongoDB for storing user profiles, logs, and analytical data.
4. **Machine Learning** – Python-based ML models for sentiment analysis and mood predictions.
5. **APIs** – For motivational content and health resource implantation.

These above methods of approaches will ensure that the system is built efficiently adheres to the best practices and address the unique mental health challenges while adapting to the global audiences.

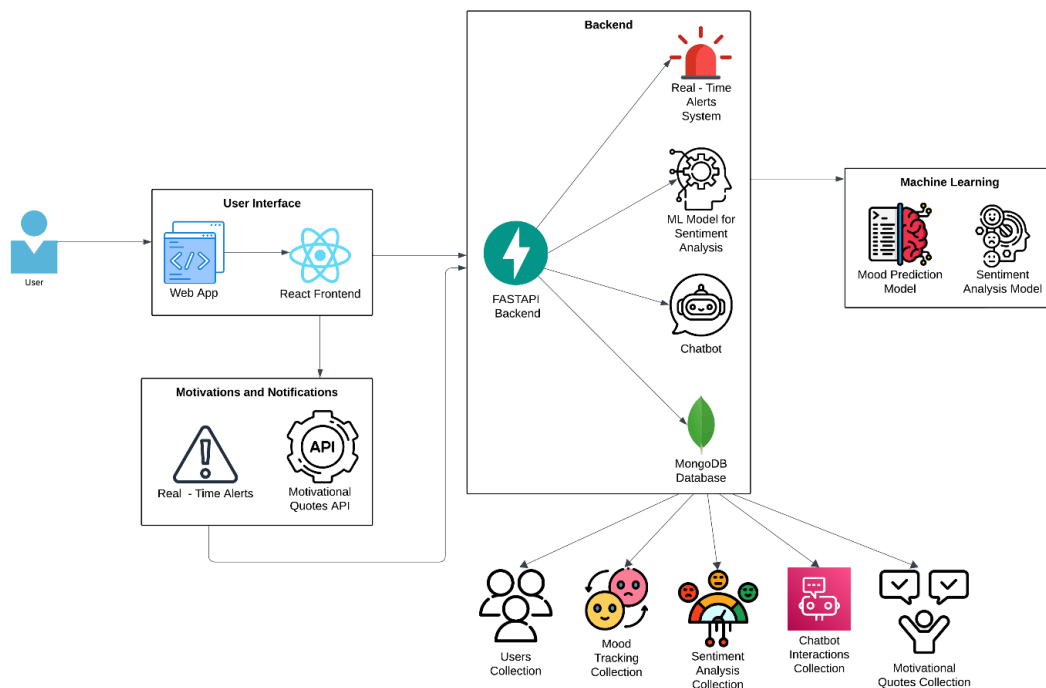
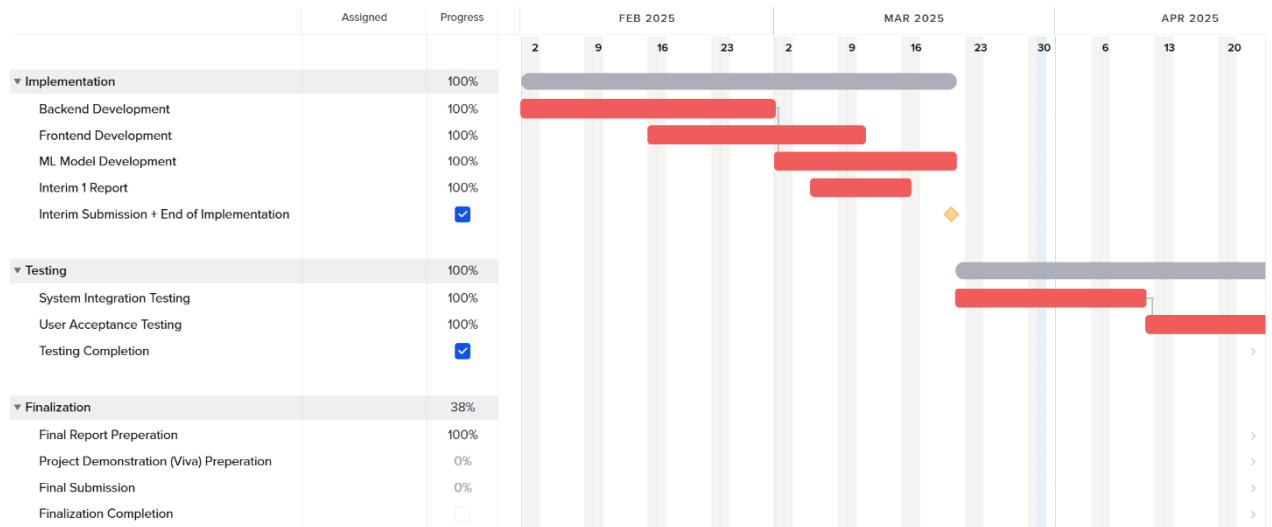
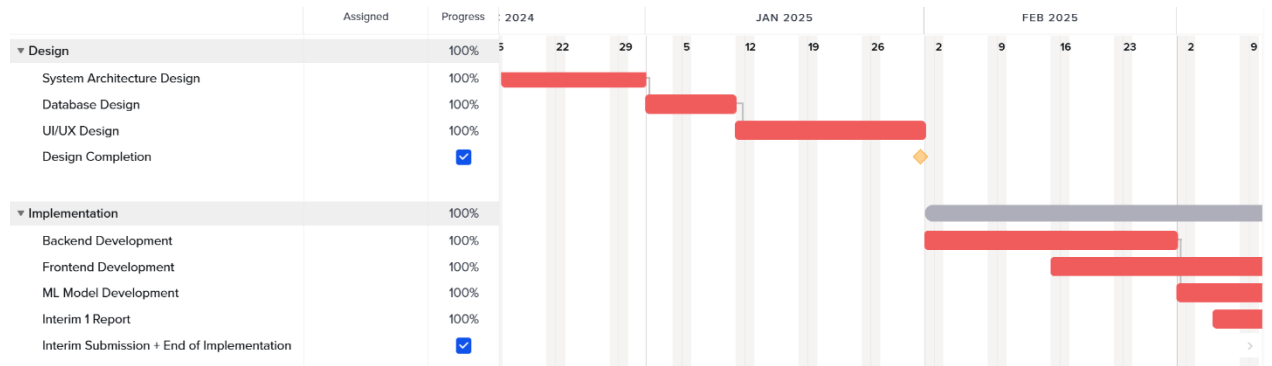
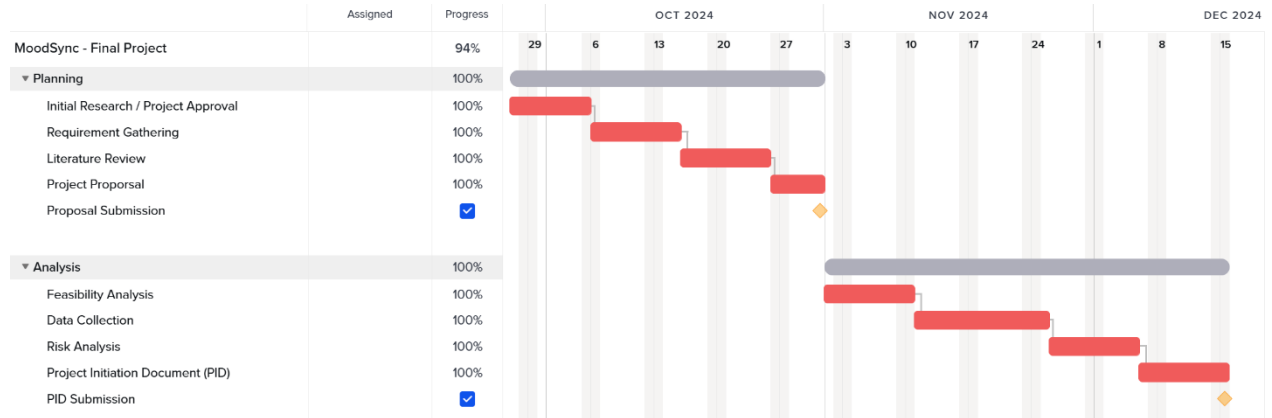


Figure 37 - High - Level Architectural Diagram

6. Initial Project Plan



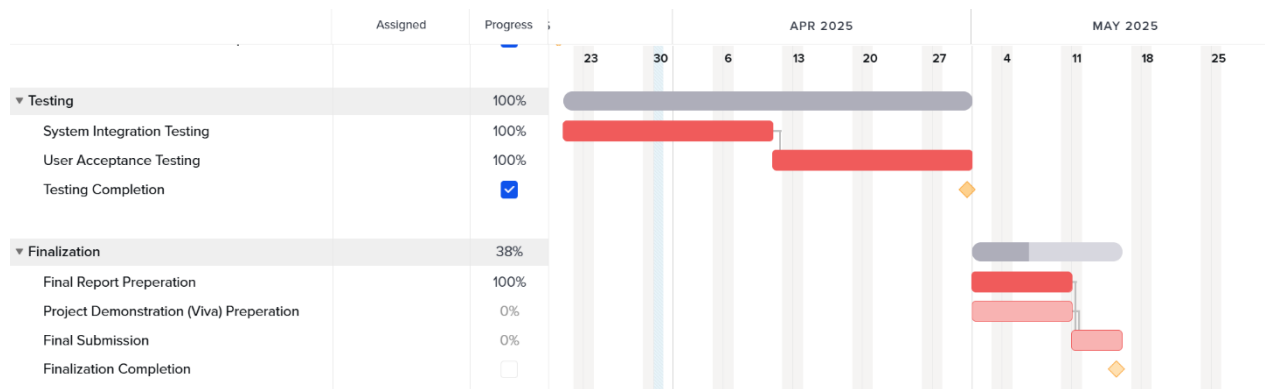


Figure 4 - Gantt Chart

7. Risk Analysis

7.1. Technical Risks

1. Sentiment Models Accuracy

- **Risk** – The NLP model may not identify user inputs to slang and cultural nuances.
- **Chance** – Medium.
- **Impact** - High.
- **Mitigation** – Using a localized dataset to train the machine learning model, test users' input samples, and tune the algorithm based on results and user feedback.

2. API Integration

- **Risk** – Delays and Failures in API responses in health data and motivational content.
- **Chance** – Medium.
- **Impact** – Medium.
- **Mitigation** – By using the fallback method and cache data in the browser to handle the API unavailability.

3. Real-Time Alert System Reliability

- **Risk** – Delayed and failed to notify emergency contacts in peak situations.
- **Chance** – Low.
- **Impact** – High.
- **Mitigation** – Using a reliable third-party SMS generator, and also to monitor how the performance works out regularly

7.2. Project Management Risks

1. Timeline Delays

- **Risk** – Missing deadlines due to technical issues and other unexpected problems.
- **Chance** – Medium.
- **Impact** – Medium.
- **Mitigation** – By using agile methodology as mentioned above to track progress in sprints and allocate resources as needed.

2. Skills Gap

- **Risk** – Lack of expertise in technologies like Natural Language Processing and Machine Learning.
- **Chance** – Medium.
- **Impact** – Medium.
- **Mitigation** – By allocating time to learning, meeting with the project supervisor to get information, and the use of online resources as guidance for correct integrations.

7.3. Financial and Resource Risks

1. Budget Constraints

- **Risk** – Additional costs for data gathering from health professionals, use of IDE's professional versions, and support that may be unplanned.
- **Chance** - Low
- **Impact** - Low
- **Mitigation** - Use of premium software services provided by the university free, use of open-source software tools, and preparing for additional expenses.

2. Resources Availability

- **Risk** – Limited access to tools and health professionals.
- **Chance** - Medium
- **Impact** - Low
- **Mitigation** - Maintain a regular connection with the professionals till the application is built.

7.4. Data Privacy and Security Risks

1. Data Breaches

- **Risk** – Unauthorized access to users' sensitive data like mood logs and emergency contact details.
- **Chance** – Low.
- **Impact** – High.
- **Mitigation** – By using end-to-end encryption, the latest authentication mechanisms like OAuth 2.0, and data protection regulations (ex: GDPR).

2. Anonymity Concern

- **Risk** – Users' identities could be exposed in anonymous chats.
- **Chance** – Low.
- **Impact** – High.
- **Mitigation** – By ensuring that the chats are encrypted and accessed securely with strict access control

7.5. User Adoption Risk

1. User Resistance

- **Risk** – Users may hesitate to use this system due to cultural stigma and lack of familiarity with the application.
- **Chance** – Medium.
- **Impact** – High.
- **Mitigation** – By designing a user-friendly interface and giving tutorials inside the web application for navigation.

2. Cultural Relevance

- **Risk** – Failure to detect language nuances by the ML model.
- **Chance** – Medium.
- **Impact** – High.
- **Mitigation** – Testing with people during the development and adjusting the model.

7.6. **Ethical Risks**

1. **User Trust and Consent**

- **Risk** – Users may not fully trust the application to give their sensitive information of themselves.
- **Chance** - Low
- **Impact** - High
- **Mitigation** – Clear communication of data usage of the application getting user consent and ensuring data transparency while handling users' data.

2. **Model Algorithm Biases**

- **Risk** – ML models may give biases that give inaccurate results and unfair results to users.
- **Chance** – Medium.
- **Impact** – High.
- **Mitigation** – Train models on different datasets and regularly check for biases.

By addressing these risks, Project – MoodSync ensures the development is culturally relevant and meets user expectations. Frequent reviews and adjustments will be made to tackle these risks and come up with a quality product.

8. Additional Sections

8.1. Stakeholder Analysis

The successful development of Project MoodSync depends on effectively identifying and engaging the stakeholders. I have classified the stakeholders into two sections,

Primary Stakeholder

- **Normal Users** – People who look out for mental health support. They are the direct beneficiaries of this system developed, who expect feedback and strategies to continuously improve their mental health.
- **Mental Health Doctors** – Therapists, counselors, and psychiatrists who will use this platform to connect with the users and offer their service with ease and track historical data.

Secondary Stakeholder

- **Families and Loved Ones** – They will receive an alert in situations where the connect user's mental health would get worse and the potential chance of getting self-harmed so families can be attentive to prevent such conditions.
- **Hospital Organizations**– Hospitals may integrate this application externally to extend their mental health services, generating good revenue for the good service.
- **Government and NGOs** – Stakeholders that focus on mental health awareness and suicide prevention as volunteers.

More Details of Data Collection in Stakeholders

Sri Lanka Sumithrayo / Sri Lankan Mental Health Foundation – Planning to get localized data from these government organizations. These organizations work to promote mental health care and provide support to people suffering from mental health illnesses.

Doctor Information (Confirmed):

- Dr. Deshantha Pathinayake – MBBS(SL), DFM(COL), MCGP, MRCGP(Int)
SLMC Reg: No. 2372.
General / Mental Health Specialist.
Wellness Medi Clinic, Moratuwa 10400
- Dr. Saman Weerawardhane – MBBS(SL)
Psychiatrist / Mental Health Specialist.
National Institute of Mental Health (NIMH)
NIMH, Mulleriyawa New Town, Angoda

Meeting minutes and the application approval evidence are included in the Appendix (Other Materials) section.

Details of the Local Hospital (Data Collection)

- Mana Suwa Piyasa (මනසුවිජයපාය) – Mental Health Clinic
Colombo South Teaching Hospital – Kalubowila
B229 Hospital Rd, Dehiwala-Mount Lavinia 10350
- Survey - [CLICK HERE!](#)
Responses - [CLICK HERE!](#)

8.2. **Ethical Considerations**

Following ethical considerations plays a major role in Project MoodSync as much user group data has been handled by this application.

- **Data Privacy** - Strict restrictions will be applied to the system to ensure data privacy. (ex: OAuth 2.0, GDPR, etc.).
- **Biases in the ML Models** – Regular testing in machine learning algorithms to prevent inaccuracies of mood predictions and guidance's with stakeholders (Psychiatrist).
- **Transparency** – This will ensure that users can interact with the web application with consent and will provide 100% transparency in retrieving personal details.

8.3. **Communication Plan**

Constantly, will be in touch with the confirmed stakeholder (Psychiatrist) to check the quality of the product, mainly results as if they abide by the health standards.

References

- Attygalle, U.R., Perera, H. & Jayamanne, B.D.W. (2020) Stigma related to mental health issues – a study among adolescents in Sri Lanka. *Sri Lanka Journal of Psychiatry*. 11 (1), 8–13. doi:10.4038/sljpsyc.v11i1.8243.
- Deivanayagam, T.A., Ní Chobhthaigh, S., Devakumar, D., Patel, K. & Rannan-Eliya, R.P. (2024) Mental health prevalence, healthcare use and access between 2018 and 2022 in Sri Lanka: an analysis of survey data. *Global Health Research*. 1–16. doi:10.3310/hjwa5078.
- Hinduja, S., Afrin, M., Mistry, S. & Krishna, A. (2022) Machine learning-based proactive social-sensor service for mental health monitoring using twitter data. *International Journal of Information Management Data Insights*. 2 (2). doi:10.1016/j.jjime.2022.100113.
- Jain, E. & Rathour, A. (2024) Understanding Mental Health through Sentiment Analysis: using Machine Learning to Analyze Social Media for Emotional Insights. In: *2024 5th International Conference on Data Intelligence and Cognitive Informatics (ICDICI)*. 18 November 2024 IEEE. pp. 782–786. doi:10.1109/ICDICI62993.2024.10810855.
- Jain, T., Jain, A., Hada, P.S., Kumar, H., Verma, V.K. & Patni, A. (2021) Machine Learning Techniques for Prediction of Mental Health. In: *Proceedings of the 3rd International Conference on Inventive Research in Computing Applications, ICIRCA 2021*. 2 September 2021 Institute of Electrical and Electronics Engineers Inc. pp. 1606–1613. doi:10.1109/ICIRCA51532.2021.9545061.
- Kumar, P., Chauhan, R., Stephan, T., Shankar, A. & Thakur, S. (2021) A machine learning implementation for mental health care. Application: Smart watch for depression detection. In: *Proceedings of the Confluence 2021: 11th International Conference on Cloud Computing, Data Science and Engineering*. 28 January 2021 Institute of Electrical and Electronics Engineers Inc. pp. 568–574. doi:10.1109/Confluence51648.2021.9377199.
- Madububambachu, U., Ukpebor, A. & Ihezue, U. (2024) Machine Learning Techniques to Predict Mental Health Diagnoses: A Systematic Literature Review. *Clinical Practice & Epidemiology in Mental Health*. 20 (1). doi:10.2174/0117450179315688240607052117.

- Mody, V. (2019) *2019 IEEE 5th International Conference for Convergence in Technology (I2CT)*. IEEE.
- Radwan, A., Amarneh, M., Alawneh, H., Ashqar, H.I., AlSobeh, A. & Magableh, A.A.A.R. (2024) Predictive Analytics in Mental Health Leveraging LLM Embeddings and Machine Learning Models for Social Media Analysis. *International Journal of Web Services Research*. 21 (1). doi:10.4018/IJWSR.338222.
- Rahman, R.A., Omar, K., Noah, S.A.M., Danuri, M.S.N.M. & Al-Garadi, M.A. (2020) Application of machine learning methods in mental health detection: A systematic review. *IEEE Access*. 8, 183952–183964. doi:10.1109/ACCESS.2020.3029154.
- Shatte, A.B.R., Hutchinson, D.M. & Teague, S.J. (2019) Machine learning in mental health: A scoping review of methods and applications. *Psychological Medicine*. 49 (9) pp.1426–1448. doi:10.1017/S0033291719000151.
- Skaik, R. & Inkpen, Di. (2021) Using Social Media for Mental Health Surveillance: A Review. *ACM Computing Surveys*. 53 (6). doi:10.1145/3422824.
- VENKATARAMANAI AH, B., REDDY, K.G., S, S.K., RAMAKRISHNAN, K., M, S. & RAJESH, S. (2024) Real Time Mental Health Monitoring System using Machine Learning. In: *2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)*. 28 August 2024 IEEE. pp. 806–811. doi:10.1109/ICoICI62503.2024.10696460.
- Wang, Z., Yang, Z., Azimi, I. & Rahmani, A.M. (2024) *Differential Private Federated Transfer Learning for Mental Health Monitoring in Everyday Settings: A Case Study on Stress Detection*. <http://arxiv.org/abs/2402.10862>.
- Wijekoon Mudiyanse, K.W., Jörg, F., Mendis, M.S.D., Fuhr, D.C. & Busse, H. (2024) Identifying contextual barriers and facilitators in implementing non-specialist interventions for mental health in Sri Lanka: A qualitative study with mental health workers and community members. *Global Mental Health*. 11. doi:10.1017/gmh.2024.75.
- Mental Health App Usage Study. (2023) Effectiveness of Mental Health Applications in South Asia. *Journal of Digital Health*, 15(3), pp. 123-135. Available at: <https://journals.sagepub.com/home/dhj> (Accessed: 5 January 2025).

- National Institute of Mental Health Sri Lanka. (2023) Sri Lanka Suicide Prevention Annual Report 2023. Colombo: National Institute of Mental Health. Available at: <http://nimh.health.gov.lk/> (Accessed: 5 January 2025).
- Sri Lanka Ministry of Health. (2023) Mental Health Access and Barriers Report. Colombo: Ministry of Health Publications. Available at: <http://www.health.gov.lk/> (Accessed: 5 January 2025).
- National Institute of Mental Health Sri Lanka. (2023) Sri Lanka Suicide Prevention Annual Report 2023. Colombo: National Institute of Mental Health. Available at: <http://nimh.health.gov.lk/> (Accessed: 5 January 2025)

Interim Report

1. Introduction

1.1. Introduction

Mental health care has become a more significant issue in the world today, especially in the modern world where daily life pressures, social expectations, and technological advancements have played a huge role in the rising of stress levels, anxiety, and depression. These above-mentioned mental health issues not only affect people's well-being but also have challenged the entire healthcare system. Although there is rising awareness of mental health problems, there remains a significant gap in personalized and accessible mental health care solutions. People frequently hesitate to get help due to the cultural stigma of mental health and the lack of accessible resources (Deivanayagam et al., 2024).

This issue is mainly common in the Asian region and in Sri Lanka, where mental health support systems are under development. Traditional methods of treatment like in-person meeting appointments with mental health professionals are not accessible to a large portion of the population (Sumathipala, Mahesh & Wijesundara, 2020). Further, the fast expansion of digital platforms has created a growing need for innovative technological solutions to monitor and improve mental health. Integrating Artificial Intelligence (AI) and Machine Learning (ML) in mental health care provides a chance to bridge this gap by providing personalized care, continuous monitoring, and real-time help (Patel et al., 2018).

This project – MoodSync's main aim is to develop a mental health monitoring system with machine learning that uses natural language processing (NLP) and sentiment analysis to track and then predict the mental health state of the users. The system will gather user input like texts, and social media posts and analyze them to detect emotions, moods, and mental health statuses. Through advanced data analytics, personalized recommendations, coping strategies, and real-time alerts for loved ones, this software aims to enhance mental health support for Sri Lanka in practical, accessible, and anonymous ways.

1.2. Problem Definition

Mental health problems are always complex, subtle, and difficult to detect without professional experience. While self-checking tools do exist, many users do not actively get help, and they face challenges when describing their illness. Additionally, in Sri Lanka, access to professional help is minimal and there is a lack of completed tools that provide unique, data-driven mental health care on a larger scale (Deivanayagam et al., 2024).

The current existing systems for mental health monitoring are very generic, lack personalization, and are not integrated with sentiment analysis or mood predictions. Also, there is a lack of real-time systems that send alerts to loved ones or professionals when users may be in danger. Further, existing systems have failed to provide consistent, personalized coping strategies or motivational content that guide users to manage mental health (Tazarv et al., 2021).

Project MoodSync addresses these challenges by developing a solution that,

1. Uses sentiment analysis and mood tracking to understand users' mental health conditions.
2. Gives personalized coping strategies and health advice based on user input.
3. Provides real-time alerts to loved ones in case of life-threatening situations.
4. It has a chat system for users who are uncomfortable with face-to-face meetings.
5. Includes motivational quotes and data visualization, which helps users to manage their mental health more effectively.

1.3. Project Objectives

The main objectives of this project are,

1. **Sentiment Analysis System** – Implement an advanced machine learning model that will analyze user text inputs to predict sentiment, mood, and to provide details about the user's mental health.
2. **Mood Tracking and Predications** – Implement a system that tracks users' moods over time and provides mood predictions to help detect early signs of mental health diseases.
3. **Personalized Mental Health Recommendations** – Based on the user's mood analysis, the application will display personalized mental health advice, coping strategies, and supporting resources.
4. **Real-Time Alerts for Loved Ones** – Implement a feature that sends real-time alerts to the given emergency contact by the user, when the system detects mood changes that show a crisis.
5. **Anonymous Chat** – This lets users use the system in an anonymous manner, which addresses the main concerns of cultural stigma and fear.
6. **Motivational Content** – Provides the user with motivational quotes, tips, and tricks to help improve mental state and encourage positive thinking.
7. **Data Visualization** – Implement graphs and charts to visualize the represented user data further by providing details for mood patterns, trends, and total mental health progress of users.

8. **Adaption to Sri Lankan Users** – Craft the system to match the needs of the Sri Lankan population, considering local cultural aspects and the mental health challenges unique to own regions to train the model.

By achieving the above-mentioned objectives, Project MoodSync mainly aims to make an accessible, innovative, and data-driven mental health monitoring system that could save lives and promote good mental well-being for people in Sri Lanka and the world.

2. System Analysis

2.1. Facts Gathering Techniques

To make sure that the application is being developed on user needs and to address the problem effectively, it is mandatory to gather existing facts and data. The methods discussed below were used to gather information,

1. **Literature Review** – Thorough research was done on existing mental health applications before developing the software, and currently continuing the research that uses machine learning, sentiment analysis, and mood prediction. Seventeen academic research papers, articles, and case studies were done to understand the project's strengths and weaknesses of the current solution in this domain.
2. **Interviews** – Surveys and interviews were done with mental health professionals and diagnosed users. This helped to identify the user's expectations, needs, and challenges they face in seeking mental health care.
3. **User Feedback** – Feedback was taken by the test users of the system when the application was under development mainly to find the features that would be beneficial and to ensure the system's ease of usability.
4. **Competitive Analysis** – Analyzed the existing mental health apps, mainly the ones that are available in Sri Lanka, which provided the details of features and functionalities that are popular and also lacking in the domain.

By using these above-mentioned methods, I got a clear understanding of the problem, which allowed the system design to match both technical requirements and users' needs.

Doctor Details

- Dr. Deshantha Pathinayake – MBBS(SL), DFM(COL), MCGP, MRCGP(Int)
SLMC Reg: No. 2372.
General / Mental Health Specialist.
Wellness Medi Clinic, Moratuwa 10400
- Dr. Saman Weerawardhane – MBBS(SL)
Psychiatrist / Mental Health Specialist.
National Institute of Mental Health (NIMH)
NIMH, Mulleriyawa New Town, Angoda

Details of the Local Hospital (Data Collection):

- Mana Suwa Piyasa (මනසුවපියස) – Mental Health Clinic
Colombo South Teaching Hospital – Kalubowila
B229 Hospital Rd, Dehiwala-Mount Lavinia 10350
- Survey - [CLICK HERE!](#)
- Responses - [CLICK HERE!](#)

2.2. Existing System

Currently, as per research, there are different types of mental health monitoring systems in the form of both native to web and mobile platforms. These systems only focus on self-reporting and tracking symptoms. However, many of these platforms are limited to their generic scope and functionalities, lacking real-time analysis, personalized advice, and in providing resources for mental health uplifting (Woodward et al., n.d.).

- **Existing Applications** - Common applications like Calm, Headspace, and Moodfit focus on generic mood tracking, mindfulness, and meditation but do not use modern advanced machine learning techniques to provide unique mental health predictions and responses. (Smith et al., 2023), (Woodward et al., n.d.).
- **Shortcomings** - These systems that were researched and explored had no integration with the other services (ex: Nearby hospitals, Professionals booking, etc.) and it also does not have an alert system for the users experiencing suicidal levels in mental health (Smith et al., 2023), (Tazarv et al., 2021).
- **Cultural / Regional Gap** - Many of the current mental health applications fail to identify the cultural differences and needs they have (ex: post war trauma, Economical conflicts etc.), especially in a country like Sri Lanka which is different from other countries where stigma and language barriers prevent the usage of the systems (Deivanayagam et al., 2024).

While there are existing systems that contribute to basic mental health awareness, they do not address all the needs for personalized, real-time monitoring mainly in the Sri Lankan context.

NOTE: Existing System Comprehensive details are available on Main Content and PID documents above. I have removed it in here to reduce repeating content.

2.3. Use Case Diagram

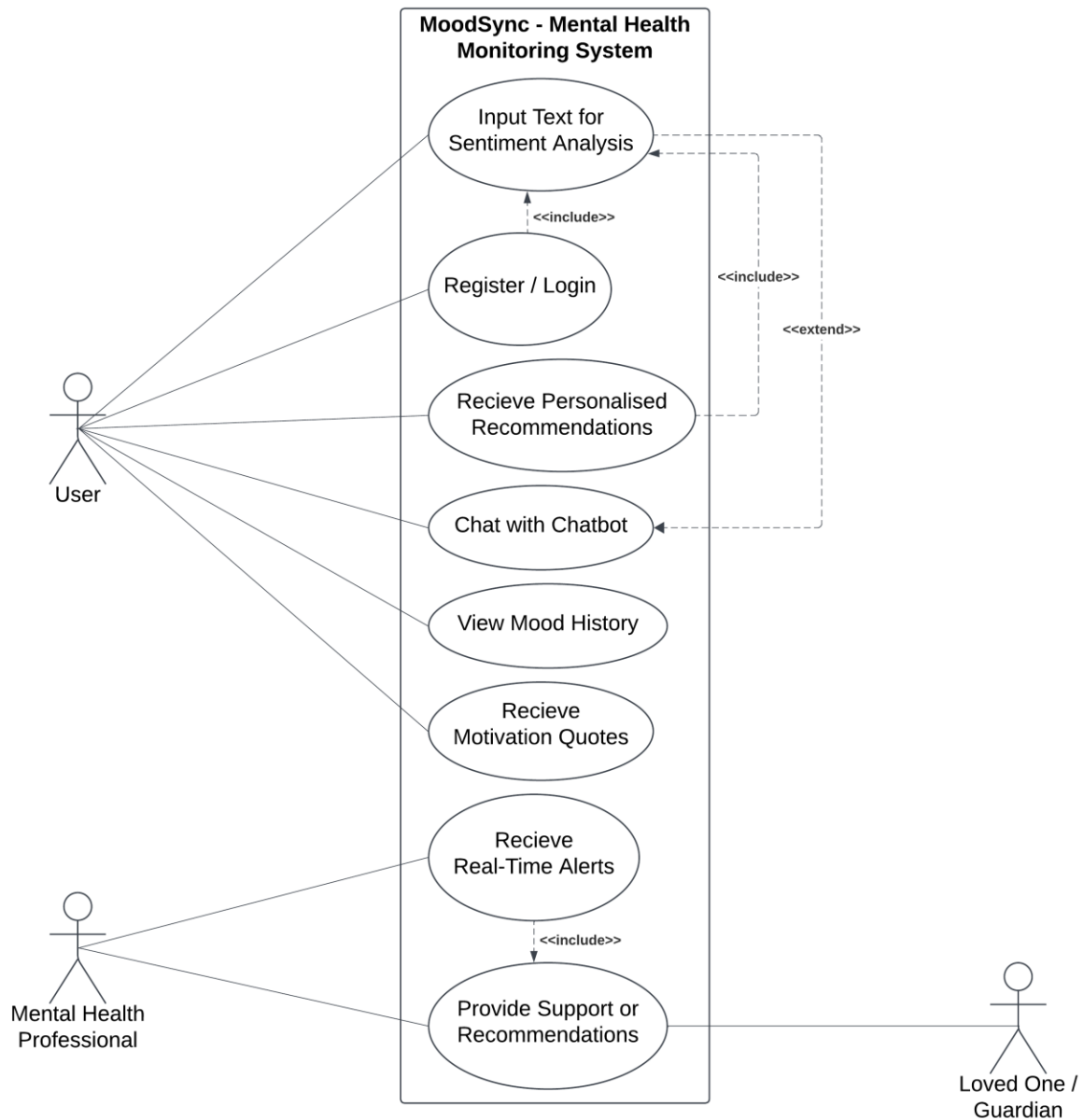


Figure 38 - Use Case Diagram

2.4. Drawbacks of the Existing System

Here are the drawbacks identified in the existing systems,

1. **Lack of Personalization** – The existing systems offer generic advice and do not consider unique mental health patterns or cultural backgrounds. They do not adjust recommendations based on users' unique mental health state.
2. **Limited Interaction** – Many of the existing systems provide only one-way testing methods to get results (ex: health quizzes), where there is limited interaction with the user. Users may benefit from a system if it is more engaging, has real-time support, especially in health breakdowns.
3. **Lack of Real-Time Alerts** – Existing systems lack real-time notification alert features for loved ones/guardians or assigned professionals when the user is in distress. In dangerous situations, immediate support given can be lifesaving, which the system lacks.
4. **No integration with near health care providers** – No current system has integrated mental health professionals or hospitals to assist. This reduces the help in need, which fails the whole product.
5. **Cultural Relevance** – Many of the systems are designed for the global audience where they do not shape to detect cultural differences in mental health understanding (ex: post-war trauma, economic situations, etc.).

This project – MoodSync addresses these issues mentioned with a more personalized, real-time, and culturally trained solution with additional features such as anonymous communication, mental health professional's connections, and real-time alerts for loved ones.

3. Requirements Specification

3.1. Functional Requirements

The functional requirements detail the essential features and functions that this system should have to meet the project objectives as I mentioned in Chapter 1. These requirements ensure that the system delivers the needed features for mental health monitoring and tracking. The functional requirements are,

10. User Registration and Authentication

- The user should be able to create an account and log in securely.
- User data needs to be stored safely, and privacy needs to be ensured by using encryption.

11. Sentiment Analysis and Mood Prediction

- The system should analyze user inputs (text, social media posts, etc.) to detect user sentiment and predict moods.

12. Mood Tracking

- The system should be able to track users' moods over time, which should record any daily input and provide insights into mood patterns.
- Users should be able to view their mood history by graphs and visualization features.

13. Unique Mental Health Recommendations

- From sentiment analysis and mood tracking, the system should give unique mental health advice, coping strategies, and resources to improve mental health.

- Recommendations given should be matched to users' emotional and mental health condition.

14. Anonymous Chat Feature

- Users should be able to communicate with the system through an anonymous chat interface.
- The chatbot should be able to provide responses, ask questions back, and give guidance based on user inputs.

15. Real-Time Alerts for Loved Ones

- The system should automatically send real-time alerts for users' loved ones when the user's mood is tracked as dangerous or self-harming.
- Alerts should be sent by text SMS.

16. Motivational Content

- The application should display motivational content like quotes, affirmations, and positive content to cheer users in difficult times.

17. Data Visualization

- The system must show users' mood data, and sentiment trends by graphs and charts, that allow the user to track their progress over time.

18. Mental Health Professionals and Hospital Connections

- This system should have an option to connect with mental health professionals or the local hospitals.

- This feature should provide contact details and emergency help for mental health support.

3.2. Non-Functional Requirements

Non-functional requirements detail this software's operational characteristics, which ensure that the system performs efficiently and addresses the user's expectations in performance, security, and usability. The non-functional requirements identified are,

9. Performance

- This application should display the real-time analysis when a user inputs the texts and generates recommendations within 5 – 10 seconds.
- This application should also have the capability to handle many users without running into any issues.

10. Scalability

- This application should be made to scale efficiently as the users increase, which allows further expansion without any performance reduction.

11. Security

- This application should ensure privacy, integrity, and the availability of the user data.
- Encryption methods should be applied for sensitive data like user login details and personal health information.

12. Usability

- This system should have a user-friendly interface to easily navigate mainly for users having less technical knowledge.
- The chat feature should be simple and easy to use.

13. Availability

- This system should be 24 * 7, ensuring that users can access mental health monitoring at any time.
- Downtime should be reduced, and it should only happen when the system is maintained only.

14. Reliability

- The system should be reliable, giving accurate sentiment analysis, mood tracking, and recommendations on user inputs.
- The system should handle unexpected errors so that the user experience will not be disturbed.

15. Maintainability

- This system should be easy for maintenance purposes, with components.

16. Compliance

- This system must follow Sri Lankan data protection regulations (2022) like GDPR and privacy laws to ensure that user data are handled ethically.

3.3. Hardware / Software Requirements

The hardware and software requirement mentioned shows the resources that are needed to develop, deploy, and run the software. This includes server needs, development tools, and platforms for the deployment.

3. Hardware Requirements

- **Server** – A cloud server (ex: AWS) to host the application backend services, machine learning models, and the database.
- **End User Devices** – The system should be responsive to work on web applications from desktops and smartphones.
- **Database Storage** – Cloud-based database storage needed (ex: MongoDB) for storing the user data securely.

4. Software Requirements

- **Frontend Development** – React.js, Vite, TypeScript, Tailwind CSS, and Flowbite library for styling.
- **Backend Development** – FastAPI with Python, Node.js with Express for the API endpoints and to handle machine learning processes.
- **Machine Learning Libraries** – Pandas, scikit-learn, imblearn, NLTK, and other Python libraries are mainly used for sentiment analysis and to predict moods.
- **Database** – MongoDB for storage of user data, feedback, mood logs, and history.
- **Cloud Hosting / Platforms** – AWS for handling computational tasks and hosting the application.

3.4. Networking Requirements

This project has less network requirements, but these are the basic requirements for system components and external services communication in the application,

6. Internet Connectivity

- This system needs a good internet connection to provide real-time updates and alerts.
- Software must be able to communicate with the backend server to process the data.

7. API Integration

- The system needs a third-party API integration for the SMS notifications and integration with healthcare providers.

8. Real-Time Communication

- For the real-time alerts feature, the system should use push notifications by WebSockets (Socket.IO) or a similar way to ensure delivery of the alerts.

4. Feasibility Study

Carrying out a feasibility study is needed for practicality and survival of the product. This chapter details the operational, technical and financial aspects of the project MoodSync, ensuring that this project can be further developed and completed with the limits of given time, resources and the budget.

4.1. Operational Feasibility

Operational feasibility mainly shows whether this proposed system can be implemented effectively and used in the real world, considering the user's needs, accessibility, and system integration.

1. User Adoption

- This system targets users going through mental health challenges, especially in a low-accessed country like Sri Lanka. It mainly focuses on good free software for people who are hesitant/shy/feared to look out for help in need. Anonymous chat and real-time alert features for loved ones increase the user's adoption of this application.
- This system will be designed with a good, featured user interface, matching for the users who have less technical knowledge. This improves the ease to use this software, and it also make sures that users can access this system well.

2. Integration of mental health professionals

- One of the key features of this system is the connection with mental health professionals. This feature makes sure that users have the professional's quick access when needed.

- By giving easy access to professionals and hospitals, this system assures that users have a solid network to support, improving this system's operational possibility.

3. Cultural Adoption

- This system will be developed mainly to meet the needs of Sri Lankan users, where features will be included with well-tuned cultural context and sensitivities.
- This system's design and functionality consider cultural details related to mental health, reducing its stigma and encouraging people to get help.

4. Real-Time Alerts

- Real-time alerts for loved ones will provide operational advantages, ensuring quick actions in case of a mental health crisis. This driven feature helps users during critical moments by reducing life-threatening situations.
- These alerts especially benefit the user who will not have quick access to healthcare providers, ensuring their safety and well-being on top.

4.2. Technical Feasibility

Technical feasibility shows if the technology required is available, reliable, and matching to successfully develop the project's solution.

1. Technology Stack

- This system uses modern, scalable technologies which are well suited to developing a system with mental health monitoring. Using MERN Stack with TypeScript for the frontend (client) and FastAPI and Node.js for the backend (server) gives an efficient solution for developing the web application for the system.
- The Python-based machine learning model uses popular libraries like scikit-learn, Pandas, and NLTK mainly for mood predictions and sentiment analysis. This tech stack is reliable and proven to build good models under different machine learning and data analysis software.

2. Machine Learning and NLP

- This system's need to analyze the user inputs and predict mood by sentiment analysis requires advanced Natural Language Processing (NLP) techniques. Python libraries such as TF-IDF and logistic regression have good documentation to refer, and it is efficient for text classification tasks.
- The SMOTE algorithm will be used to handle the dataset's imbalances, making sure that the model performs well even when some classes (Ex: mental health disorders) are not present in the trained dataset.

3. Cloud Infrastructure

- This system will be hosted on the AWS (Amazon Web Services) cloud platform EC2 Service (free tier) to temporarily provide the application to users.

- AWS EC2 hosting provides higher availability, a secure database, and flexibility in accessing the software.

4. Security and Data Protection

- Given the sensitive domain of health data, modern strict security measures will be used like data encryption in chats, chat protocols and safety practices to protect user privacy. (ex: OAuth 2.0, JWT, Bcrypt).
- This system will match the industry's best practices to ensure that the user's data is kept secure.

4.3. Outline Budget

This project – MoodSync does not include any additional charges and given charges would be taken care of by my personal funds.

1. **Mental Health Doctor** – Appointment charge of Rs. 5,500 (£14.85) for data gathering and an additional Rs. 1000 (£2.62) for every visit to check the application quality changes and inspections.
2. **Development** – Gets paid versions of development tools free facilitated by University of Plymouth student account for JetBrains, PyCharm Ultimate Edition IDE.
3. **Cloud Hosting** – Use of free tier of cloud services (AWS) at present, domain and SSL certificate and charges will be applied for further hosting.

5. System Architecture

- Link for all diagrams for a clear view - [CLICK HERE!](#)

5.1. Class Diagram of Proposed System

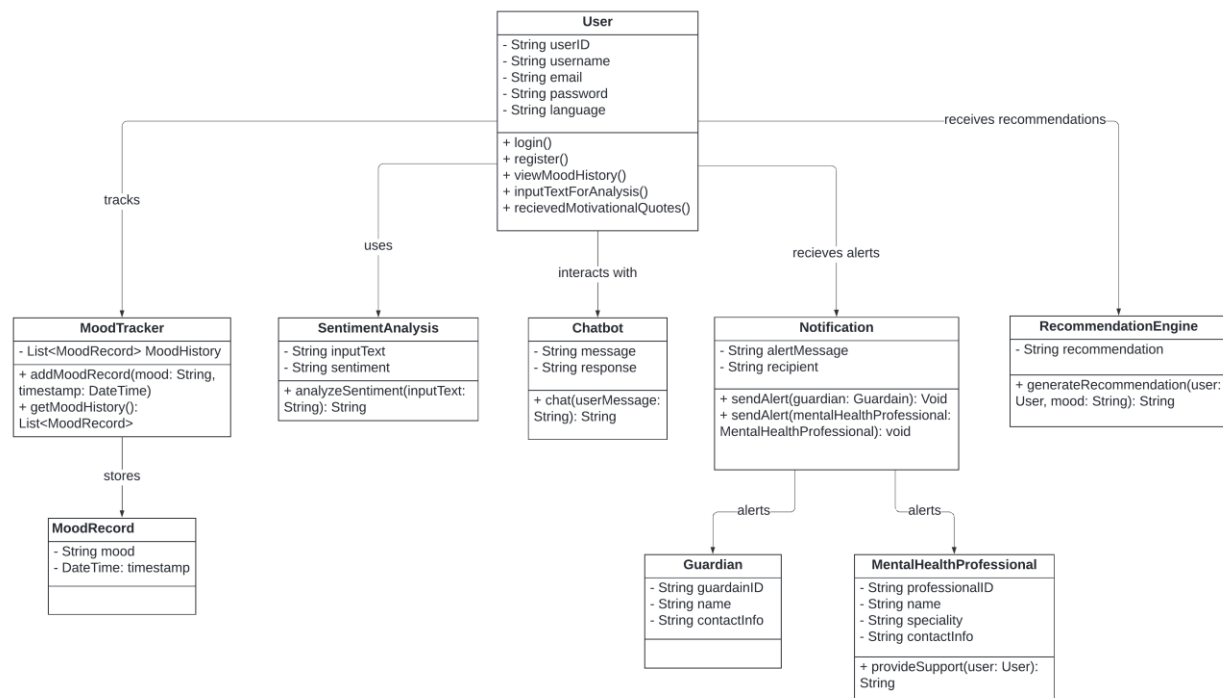


Figure 39 - Class Diagram

5.2. Database Diagram

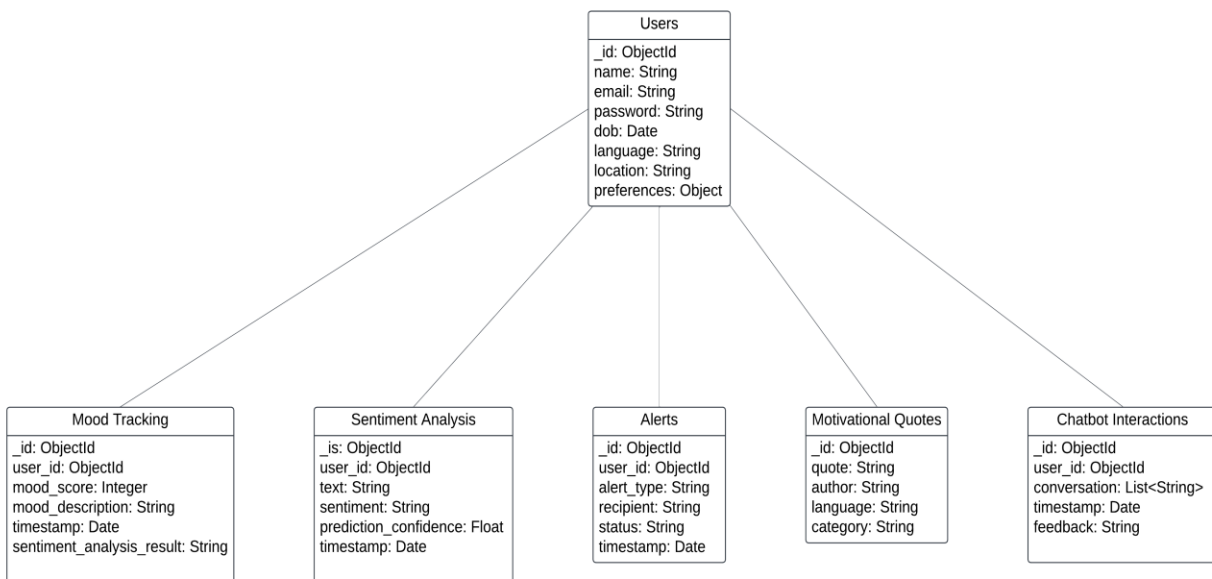


Figure 40 - Database Design

- **NOTE** – ER Diagram cannot be drawn for this software project as it uses NoSQL unstructured database. The database design diagram is drawn above showing how the data collections are managed, and how their connections interact.

5.3. High – Level Architectural Diagram

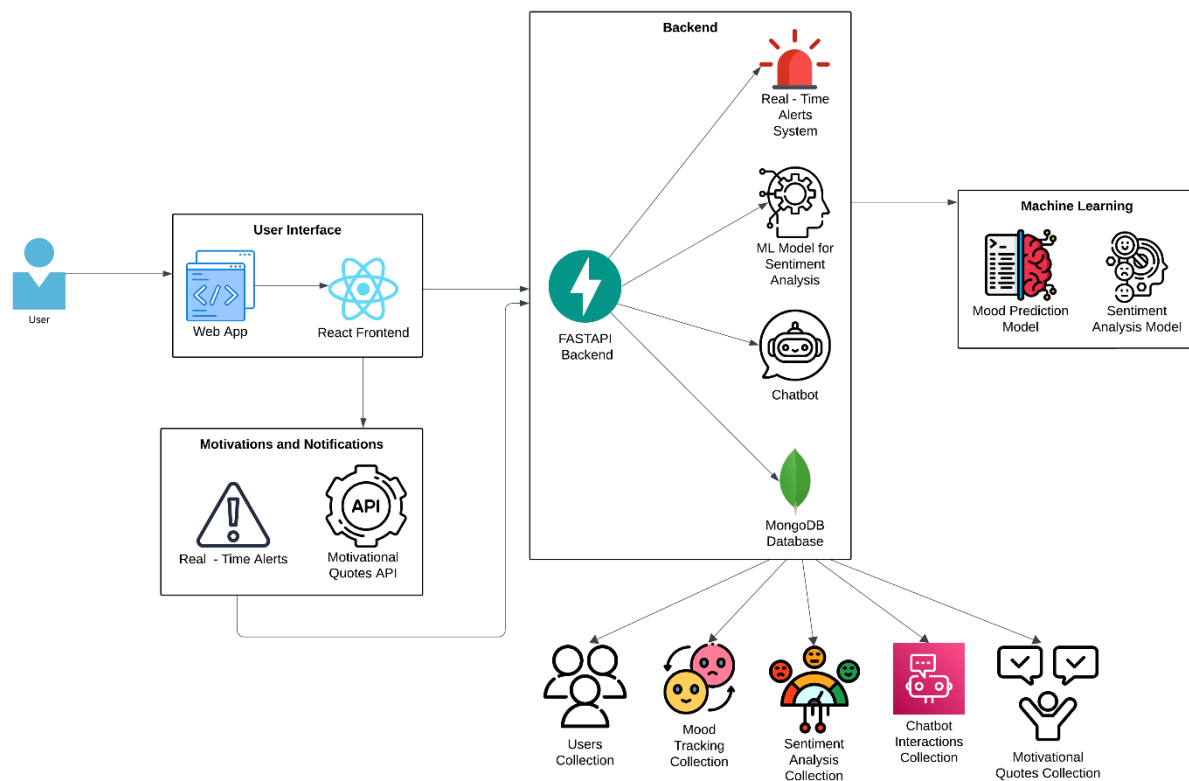


Figure 41 - High - Level Architectural Diagram

5.4. Networking Diagram

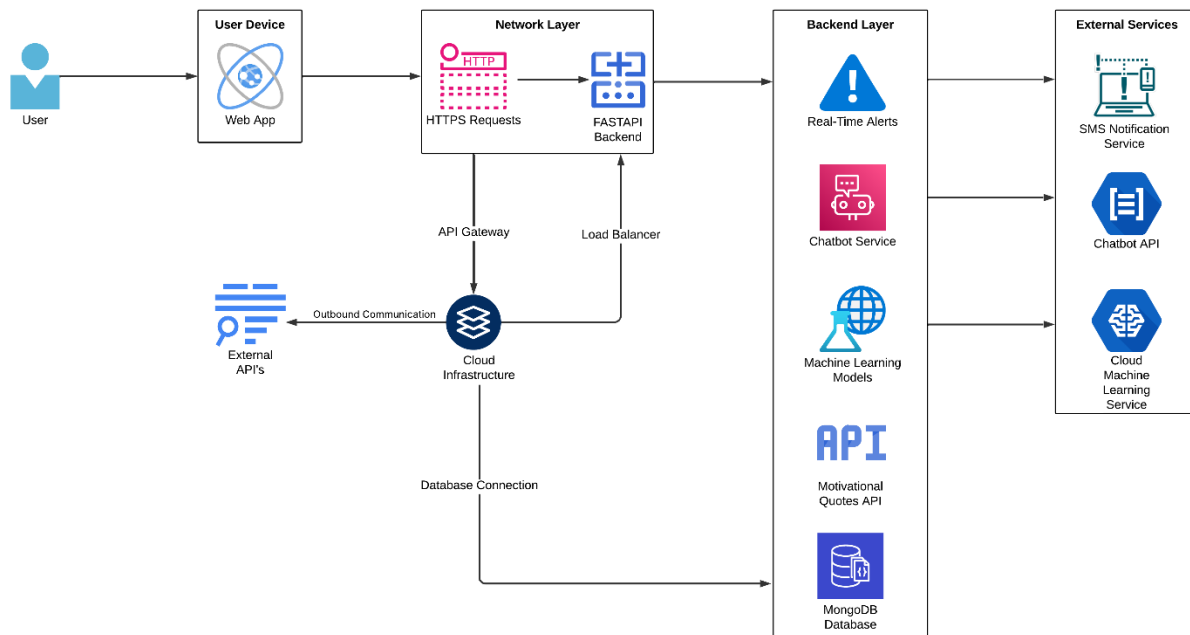


Figure 42 - Networking Diagram

6. Development Tools and Techniques

The development of the project MoodSync has been done with modern technologies and tools to create a robust, adaptable and user-friendly application. This chapter shows the main tools, libraries, frameworks and methodologies used in the development of this software. These tools and technologies were chosen to ensure that the application matches the given functional and non-functional requirements while providing good user experience.

6.1. Development Methodology

The development methodology selected for this project is Agile, though this is an individual project, it allows a flexible and repetitive process. The agile methodology focuses on delivering functional progress of the application regularly, with continuous feedback from the test users and mental health professionals which is considered for the development process. By using this approach, it helps to ensure that this software is developed based on the real user's needs and can suit the change of requirements over time.

- **Sprint-Based Development** – This project is divided into sprints, which last for 2 weeks. Every sprint is focused on separate functionalities, like user authentication, sentiment analysis or mood-based tracking.
- **Continuous Integration and Testing** – Agile prioritize the importance of testing in the development process itself. Automated testing tools and continuous integration pipelines ensure that the new features are tested frequently and integrated into the application.
- **User Feedback** – After every sprint, feedback is taken from a set of test users to make sure that the application follows the expectations and solves the expected problem effectively.

6.2. Programming Languages and Tools

This system is built using different programming languages and tools to ensure that both frontend and backend components work smoothly, where machine learning is used for sentiment analysis and mood prediction.

Frontend Development

- **React.js** – React.js is a JavaScript library that is used to build the user interfaces (UI) for this project MoodSync. Reacts allow the development of dynamic, responsive & interactive UIs where components can be reused (ex: Navigation Bar, Footer etc.). It is a fast and efficient way to develop maintainable and scalable software.
- **Vite** – Vite is used as for both the development server and the build tool. This ensures fast hot module reload (HMR) and builds optimization, which improves the development flow, making it fifteen times faster than using react alone.
- **TypeScript** – TypeScript is used for modern frontend and backend development to add static typing, which will reduce errors and improve the quality by ease of catching bugs in the development stage itself.
- **Tailwind CSS** – TailwindCSS is mainly used for front-end styling. The first utility CSS framework that allows for a fast, responsive, and user-customized design. It simplifies the design processes by giving a set of classes that can be used with HTML.
- **Flowbite** – Flowbite is a component UI library used with the Tailwind CSS framework that gives pre-designed, responsive UI components like modals, buttons, and form controls, which makes the UI development faster and more consistent.

Backend Development

- **FastAPI** – Is used to develop the main backend of the software. It is the latest fast-performance web framework for building APIs with the use of Python. It supports auto-generation of the API documents and also it comes up with built-in features like dependency injection, making it more of a best choice for developing RESTful APIs. Also, its support for asynchronous programming makes sure that the backend will handle multiple requests at the same time, improving performance and scalability.
- **Python** – Python is the main programming language used in this project which is for backend and machine learning developments in the software. Python is mainly used in data analytics, machine learning, and web development, which makes it the best choice for this project.
- **Node.js** – Node.js backend is also used apart from FastAPI for handling normal third-party APIs (ex: Motivational Content) as some APIs don't support python environments. Express.js will also be used as the middleware.

Database

- **MongoDB** – A NoSQL database used for this project to store users' data, mood tracking records, and the software logs. It is mainly chosen for its scalability, flexibility, and easy integration with the software. It allows the software to store unstructured data which will be essential to handling different dynamic natures of user input data.

Machine Learning

- **scikit-learn** – The scikit-learn library is used for machine learning development in this software. It provides efficient tools for sentiment analysis and mood prediction model building. The logistic regression model in scikit-learn is used to separately identify sentiments from users' text inputs and help in predicting mental health conditions.
- **Pandas** – Used for data manipulation and preprocessing. It does provide data structures that allow to handle data efficiently and for transformation, which is important for purifying and using the input data for the machine learning models.
- **NLTK** – The Natural Language Toolkit (NLTK) is used for text preprocessing, including tokenization, lemmatization and to remove the stopwords. These NLP methods help in normalizing and cleaning the user input data, which does a big role in improving the accuracy of sentiment analysis and mood predictions.
- **Imbalanced-learn (imblearn)** – This learning library is used mainly for handling dataset imbalances with the usage of techniques like SMOTE (Synthetic Minority Oversampling Technique) which ensures that the machine learning model performs well when the mental health conditions are not fully represented in the dataset that is used to train the model.

6.3. Third Party Components and Libraries

To improve the functionalities and efficiency of this application, third-party libraries and tools have been used to build this application. These below-mentioned components allow the software to have advanced features and largely reduce its development time.

1. **Twilio (SMS Notification)** – This is integrated with the application to send real-time SMS alerts to users' loved ones and mental health professionals. This software uses this API to send SMS messages when triggered automatically by the system, making sure that alerts are delivered quickly.
2. **Chart.js** – This is used to create interactive and dynamic data visualization. It helps to display users' tracking details and sentiment analysis results display in graphs, where users and doctors can use them to understand their mental health over time.
3. **Socket.IO** – This is used for real-time communication between the backend (server) and the frontend (client) which lets features like messaging in the chatbot and real-time alerts to users' loved ones and to communicate to health professionals.
4. **Passport.js** – This is used for user authentication. It supports many authentication methods such as local login, email-based login, and OAuth2.0 making it an easier task to develop secure and custom authentication methods.
5. **Bcrypt** – This is used mainly for password hashing purposes. It ensures that the user's password is stored safely, and it also cuts off unauthorized access to sensitive information.

6.4. Algorithms

1. **Logistic Regression** – This is used as the primary algorithm to classify sentiments from user input. It helps to predict the mental state of the user by text data that is collected from text input or by chat.
2. **TF – IDF (Term Frequency – Inverse Document Frequency)** – It is mainly used to convert text data to numerical features so that it can be easily processed in the trained machine learning models. It also measures the importance of words in context by the user's input and helped to improve the performance of sentiment analysis.
3. **SMOTE (Synthetic Minority Oversampling Technique)** – This is applied to the dataset to handle class imbalances. It generates synthetic examples in the minority class, making sure that the model can learn well from all given classes.
4. **K-Fold Cross Validation** – This is used to evaluate the machine learning model performance. The dataset will be split into K subsets, where training and testing will happen K times to ensure accuracy and adaptability.

7. Discussion

7.1. Overview of the Interim Report

This interim report shows the current progress and state of the Project MoodSync. This project aims to provide innovative and accessible software to monitor users' mental health through sentiment analysis, mood tracking, and personalized recommendations. By using machine learning technologies, the system tracks mental health patterns by user inputs and gives real-time intervention when needed. This report also covers the system's requirements, feasibility analysis and technologies that were used to build this software.

7.2. Summary of the Report

This report provides an in-depth analysis of the problem in mental health monitoring, mainly in the Sri Lankan context. It shows the system's functional and non-functional requirements, also the technological needs to support this solution. It details different challenges the applications face, like lack of personalization, real-time intervention, and cultural inclusivity, which this project addresses. Additionally, the report details the tools, technologies, and machine learning algorithms used in this application, ensuring an software that is a secure and adaptable solution for the users.

7.3. Challenges Faced

In the development of this software, some challenges were faced like,

1. **Data Quality and Preprocessing** – The input dataset used to train the model needed thorough cleaning and preprocessing to ensure that it was suitable for mood predictions and sentiment analysis. Balancing the dataset using methods like SMOTE was complex and essential to improve the model's output accuracy.
2. **Real-Time Alerting** – Developing a reliable real-time alert system for loved ones was a challenge, mainly to ensure that the notifications were delivered in real-time effectively processed within backend without a delay.

3. **Cultural Adaption** – Shaping the system to maintain high accuracy in sentiment analysis was a major technical challenge with the cultural inclusions, which needed to include regionally native text processing methods.
4. **User Privacy** – Making sure that the data stored and processed are without changed privacy concerns was challenging, which needed robust encryption methods (ex: OAuth 2.0, JWT & Bcrypt) and followed with Sri Lankan data protection acts.

7.4. Future Plans / Upcoming Work

The next further step in the project is to include,

1. **Model Optimization** – To tune the machine learning model further to improve its accuracy and robustness, mainly in managing real-time users' data and different types of inputs.
2. **Feature Enhancement** – Adding extra features like integration with wearable devices (for tracking physical and health data) and expanding the emergency alert system for more precise locations for better user safety.
3. **Deployment** – Prepare the application to publish on a large scale, with a focus on cloud integration to make sure it can handle a growing user base concurrently providing real-time support without running into performance issues.
4. **User Testing** – Keep up through the user testing, to get feedback on the application's core usability and effectiveness, and make sure that it follows the user's expectations.

References

- Deivanayagam, T.A., Ní Chobhthaigh, S., Devakumar, D., Patel, K. & Rannan-Eliya, R.P. (2024) Mental health prevalence, healthcare use and access between 2018 and 2022 in Sri Lanka: an analysis of survey data. *Global Health Research*. 1–16. doi:10.3310/hjwa5078.
- Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., et al. (2018) The Lancet Commission on global mental health and sustainable development. *Lancet (London, England)*. 392 (10157), 1553–1598. doi:10.1016/S0140-6736(18)31612-X.
- Smith, K.A., Blease, C., Faurholt-Jepsen, M., Firth, J., Van Daele, T., Moreno, C., Carlbring, P., Ebner-Priemer, U.W., Koutsouleris, N., Riper, H., Mouchabac, S., Torous, J. & Cipriani, A. (2023) Digital mental health: challenges and next steps. *BMJ mental health*.26 (1). doi:10.1136/bmjment-2023-300670.
- Sumathipala, A., Mahesh, D. & Wijesundara, H. (2020) *Challenges of Research in Rural Mental Health in Sri Lanka*. In: pp. 195–220. doi:10.1007/978-981-10-2345-3_30.
- Tazarv, A., Labbaf, S., Reich, S.M., Dutt, N., Rahmani, A.M. & Levorato, M. (2021) Personalized Stress Monitoring using Wearable Sensors in Everyday Settings. In: *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. 2021 pp. 7332–7335. doi:10.1109/EMBC46164.2021.9630224.
- Woodward, K., Kanjo, E., Brown, D.J., McGinnity, T.M., Inkster, B., Macintyre, D.J. & Tsanas, A. (n.d.) *Beyond Mobile Apps: a Survey of Technologies for Mental Well-being*.
- Calm - <https://www.calm.com/>
- Headspace - <https://www.headspace.com/>
- Moodfit - <https://www.getmoodfit.com/>

Records of Supervisory Meetings



PUSL3190 – Computing Individual Project
Student Progression Report
[Student Copy]

1. Student Name: Samarasinghe Methmal.....
2. Plymouth Index Number: 10898561.....
3. Degree Program: BSc. (Hons) Software Engineering.....
4. Supervisor Name: Mr. Gayan Perera.....
5. Project Title: Mental Health Monitoring System.....

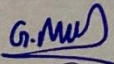
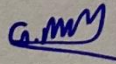
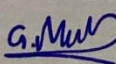
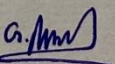
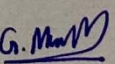
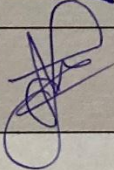
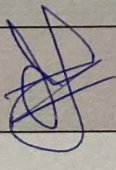
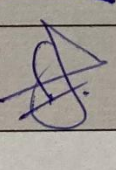
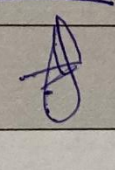
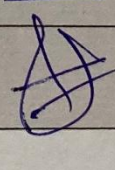
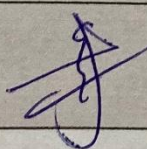
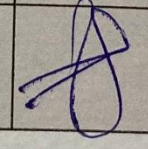
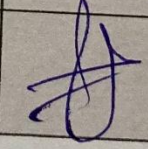
Meeting Number	Meeting 01	Meeting 02	Meeting 03	Meeting 04	Meeting 05	Meeting 06	Meeting 07
Date	13/9/24	27/9/24	11/10/2024	11/11/2024	26/02/2025		
Student Signature							
Supervisor Signature							

Figure 43 - Supervisory Meetings

Documentations	Proposal	PID	Interim 01	Interim 02	Re A
Date	31/10/2024	06/10/25	9/3/25		
Approved (Yes / No)	Yes	Yes	Yes		
Student Signature	<u>G. M. M.</u>	<u>G. M. M.</u>	<u>G. M. M.</u>		
Supervisor Signature					

Other Comments (Supervisor use only)

Final Year Project – Supervisory Meeting Minutes

Meeting No: 01

Date : 13/09/2024
Project Title : Mental Health Monitoring System
Name of the Student : Samarasinghe Methmal
Student's ID : 10898561
Name of the Supervisor : Mr. Gayan Perera

Items discussed:

1. Project Idea approval.
2. Discussed about Overall Project representation and objectives basic.
3. Natural language Processing (NLP) and how it can be applied to the Project.

Items to be completed before the next supervisory meeting:

1. Train a hotel review dataset using Natural language processing (NLP) to give a feedback on reviews.
2. Write the Project Scope.


(27/09/2024)
Supervisor (Signature & Date)

Instruction to the Supervisor: **Do not sign** if the above boxes are blank.

Final Year Project – Supervisory Meeting Minutes

Meeting No: 02

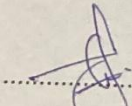
Date : 27/09/2024
Project Title : Mental Health Monitoring System
Name of the Student : Samarasinghe Methmal
Student's ID : 10898561
Name of the Supervisor : Mr. Gayan Perera

Items discussed:

1. checked the developed ML model and tested.
2. checked the Scope of the project and approved.
3. ways to continue the project further with submissions.

Items to be completed before the next supervisory meeting:

1. Write a literature review researching 10 papers published.
2. Increase the repository of the ML Model adding more vocabulary to analyse.

 (11/10/2024)
Supervisor (Signature & Date)

Instruction to the Supervisor: **Do not sign** if the above boxes are blank.

Final Year Project – Supervisory Meeting Minutes

Meeting No: 03

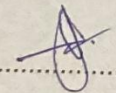
Date : 11/10/2024
Project Title : Mental Health Monitoring System
Name of the Student : Samarasinghe Methmal
Student's ID : 10898561
Name of the Supervisor : Mr. Gayan Perera

Items discussed:

1. Checked the literature review, instructed to add the complete version to the project proposal.
2. Checked the identified unique features to the 2nd phase of the project.
3. Checked out the identified gaps and how it can be fulfilled.

Items to be completed before the next supervisory meeting:

1. Add a weighting to the ML Model and convert it to mental health related.
2. Meet a psychiatrist and know how the result/Score calculation is done.

 (11/11/2024)
Supervisor (Signature & Date)

Instruction to the Supervisor: Do not sign if the above boxes are blank.

Final Year Project – Supervisory Meeting Minutes

Meeting No: 04

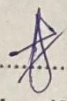
Date : 11/11/2024
Project Title : Mental Health Monitoring System
Name of the Student : Samarasinghe Methmal
Student's ID : 10898561
Name of the Supervisor : Mr. Gayan Perera

Items discussed:

1. checked the updated ML Model.
2. checked out the findings from Professionals to get an overall score.
3. Guided on creations of UI's and chatbot.

Items to be completed before the next supervisory meeting:

1. Using the model to build a chatbot.
2. Document the overall score calculation.
3. Creation of UI's.


Supervisor (Signature & Date)

Instruction to the Supervisor: Do not sign if the above boxes are blank.

Final Year Project – Supervisory Meeting Minutes

Meeting No: 05

Date : 26/02/2025
Project Title : Mental Health Monitoring System
Name of the Student : Samarasinghe Methmal
Student's ID : 10898561
Name of the Supervisor : Mr. Gayan Perera

Items discussed:

- x Backend Integrations.
- x Research discussion.
- x How to shape the project

Items to be completed before the next supervisory meeting:

- ~~x Frontend Completion.~~
- x Completion of the project.


.....
Supervisor (Signature & Date)

Instruction to the Supervisor: Do not sign if the above boxes are blank.

Other Materials

Doctor Approval Letter

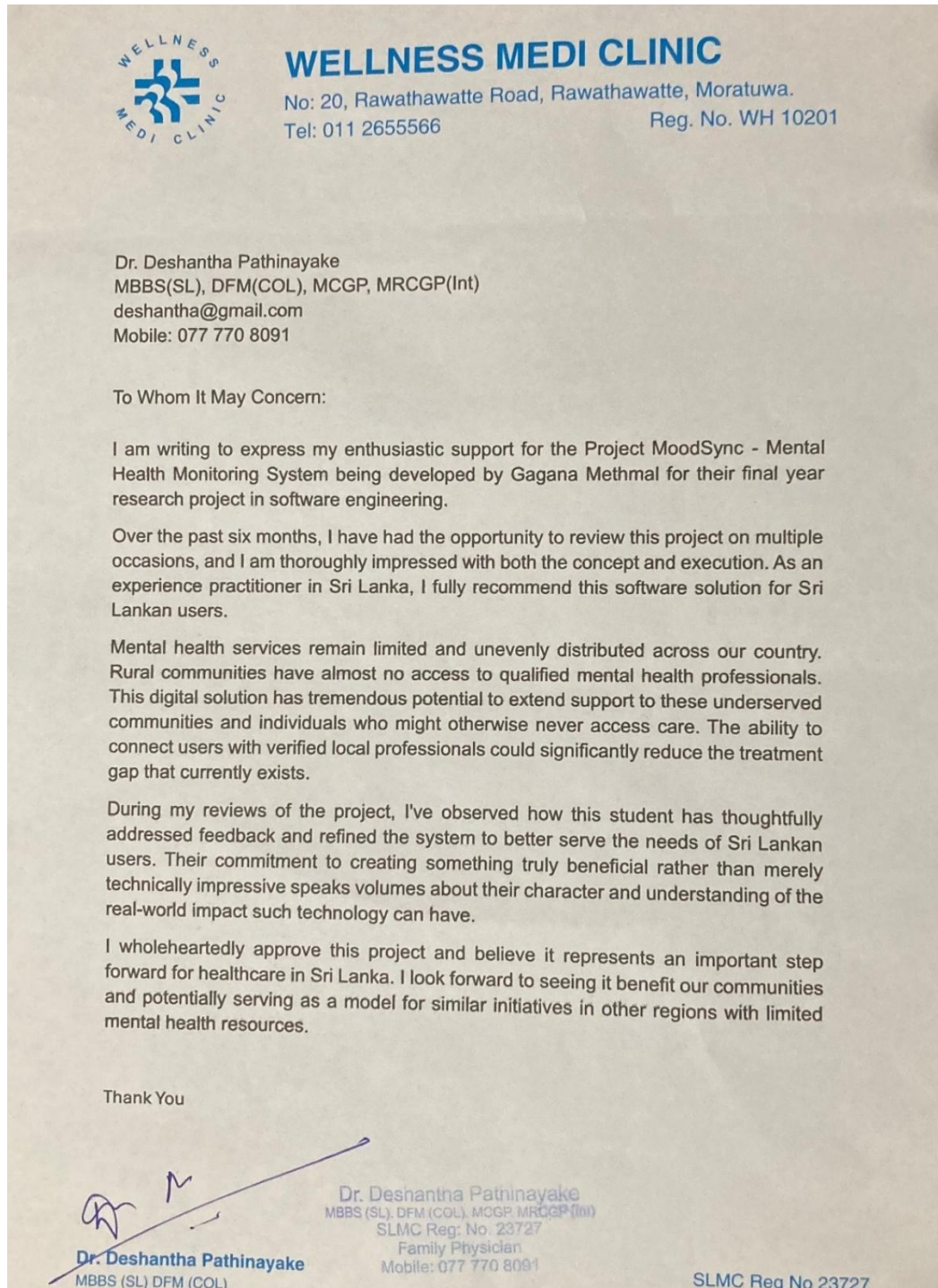


Figure 44 - Dr. Deshantha's (Approval of the Software)

Meeting Minutes

15/11/2024 (first day)

Date: 25/04/2025 (last day).

Time: 7.15 PM.

Location: Wellness Medi Clinic, Moratuwa.

Attendees: Dr. Deshantha Pathinayake, Samarasinghe Methmal.

Discussion Points:

1. Model Validation Results

- 93% accuracy achieved in sentiment analysis model.
- Clinical thresholds confirmed appropriate for Sri Lankan context.
- Post-war trauma indicators successfully implemented.

2. User Interface Check

- Patient interface approved for accessibility and cultural sensitivity.
- Professional dashboard meets clinical workflow requirements.
- Anonymous communication system validated for privacy compliance.

3. Crisis Intervention

- Emergency alert system thresholds verified.

Conclusions:

- Project MoodSync system meets healthcare standards and clinical requirements.
- Cultural adaptations successfully address Sri Lankan mental health context.
- System ready for implementation in healthcare settings.

Next Steps:

- Formal approval letter to be issued by Dr. Pathinayake.
- Preparation for deployment phase to begin.

Approved by: Dr. Deshantha Pathinayake



Figure 45 - Dr. Deshantha Pathinayake (Showcase of the final product).

Date: 7/12/2024.

Time: 11.45 AM.

Location: By Zoom.

Attendees: Dr. Saman Weerawardhane, Samarasinghe Methmal

Discussion Points:

1. Clinical Algorithm Assessment

- Review of machine learning model performance with datasets.
- Validation of sentiment analysis accuracy for depression and anxiety indicators.
- Recommendations for enhancing suicide risk detection parameters.

2. Professional Interface Evaluation

- Assessment of patient analytics visualization for clinical decision-making.
- Review of anonymous communication protocols for effectiveness.
- Feedback on doctor dashboard functionality for high-risk patient management.

3. Cultural Adaptation Review

- Evaluation of language processing abilities.
- Assessment of culturally appropriate intervention recommendations.
- Verification of economic stress indicators needed to local context.

Conclusions:

- The system shows the possibility for integration with existing mental health services.
- Anonymous communication feature addresses cultural stigma barriers.
- Alert system meets psychiatric emergency intervention standards.

Next Steps:

- Implementation of suggested changes to suicide risk algorithms.
- Report of validation for regulations.
- Planning for pilot program at NIMH services.

Approved by: Dr. Saman Weerawardhane

Wireframes UI Designs

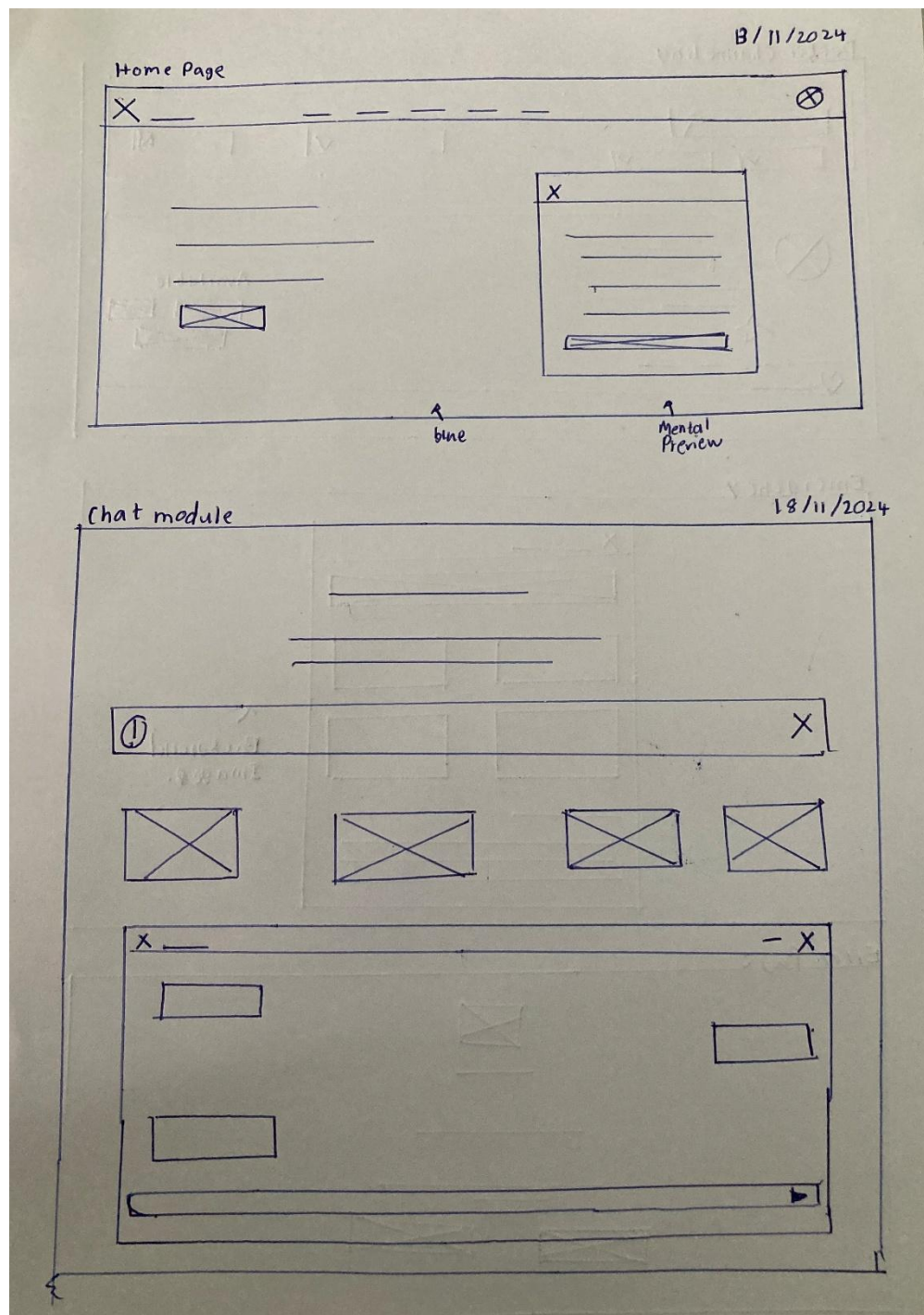
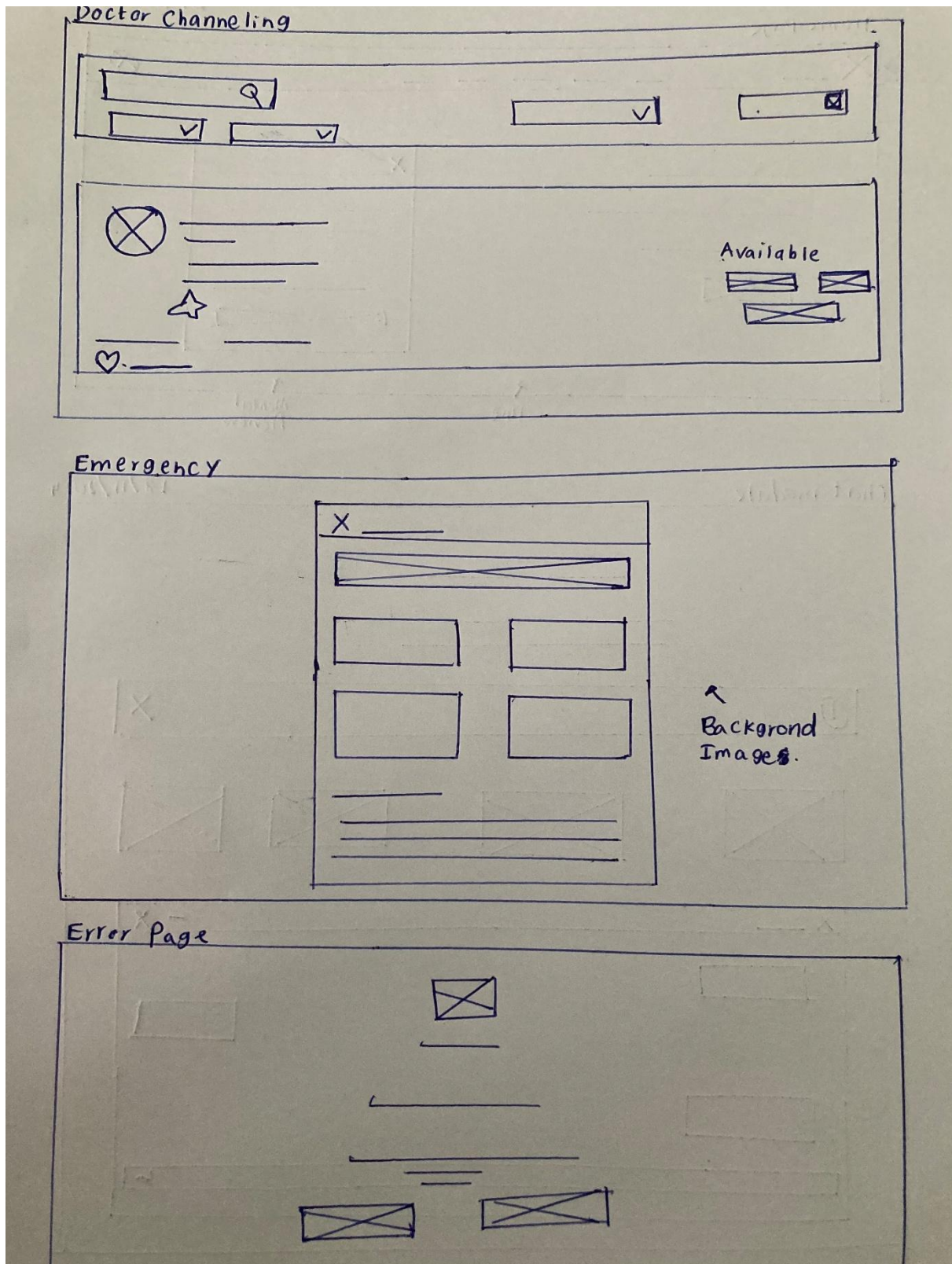


Figure 46 - Wireframes UI Design



Go to the Figma link given below for more designs (Low – High Fidelity).

Test Cases

Test ID	Test Description	Prerequisites	Test Steps	Expected Result	Actual Result	Result
TC01	User Registration - Successful registration	None	1. Open the app. 2. Navigate to the registration page. 3. Enter valid user details. 4. Submit the registration form.	User should be registered successfully and redirected to the home screen	User was registered successfully and redirected to the home screen	Passed
TC02	User Login - Valid credentials	Registered user account	1. Open the app 2. Navigate to login page 3. Enter valid credentials 4. Click login button	User should be authenticated and directed to the dashboard	User was authenticated and directed to the dashboard with all components loaded	Passed
TC03	Sentiment Analysis - Journal entry	Authenticated user	1. Navigate to journal section 2. Enter text with negative sentiment 3. Submit entry 4. View analysis results	System should correctly analyze and categorize negative sentiment with appropriate score	System correctly identified negative sentiment with a score of 0.78 and highlighted concerning phrases	Passed
TC04	Social Media Integration - Connect Twitter/X	Authenticated user with Twitter account	1. Go to profile settings 2. Select "Connect Social Media" 3. Choose Twitter/X 4. Complete OAuth authentication	Account should be successfully connected and show up in connected accounts list	Twitter account was connected successfully and displayed in the connected accounts list with last synced timestamp	Passed
TC05	Social Media Analysis - Fetch and analyze posts	Connected social media accounts	1. Navigate to insights dashboard 2. Select "Social Media Analysis" 3. Choose time period 4. View results	System should display sentiment analysis of recent social media posts with trend visualization	System displayed sentiment analysis but visualization failed to load for weekly trend view	Failed

Figure 47 - Test Cases

25 Test Cases Link – [CLICK HERE!](#)

Hyperparameter Tuning

Splitting for Training and Testing

```
[8]: # Splitting Dataset.  
X_train, X_test, y_train, y_test = train_test_split(X_resampled, y_resampled, test_size=0.2, random_state=42)
```

Hyperparameter Tuning

```
[9]: # Hyperparameter Tuning for Logistic Regression.  
param_grid = {  
    'C': [0.1, 1, 10],  
    'solver': ['lbfgs', 'liblinear'],  
    'penalty': ['l2'],  
}  
grid_search = GridSearchCV(LogisticRegression(max_iter=1000), param_grid, cv=5, scoring='accuracy')  
grid_search.fit(X_train, y_train)  
  
best_model = grid_search.best_estimator_  
print("\nBest Parameters:", grid_search.best_params_)
```

```
Best Parameters: {'C': 10, 'penalty': 'l2', 'solver': 'liblinear'}
```

Figure 48 - Hyperparameter Tuning

Data split to 80% training and 20% for testing. Workflow for finding optimal model configuration before final evaluation on test data.

Project Links

- All resources (diagrams, documents, test cases etc.) - [CLICK HERE!](#)